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Financial Markets and Products 目录

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1. Consider an exposure with a value of \$30.0 million. In L(+)/P(-) format, its expected loss is 4.0% with volatility of 15.0%; i.e., $\mu(L/P) = +4.0\%$, $\sigma = 15.0\%$. Conveniently, the exposure happens to have a normal distribution. Which is **nearest** to the single-period 97.5% expected shortfall (ES)?

考虑一笔名义金额为 \$30.0 百万美元的风险敞口。在 L(+)/P(-) 格式下，其预期损失为 4.0%，波动率为 15.0%；即 $\mu(L/P) = +4.0\%$, $\sigma = 15.0\%$ 。为方便起见，假定该敞口的分布为正态分布。单期 97.5% 期望损失（ES）中，哪一个数值与其**最接近**？

- A. 6.60 million
六百六十万
- B. 9.38 million
九百三十八万
- C. 11.72 million
一千一百七十二万
- D. 14.50 million
一千四百五十万

答案:C

解析:

C 选项正确。 If the distribution is normal, the 97.5% ES is approximated by the 99.0% VaR. In this case, we can use $\$30.0 * (4.0\% + 15\% * 2.33) = \$30.0 * 38.950\% = \$11.69$. The exact 97.5% ES is given by \$11.7210, so that is given as the correct choice.

C 选项正确。 若分布为正态分布，则 97.5% ES 可由 99.0% VaR 近似。此时，我们可以使用 $\$30.0 * (4.0\% + 15\% * 2.33) = \$30.0 * 38.950\% = \$11.69$ 。精确的 97.5% ES 为 \$11.7210，因此该数值被作为正确选项。

考察: Market Risk 重要程度: [■■■□]

科目: Quantitative Analysis

2. Among its financial positions, Zapex Financial Corporation has written a call, entered into an interest rate swap, and written a credit default swap. Among these (the following) four positions, which is the BEST example of wrong-way risk?

在 Zapex Financial Corporation 的金融头寸中，它写了看涨期权、进入了利率互换并写了信用违约互换。在这些（以下）四个头寸中，哪一个是最典型的反向风险？

- A. Zapex has written (aka, sold) a call option to a counterparty in the same industry
Zapex 向同一行业的对手方出售了看涨期权
- B. Zapex is the fixed-rate payer in an interest rate swap during an economic recovery
Zapex 是利率互换中的固定利率支付方，在经济复苏期间
- C. Zapex has written (aka, sold) a credit default swap (CDS) to counterparty whose credit quality is positively correlated with the reference entity
Zapex 向信用质量与参考实体正相关的对手方出售了信用违约互换
- D. Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty’s default probability
Zapex 在远期货币合约中持有正相关于对手方违约概率的标记市值收益

答案:D

解析:

D 选项正确。 Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty’s default probability.

Wrong-way (credit) risk is adverse correlation between the exposure and the credit quality, so wrong-way risk is a positive (negative) correlation between exposure and default probability (credit quality). This is wrong-way risk because greater exposure is associated with a lower likelihood of payment. Algebraically, wrong-way (or right-way) risk is some dependence between exposure/EAD and PD. We normally write $EL\% = PD \times EAD$ but that assumes independence between PD and EAD, so a way to think about WWR is simply when $EL > PD \times EAD$, because $EL \neq PD \times EAD$ is also a condition for non-independence.

In regard to (A), (B) and (C), each is an example of RIGHT-WAY risk.

考察: Credit Risk 重要程度: [■ ■ □]

科目: Financial Markets and Products

3. In December 2017, the Basel Committee (BCBS) made several revisions to the Basel III framework. This included revisions to both the standardized approach to credit risk (“Credit risk accounts for the bulk of most banks’ risk-taking activities and hence their regulatory capital requirements. The standardized approach is used by the majority of banks around the world, including in non-Basel Committee jurisdictions,” wrote the Committee) and the internal ratings-based (IRB) approaches to credit risk (“The financial crisis highlighted a number of shortcomings related to the use of internally modeled approaches for regulatory capital, including the IRB approaches to credit risk. These shortcomings include the excessive complexity of the IRB approaches, the lack of comparability in banks’ internally modeled IRB capital requirements and the lack of robustness in modelling certain asset classes,” wrote the Committee).¹

In regard to these December 2017 revisions to credit risk approaches in Basel III, which of the following statements is TRUE?

¹ “High-level summary of Basel III reforms,” (Basel Committee on Banking Supervision Publication, December 2017)

2017 年 12 月，巴塞尔委员会（BCBS）对巴塞尔协议 III 框架进行了多项修订。其中包括对信用风险标准法的修订（委员会写道：“信用风险占据了大多数银行风险承担活动以及相应监管资本要求的主要部分。标准法被世界上大多数银行采用，包括那些不在巴塞尔委员会成员辖区内的银行”），以及对信用风险内部评级法（IRB）方法的修订（委员会写道：“金融危机凸显了与监管资本内部模型方法使用相关的一些缺陷，其中包括信用风险内部评级法（IRB）方法。这些缺陷包括 IRB 方法的过度复杂性、各银行内部模型化的 IRB 资本要求缺乏可比性，以及在对某些资产类别建模时缺乏稳健性”）。¹ 关于 2017 年 12 月对巴塞尔 III 中信用风险计量方法所作的这些修订，以下哪一项表述是正确的？¹ “High-level summary of Basel III reforms”（《巴塞尔协议 III 改革的高层次概要》），巴塞尔银行监管委员会出版物，2017 年 12 月

- A. In the standardized approach (SA) for residential real estate exposures, the risk weights are lower if the loan-to-value (LTV) falls into a lower band
在住宅房地产敞口中，标准法（SA）下，如果贷款价值比（LTV）落入较低的区间，则风险权重较低
- B. The standardized approach (SA) eliminated the treatment of unrated exposures because all exposures are required to have an external credit rating
标准法（SA）消除了对无评级敞口的治疗，因为所有敞口都必须具有外部信用评级
- C. The internal ratings-based (IRB) approach removed the option to use the foundation IRB (F-IRB) because the advanced IRB (A-IRB) must be used for all asset classes
内部评级法（IRB）消除了使用基础 IRB（F-IRB）的选项，因为所有资产类别都必须使用高级 IRB（A-IRB）
- D. The internal ratings-based (IRB) approach removed the parameter input floors (in the case of PD, LGD, and EAD) to increase RWA variability across different banks
内部评级法（IRB）消除了参数输入地板（在 PD、LGD 和 EAD 的情况下），以增加不同银行之间的 RWA 可变性

答案:A

解析:

Option A is correct. In the standardized approach (SA) for residential real estate exposures, the risk weights is lower if the loan-to-value (LTV) falls into a lower band.

See below. Writes BCBS (High-level summary of Basel III reforms, December 2017): “For residential real estate exposures, more risk-sensitive approaches have been developed, whereby risk weights vary based on the LTV ratio of the mortgage (instead of the existing single risk weight) and in ways that better reflect differences in market structures.”

Residential real estate exposures								
LTV bands	Below 50%	50% to 60%	60% to 70%	70% to 80%	80% to 90%	90% to 100%	above 100%	Criteria not met
General RRE								
Whole loan approach RW	20%	25%	30%		40%	50%	70%	RW of counterparty
Loan-splitting approach ² RW	20%		RW of counterparty					RW of counterparty
Income-producing residential real estate (IPRRE)								
Whole loan approach RW	30%	35%	45%		60%	75%	105%	150%

In regard to (B), (C) and (D), each is FALSE. Instead, the following are true:

- The Basel III reforms included a more granular approach to unrated exposures and reduced the “mechanistic reliance on credit ratings”
- In regard to the internal ratings-based approaches, GARP writes, “the financial crisis highlighted a number of shortcomings related to the use of internally modeled approaches for regulatory capital, including the IRB approaches to credit risk. These shortcomings include the excessive complexity of the IRB approaches, the lack of comparability in banks’ internally modeled IRB capital requirements and the lack of robustness in modelling certain asset classes. To address these shortcomings, the Committee has made the following revisions to the IRB approaches: (i) removed the option to use the advanced IRB (A-IRB) approach for certain asset classes; (ii) adopted “input” floors (for metrics such as probabilities of default (PD) and loss-given-default (LGD)) to ensure a minimum level of conservatism in model parameters for asset classes where the IRB approaches remain available; and (iii) provided greater specification of parameter estimation practices to reduce RWA variability.” — Source: 2020 FRM Part II: Operational Risk and Resiliency, 10th Edition.
- In regard to the Specification of Input Floors: “The revised IRB framework also introduces minimum floor values for bank-estimated IRB parameters that are used as inputs to the calculation of RWA. These include PD floors for both the F-IRB and A-IRB approaches, and LGD and EAD floors for the A-IRB approach. In some cases, these floors consist of recalibrated values of the existing Basel II floors. In other cases, the floors represent new constraints for banks’ IRB models.” — Source: 2020 FRM Part II: Operational Risk and Resiliency, 10th Edition.

考察: Operational Risk and Resiliency 重要程度: [■■■□]

科目: Quantitative Analysis

4. An asset that is not very liquid is quoted bid \$89.00, offer \$96.00. Which is **nearest** to the proportional bid-offer spread?
一笔不太流动的资产的报价为买入价 \$89.00，卖出价 \$96.00。哪一个数值与其**最接近**的相对买卖价差？

- A. 0.0378
- B. 0.0757
- C. 0.1538
- D. 7.000

答案:B

解析:

B 选项正确。 0.0757 . Because $(96.00 - 89.00)/92.50 = 0.075676$

According to Hull (24.1.2. Measuring Market Liquidity), the two measures of bid-offer spread are:

- Dollar bid-offer spread = p = Offer price – Bid price
- Proportional bid-offer spread = s = (Offer price – Bid price) / Mid-market price

(Source: John C. Hull, Risk Management and Financial Institutions, 5th Edition (Hoboken, NJ: John Wiley & Sons, 2018))

考察: Market Liquidity 重要程度: [■■■□]

科目: Quantitative Analysis

5. Richard conducted a factor regression for his firm’s Flagship Portfolio. He used the three-factor Fama-French specification where the factors are MKT (the market), SMB (small stocks minus big stocks), and HML (value stocks minus growth stocks). His factor regression results are displayed below.

***Flagship excess returns, $R(p) - R_f$,
Fama-French benchmark (monthly)***

	Coeffi- cient	Standard Error (SE)
Alpha, α	0.300%	0.033%
MKT Loading	1.250	0.096
SMB Loading	0.640	0.043
HML Loading	-0.920	0.066
Adjusted R^2	0.290	

Please note the regression is based on monthly returns. Each of the following statements is TRUE about these factor regression results EXCEPT which is false?

Richard 对其公司旗舰组合进行了因子回归。他采用了三因子 Fama-French 模型设定，其中因子包括 MKT（市场）、SMB（小盘股减大盘股）以及 HML（价值股减成长股）。其因子回归结果如下所示。

***Flagship excess returns, $R(p) - R_f$,
Fama-French benchmark (monthly)***

	Coeffi- cient	Standard Error (SE)
Alpha, α	0.300%	0.033%
MKT Loading	1.250	0.096
SMB Loading	0.640	0.043
HML Loading	-0.920	0.066
Adjusted R^2	0.290	

请注意，该回归是基于月度收益率进行的。以下关于这些因子回归结果的陈述中，除了哪一项为假，其余均为真

- A. The portfolio's style appears to growth-oriented, small-cap
该组合的风格似乎是成长型、小盘股
- B. This portfolio manager(s) appear to be skillful; that is, the alpha is significant
该组合经理似乎很擅长；即，alpha 是显著的
- C. The low adjusted R^2 and low standard errors imply these factors are not appropriate to this portfolio
低调整后的 R^2 和低标准误差表明这些因子不适合该组合
- D. The implied CAPM benchmark is leveraged as it borrows cash (i.e., T-bills) to invest more than 100% (i.e., > 1.0) in the market portfolio
隐含的 CAPM 基准被杠杆化，因为它借入现金（即，T-bills）以投资超过 100%（即， > 1.0 ）的市场组合

答案:C

解析:

C 选项正确。C. FALSE: The factors are each significant and the adjusted R^2 is relatively high for empirical finance. In regard to (A), (B) and (D), each is true.

考察: Finance: Portfolio Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

6. The riskfree rate is 5.0% while the current stock price is \$80.00. Two most liquid options both have a strike price of \$70.00 such that the ratio $K/S(0) = 0.90$. In addition to the same strike price, both options have the same maturity of three months, $T = 0.25$ years. The implied volatility of the call is 30.0%, and the implied volatility of the put is 38.0%. What is the trade that exploits this scenario, if any?

无风险利率为 5.0%，当前股票价格为 \$80.00。两种最流动的期权都以 \$70.00 的执行价格，使得比率 $K/S(0) = 0.90$ 。此外，两种期权都具有相同的到期时间，即三个月， $T = 0.25$ 年。看涨期权的隐含波动率为 30.0%，看跌期权的隐含波动率为 38.0%。如果存在这样的交易机会，请指出哪种交易策略最合适？

- A. Buy call, sell put, and short the stock
买入看涨期权，卖出看跌期权，并卖空股票
- B. Buy put, sell call, and buy the stock
买入看跌期权，卖出看涨期权，并买入股票
- C. Buy the call, buy the put, and short a straddle
买入看涨期权，买入看跌期权，并卖空跨式期权
- D. Sell the call, sell the put, and buy a straddle
卖出看涨期权，卖出看跌期权，并买入跨式期权

答案:A

解析:

A 选项正确。True: Buy call, sell put, and short the stock

考察: Derivatives: Options 重要程度: [■ ■ □]

科目: Quantitative Analysis

7. Consider a netting set with two derivative positions: an interest rate swap (IRS) with a marginal CVA of -1.450 and a cross-currency swap (CCS) with a marginal CVA of -1.550 . If the first transaction is the cross-currency swap (CCS) and its incremental CVA is -2.100 , then what is the incremental CVA of the IRS?

考虑一个净额结算集，其中包含两个衍生品头寸：一个利率互换（IRS），其边际 CVA 为 -1.450 ，一个交叉货币互换（CCS），其边际 CVA 为 -1.550 。如果第一个交易是交叉货币互换（CCS），其增量 CVA 为 -2.100 ，那么利率互换（IRS）的增量 CVA 是多少？

- A. -0.900
- B. -1.100
- C. -1.550
- D. -2.220

答案:A

解析:

A 选项正确。The total CVA is the sum of marginal CVAs and therefore must be $-1.450 + (-1.550) = -3.000$. The incremental CVAs must sum to the total (netted) CVA such that, if the IRS is second in sequence, its incremental CVA is equal to $-3.000 - (-2.100) = -0.900$.

考察: Credit Risk 重要程度: [■ □ □ □ □]

8. The central feature of Basel III's first pillar is the regulatory capital (i.e., CET1, Tier 1, and total capital) requirements which require minimum fractions of risk-weighted assets (RWA). In addition to these minimums, as of early 2019 (when the buffers were fully phased in), Basel specifies three buffers: the capital conservation buffer (CCB), the G-SIB buffer, and the countercyclical buffer (CCyB). This is all very confusing, so you ask your colleague Mary to summarize these additional buffers and their motivations. She makes the following five points:

- I. For all three, the additional buffer must be CET1
- II. For all three, breach of the buffer requirements implies the bank's ability to pay dividends will be restricted
- III. The CCB requires an additional 2.5% of CET1 capital and is meant to ensure that banks have an additional layer of usable capital that can be drawn down when losses are incurred.
- IV. The additional G-SIB requirement includes five buckets {1.0%, 1.5%, 2.0%, 2.5%, or 3.5%} https://www.bis.org/fsi/fsisummary_g-sib_framework.htm and is meant to reduce the likelihood and severity of the failure of a global systemically important financial institution
- V. The CCyB requirement varies between zero and 2.5% and is meant to protect the banking sector from periods of excess aggregate credit growth that have often been associated with the build-up of system-wide risks

Is Mary correct in her summary?

巴塞尔 III 第一支柱的核心特征是监管资本（即 CET1、一级资本和总资本）要求，这些要求规定了相对于风险加权资产（RWA）的最低资本比例。除这些最低要求之外，截至 2019 年初（当各类缓冲全部过渡到位时），巴塞尔还规定了三种缓冲：资本留存缓冲（CCB）、全球系统重要性银行缓冲（G-SIB 缓冲）以及逆周期资本缓冲（CCyB）。这些内容非常容易让人困惑，因此你请同事 Mary 总结这些额外缓冲及其设立动机。她给出了以下五点：

- I. 对这三类缓冲而言，额外缓冲部分都必须由 CET1 构成
- II. 对这三类缓冲而言，一旦未满足缓冲要求，银行支付股利的能力将受到限制
- III. CCB 要求额外 2.5% 的 CET1 资本，旨在确保银行拥有一层在发生损失时可以被动用的额外可用资本。
- IV. 额外的 G-SIB 要求包含五个档次 {1.0%, 1.5%, 2.0%, 2.5%, or 3.5%} <https://www.bis.org/fsi/fsisummaries/g-sibframework.htm>，旨在降低全球系统重要性金融机构倒闭的可能性及其严重程度
- V. CCyB 要求在零和 2.5% 之间变动，旨在保护银行体系免受总量信贷过度增长时期的影响，而这种过度增长往往伴随着系统性风险的积聚

Mary 在她的总结中是否正确？

- A. No, unfortunately, none of her statements is correct
不幸的是，她的陈述没有一项是正确的
- B. I. and II. are correct, but the others (III., IV., and V.) are wrong
I. 和 II. 是正确的，但其他（III.、IV. 和 V.）是错误的
- C. III., IV., and V. are correct, but the others (I. and II.) are wrong
III.、IV. 和 V. 是正确的，但其他（I. 和 II.）是错误的
- D. Yes, all five statements are correct
是的，所有五项陈述都是正确的

答案:D

解析:

D 选项正确。 Yes, all five statements are correct

- In regard to true statements (I.) and (II.), the capital conservation buffer (CCB), the G-SIB buffer, and the countercyclical buffer (CCyB) each require common equity Tier 1 (CET1) which is a buffer of the highest quality. Further, breach of any of them implies restrictions on dividend payments (in addition to other supervisor tools such as restriction on executive bonuses)
- In regard to true statement (III.), the BCBS says that “the capital conservation buffer [CCB] was introduced to ensure that banks have an additional layer of usable capital that can be drawn down when losses are incurred” (see https://www.bis.org/fsi/fsisummaries/b3_capital.htm) and GARP says “In the case of the CCB, the rationale roughly follows that for the Prompt Corrective Action (PCA) system built into U.S. capital regulation beginning in 199; i.e., a bank with ratios that begin to approach the minimums should be subject to increasingly stringent supervisory intervention in order to induce a return to well-capitalized status).” -2020 FRM Part II: Operational Risk and Resiliency, 10th Edition.
- In regard to true statement (IV.), the G-SIB requirement includes five buckets {1.0%, 1.5%, 2.0%, 2.5%, or 3.5%} and, explains GARP, “In the case of the G-SIB buffer, the rationale is similar to that for the CCB but also recognizes the very large costs to society of distress at G-SIBs (and the higher volatility of losses at some of them). Thus, larger buffers are specified to further reduce the chance of failure.”
- In regard to true statement (V.), “The CCyB [countercyclical capital buffer] has two rationales. One is to provide an instrument for macroprudential restraint of overheating; the other is attentive to the cost of capital,” says GARP. Further, explains BCBS, “The countercyclical capital buffer [CCyB] aims to ensure that banking sector capital requirements take account of the macro-financial environment in which banks operate. Its primary objective is to use a buffer of capital to achieve the broader macroprudential goal of protecting the banking sector from periods of excess aggregate credit growth that have often been associated with the build-up of system-wide risk. Due to its countercyclical nature, the countercyclical capital buffer regime may also help to lean against the build-up phase of the credit cycle in the first place. In downturns, the regime should help to reduce the risk that the supply of credit will be constrained by regulatory capital requirements that could undermine the performance of the real economy and result in additional credit losses in the banking system.” See <https://www.bis.org/bcbs/ccyb/>

Keep in mind the basic requirements of these three additional buffers: the CCB requires +2.5% CET1; the G-SIB requires either {+1.0%, +1.5%, +2.0%, +2.5%, or +3.5%} depending on the bank’s score (see <https://trtl.bz/fsb-g-sibs>); and the CCyB requires between zero and 2.5%.

考察: Finance: Regulatory Capital 重要程度: ■■□

科目: Quantitative Analysis

9. A portfolio holds 100,000 shares of a stock and this single position has a value of \$3.0 million. The stock is quoted bid \$29.00, offer \$31.00. The stock’s daily volatility is 1.43% or 143 basis points. For purposes of value at risk (VaR), we will assume the stock’s arithmetic returns are normally distributed (aka, normal VaR) and the expected daily return rounds to zero (under these assumptions absolute VaR is identical to relative VaR). Which is **NEAREST** to the position’s one-day 99.0% confident liquidity-adjusted value at risk (LVaR)?

一个投资组合持有 100,000 股股票，该单一头寸的价值为 300 万美元。股票的买卖报价为 29 美元/股和 31 美元/股。股票的日波动率为 1.43% 或 143 个基点。为了计算价值风险 (VaR)，我们假设股票的算术回报率是正态分布的 (即，正态 VaR)，并且预期日回报率近似为零 (在这些假设下，绝对 VaR 与相对 VaR 相同)。哪一个选项最接近该头寸的单日 99.0% 置信水平的流动性调整价值风险 (LVaR)?

- A. \$15,000
- B. \$80,000
- C. \$200,000
- D. \$1.0 million

答案:C

解析:

Option C is correct. The one-day 99.0% VaR is given by $-0 + 1.43\% * 2.326 * \$3.0 \text{ million} = \$99,800$.

The liquidity cost (LC) is given by $(31.00 - 29.00)/30.00 * 0.5 * \$3.0 \text{ million} = \$100,000$.

The one-day 99.0% VaR is therefore $99,800 + 100,000 = 199,800$

考察: Market Risk: Liquidity Risk 重要程度: [■■■□]

科目: Quantitative Analysis

10. According to Grinold, typical or illustrative information ratios (IR) include: 0.500 is good, 0.750 is very good and 1.00 is exceptional. In regard to risk aversion, λ , 0.150 is restrained, 0.100 is moderate, and 0.050 is aggressive. Patricia is a manager with a “very good” information ratio of 0.750 and a “moderate” risk aversion of 0.10. If she shifts from this moderate risk aversion to an aggressive risk aversion of 0.050, what is the impact on the optimal level of active risk?

根据 Grinold 的说法，典型的或说明性的信息比率（IR）包括：0.500 是好的，0.750 是非常好的，1.00 是异常的。在风险厌恶程度上， λ , 0.150 是抑制的，0.100 是温和的，0.050 是激进的。帕特丽夏是一位经理，她有一个“非常好”的信息比率 0.750 和一个“温和”的风险厌恶 0.10。如果她从这种温和的风险厌恶转变为激进的风险厌恶 0.050，对最优主动风险水平有什么影响？

- A. No effect
没有影响
- B. Optimal active risk reduces by one-half
最优主动风险减少一半
- C. Optimal active risk Increases by about +41%
最优主动风险增加约 +41%
- D. Optimal active risk doubles
最优主动风险翻倍

答案:D

解析:

D 选项正确。 True: Optimal active risk doubles As the optimal level of active risk is given by $IR/(2 * \lambda)$, Patricia doubles her optimal level when she goes from $0.750/(2 * 0.10) = 3.750$ to $0.750/(2 * 0.050) = 7.500$

考察: Portfolio Management: Fundamental Law of Active Management 重要程度: [■■■□]

科目: Quantitative Analysis

11. For his term structure model, Peter considered the Vasicek model, but on reflection, he prefers the Cox-Ingersoll-Ross (CIR). In regard to the CIR, which of the following statements is TRUE?

彼得考虑使用 Vasicek 模型来构建他的期限结构模型，但他后来更喜欢 Cox-Ingersoll-Ross（CIR）模型。在 CIR 方面，以下哪个陈述是正确的？

- A. A drawback of the CIR is that it does not mean revert to a long-term rate
一个缺点是 CIR 不能恢复到长期利率
- B. In the CIR, the drift is time-dependent such that CIR is a no-arbitrage model
CIR 的漂移是时间依赖的，因此 CIR 是一个无套利模型
- C. An advantage of the CIR is that the terminal distribution of the short rate is eventually (e.g., > 10 years) normally distributed
CIR 的一个优点是，短期利率的最终分布（例如，>10 年）最终是正态分布的
- D. Unlike the Vasicek model’s constant volatility, the CIR’s basis point volatility is not constant, and this ensures the short rate cannot become negative
与 Vasicek 模型的恒定波动率不同，CIR 的基点波动率不是恒定的，这确保了短期利率不会变为负数

答案:D

解析:

D 选项正确。 True:Unlike the Vasicek model’s constant volatility, the CIR’s basis point volatility is not constant, and this ensures the short rate cannot become negative.

In regard to (A), (B), and (C), each is FALSE.

考察: Term Structure Models 重要程度: [■■■□]

科目: Quantitative Analysis

12. Jason is a small business owner specializing in the wholesale of steel. Jason worked hard on his business, so he purchased insurance to help safeguard his company on January 4th, 2023. Later that month, on January 31st, he made a significant payment to secure a shipment of steel that will arrive on February 2nd, 2023. Jason delivered the steel to two clients on February 8th and 9th; however, he did not receive payment from the clients until February 13th. After receiving the proceeds Jason deposited the money at ABC bank.

Among these transaction dates, on which did Jason increase his credit risk?

贾森是一家钢铁批发商。贾森努力经营他的业务，所以他在 2023 年 1 月 4 日购买了保险，以帮助保护他的公司在 1 月 31 日，他支付了一大笔钱以确保在 2 月 2 日到货的钢铁。贾森在 2 月 8 日和 2 月 9 日向两个客户交付了钢铁；然而，他直到 2 月 13 日才收到客户的付款。收到款项后，贾森将钱存入 ABC 银行。在这些交易日期中，哪一天贾森增加了他的信用风险？

A. None of the transactions increase his credit risk

这些交易没有增加他的信用风险

B. His credit risk increased on the Jan dates (Jan 4th and 31st) but was unchanged on Feb dates (Feb 8th, 9th and 13th)

他的信用风险在 1 月日期（1 月 4 日和 31 日）增加，但在 2 月日期（2 月 8 日、9 日和 13 日）保持不变

C. His credit risk increased on the Feb dates (Feb 8th, 9th, and 13th) but was unchanged on Jan dates (Jan 4th and 31st)

他的信用风险在 2 月日期（2 月 8 日、9 日和 13 日）增加，但在 1 月日期（1 月 4 日和 31 日）保持不变

D. His credit risk increased on all of the cited transaction dates; i.e., both Jan dates (4th and 31st) and all Feb dates (8th, 9th, and 13th)

他的信用风险在所有引用的交易日期增加；即，1 月日期（4 日和 31 日）和所有 2 月日期（8 日、9 日和 13 日）

答案:D

解析:

D 选项正确。 True: His credit risk increased on all of the cited transaction dates; i.e., both Jan dates (4th and 31st) and all Feb dates (8th, 9th, and 13th)

- Jan 4th: Purchasing insurance increases credit risk because Jason is buying a policy that will not be delivered if ever once an event occurs. *Contingent claims* are a type of credit risk. According to the text, “A sixth type of transaction is a special case of a claim on an asset—a contingent claim. The claim is contingent on certain events occurring, such as a loss covered by an insurance policy. At policy inception, the policyholder has no claim on the insurer. However, once the insured suffers a covered loss, the insured has a claim. If the insurer fails to pay the claim, this would constitute a credit loss.” (†)
- Jan 31st: He prepays for an order and thus has the possibility of his counterpart not delivering on their obligation. According to the text, “*Prepayment of goods and services* is a fourth type of transaction that involves credit risk. Delivery is expected at a certain time and of a certain quality and/or performance, and the failure of the counterparty may lead to the loss of the advanced payments and also generates business interruption costs.” (†)
- February 8th and 9th: Jason delivered an order of steel to his clients. Either of his clients could have not delivered on their obligation to pay him, and thus, Jason increased his credit risk. According to the text, “The third type is the sale of a product or a service without immediate cash payment. The seller sends an invoice to the buyer after the product has been shipped or the service performed, and the buyer has a few weeks to pay. This is known as an *account receivable*.” (†)
- February 13th: He deposited the proceeds of his sale and his earnings from last quarter at ABC Bank. Jason granted a third-party custody over his assets, and thus, he has the risk of the bank not being able to repay his assets. Per the text, “A fifth type of transaction that creates credit risk involves a party’s claim on an asset in the custody of or under the management of another party, such as a bank deposit.” (†)

Credit risk can be generated in myriad ways. The text(†) lists seven:

1. Lending: Risk of non-repayment from the borrower to the lender.
2. Leases: Risk that the lessee will fail to make future payments for the use of an asset.
3. Accounts Receivable (Sale of Product/Service without Immediate Payment): Risk that the buyer will not pay after receiving the product or service.
4. Prepayment of Goods and Services: Risk that the counterparty will not deliver or meet the quality/performance after receiving payment.
5. Deposits and Bank Balances: Risk that the financial institution holding deposits or managing assets will default.
6. Contingent Claims (e.g., Insurance, Pension Fund Claims): Risk that the insurer or sponsor will not be able to pay out the claim or pension liabilities.
7. Derivative Exposures: Risk from future payment obligations in derivative transactions like swaps or futures, which are dependent on the underlying asset's market value.

(†) Sylvain Bouteille and Diane Coogan-Pushner, **The Handbook of Credit Risk Management: Originating, Assessing, and Managing Credit Exposures** (2nd Edition, Hoboken, NJ: John Wiley & Sons, 2022).

考察: Finance: Credit Risk 重要程度: [■□□]

科目: Quantitative Analysis

13. Peter is studying for the Financial Risk Manager (FRM) exam, and he thinks the historical evolution of the Basel regulations is confusing. In particular, he wants to better understand the difference between the original Basel I accord (aka, the 1988 Basel Accord that was implemented by 1992) and Basel II, which was finally published over ten years later in 2004. His colleague Mary explains that, in comparison to the original accord, Basel II contained four significant innovations. Specifically, in comparison to the original Basel I accord (which do include the 1995/1996 Amendments) that came before, Mary says that significant innovations in Basel II included the following:

- I. In addition to credit risk and market risk, Basel II required capital for operational risk
- II. Basel II eliminated risk weights and risk-weighted assets (RWA) and replaced them with direct calculation of risk charges
- III. Basel II contained specific requirements for supervision related to capital and risk management (Pillar 2) and required public disclosures (Pillar 3)
- IV. To fine-tune the accord's design, Basel II made repeated use of Quantitative Impact Studies (QIS) to which banks contributed data (i.e., feedback) that was analyzed by supervisors

Is Mary correct?

Peter 正在备考金融风险管理师 (FRM) 考试, 他觉得巴塞尔法规的历史演进非常令人困惑。具体来说, 他想更好地理解最初的巴塞尔 I 协议 (又称 1988 年巴塞尔协议, 并在 1992 年前后实施) 与巴塞尔 II (于十多年后的 2004 年发布) 之间的差异。他的同事 Mary 解释说, 与最初的协议相比, 巴塞尔 II 包含了四项重要创新。特别是, 相较之前的巴塞尔 I 原始协议 (其中包括 1995/1996 年的修订), Mary 认为巴塞尔 II 的重大创新包括以下内容:

- I. 除了信用风险和市场风险之外, 巴塞尔 II 还要求对操作风险计提资本
- II. 巴塞尔 II 取消了风险权重和风险加权资产 (RWA), 改为直接计算风险资本占用
- III. 巴塞尔 II 包含与资本和风险管理相关的监管要求 (第二支柱), 并要求公开披露 (第三支柱)
- IV. 为了对协议设计进行微调, 巴塞尔 II 多次采用量化影响研究 (QIS), 银行提供数据 (反馈), 由监管方进行分析

Mary 的说法正确吗?

- A. Yes, all four statements are correct
是的, 所有四个陈述都是正确的
- B. I. and II. are correct but III and IV are inaccurate
I. 和 II. 是正确的, 但 III 和 IV 是不准确的

- C. Only II. is inaccurate but I, III, and IV are correct
只有 II. 是不准确的，但 I, III, 和 IV 是正确的
- D. Unfortunately, none of her statements are correct
不幸的是，她的陈述都不正确

答案:C

解析:

Option C is correct. Basel II did NOT eliminate risk-weighted assets (or risk weights); risk-weighted assets (RWA) do continue in Basel III/IV. According to GARP’ s Mark Carey, the four significant innovations of Basel II (as an improvement over the original Basel I accord) were: more sophisticated risk weight formulas; the addition of an operational risk capital charge; Pillars 2 and 3; and the repeated use of the Quantitative Impact Studies (QIS). Many experts consider the additional Pillars to be a highly significant development. As Carey explains, “The three pillars represented a push toward convergence of national practices [especially in the context of the “rules versus principles” disparity]. Specifically, Pillar 2 mandated that supervisors require banks to have more than the minimum amount of capital as well as internal capital adequacy and assessment processes (ICAAP) that take their risk profile into account ... National discretion regarding enforcement of the accord’ s provisions was reduced, and national regulators were to be transparent about their implementation efforts, including those concerning the requirements in excess of the minimums. Pillar 3 required more qualitative and quantitative disclosures, in the hope that pressure from market participants would help improve banks’ practices [aka, market discipline].” (Source: 2020 FRM Part II: Operational Risk and Resiliency, 10th Edition. Pearson Learning Solutions)

考察: Finance: Basel Accords 重要程度: [■■□]

科目: Quantitative Analysis

14. Assume the spot foreign exchange (FX) rate between the US dollar and the Japanese Yen is 110.00 which can be represented as USDJPY 110.00 or JPY 110.00 where USD is the base currency and JPY is the quote currency; aka, quote 110.00 yen per one dollar of the base or “per unit” currency. The USD interest rate is 2.50% and the JPY interest rate is 0.450%. The period is one year, and the compound frequency is annual. If the covered interest rate (CIP) arbitrage framework enforces a perfectly accurate one-year forward exchange rate, then what is the implied one-year “swap rate” which is here simply defined as the difference between the forward FX rate and the spot FX rate; put simply, what is the difference, $F(\text{USDJPY}) - S(\text{USDJPY})$, or $F(\text{USDJPY}) - 110.00$?

假设美元/日元即期汇率是 110.00，可以表示为 USDJPY 110.00 或 JPY 110.00，其中 USD 是基础货币，JPY 是报价货币；即，报价 110.00 日元每单位基础货币美元。美元利率为 2.50%，日元利率为 0.450%。期限为一年，复利频率为年复利。如果套利框架中的覆盖利率（CIP）强制执行一个完美准确的远期汇率，那么什么是隐含的一年“掉期率”，这里 simply defined as the difference between the forward FX rate and the spot FX rate; put simply, what is the difference, $F(\text{USDJPY}) - S(\text{USDJPY})$, or $F(\text{USDJPY}) - 110.00$?

- A. -2.20 JPY (or -2.20 USDJPY)
- B. Zero
- C. + 5.80 JPY
- D. + 13.70 JPY

答案:A

解析:

A 选项正确。 True: $F - S = -2.20 \text{ USDJPY}$

If CIP is enforced, then $F - S = S \times [(1 + \text{Rf_JPY}) / (1 + \text{Rf_USD}) - 1] = 110.00 \times [(1 + 0.45\%) / (1 + 2.50\%) - 1] = -2.20$.

考察: Financial Markets and Products 重要程度: [■■□]

科目: Quantitative Analysis

15. Volatility is an important risk factor according to Andrew Ang. The strong negative relationship between VIX changes and stock returns is explained by (at least) two dynamics. The first is called the leverage effect. The “second channel is a time-varying risk premium story and is the one the basic CAPM [capital asset pricing model] advocates,” explains Ang.(†)

Assume the riskfree rate is 3.0% while the equity risk premium (EPR; aka, the market’s expected excess return) is 6.0% because the market’s expected return is 9.0%. According to the CAPM, and consistent with Ang’s theory on the transmission channel between volatility and stock returns, which of the following should cause a DECLINE in a stock’s price?
根据 Andrew Ang 的说法，波动性是一个重要的风险因素。VIX 变化和股票回报之间的强负相关关系至少由两个动态解释。第一个被称为杠杆效应。第二个通道是一个时间变化的风险溢价故事，是基本 CAPM（资本资产定价模型）倡导的。Ang 解释说：假设无风险利率为 3.0%，而股权风险溢价（EPR；即，市场预期超额回报）为 6.0%，因为市场预期回报为 9.0%。根据 CAPM，并且与 Ang 关于波动率和股票回报之间的传导渠道的理论一致，以下哪个应该导致股票价格下降？

- A. The VIX Index declines from 21.00 to 19.00
VIX 指数从 21.00 下降到 19.00
- B. The market’s volatility decreases and the stock’s volatility decreases along with it
市场波动率下降，股票波动率也随之下降
- C. The stock’s volatility decreases, which corresponds to an increase in its leverage via the leverage effect
股票波动率下降，对应杠杆效应增加
- D. The stock’s volatility increases (along with an increase in its correlation to the market), while the market’s volatility is unchanged
股票波动率增加（伴随着与市场的相关性增加），而市场波动率保持不变

答案:D

解析:

Option D is correct. The stock’s volatility increases (along with an increase in its correlation to the market), while the market’s volatility is unchanged.

An increase in the stock’s volatility and/or increase in its correlation to the market, while the market’s volatility is unchanged (aka, ceteris paribus), implies an increase in the stock’s beta. This increases the stock’s expected return. Consequently, the stock’s price should decline.

Ang (emphasis ours): “The correlation between VIX changes and stock returns is -39% , so stocks do badly when volatility is rising. The negative relation between volatility and returns is called the leverage effect. When stock returns drop, the financial leverage of firms increases since debt is approximately constant while the market value of equity has fallen. This makes equities riskier and increases their volatilities. There is another channel where high volatilities lead to low stock returns: **an increase in volatility raises the required return on equity demanded by investors, also leading to a decline in stock prices.** This second channel is a time-varying risk premium story and is the one that the basic CAPM advocates: as market volatility increases, discount rates increase and stock prices must decline today so that future stock returns can be high.” (†)

In regard to (A), (B), and (C), each is false.

(†) Andrew Ang, Asset Management: A Systematic Approach to Factor Investing (New York: Oxford University Press, 2014).

考察: Finance: CAPM 重要程度: ■■■□

科目: Quantitative Analysis

16. An analyst fits a Vasicek model when the current short-term rate is 4.60%. If the theta, θ , parameter is 13.50%, then each of the following statements is true EXCEPT which is false?

分析师在当前短期利率为 4.60% 时拟合 Vasicek 模型。如果 theta 参数为 13.50%，那么以下哪个陈述是正确的，除了哪个是错误的？

- A. A key advantage of the Vasicek model is that it is a relatively simple model
Vasicek 模型的一个关键优势是它是一个相对简单的模型
- B. If the half-life is nine years, then we might expect the short rate to reach $\sim 9.0\%$ in nine years or so
如果半衰期为九年，那么我们可能期望短期利率在九年内达到 $\sim 9.0\%$
- C. The standard deviation of the short rate in ten years is less than it would be without mean reversion
十年期短期利率的标准差小于没有均值回归的情况

D. Because it is a multi-factor model, the Vasicek is popularly applied to hedge portfolios that contain short-, medium- and long-term bonds

因为它是一个多因素模型，Vasicek 被广泛应用于对包含短期、中期和长期债券的对冲组合

答案:D

解析:

D 选项正确。False. Vasicek is a single-factor model; to hedge short-, medium- and long-term bonds would require, for example, the three-factor Gauss+ model.

In regard to (A), (B), and (C), each is TRUE.

考察: Fixed Income: Vasicek Model 重要程度: [■■□]

科目: Quantitative Analysis

17. Firm A, Firm B, and Firm C are three companies in the bottled water industry. All companies have a mix of debt and equity in their capital structure. In 2022 all three companies had credit losses that lowered EBIT. Below is a summary of their financial information for the year 2022.

	Firm A	Firm B	Firm C
Total Assets	\$5,500,000	\$5,000,000	\$3,900,000
Total Debt	\$2,500,000	\$1,000,000	\$1,000,000
Debt Interest Rate	3.40%	3.40%	3.40%
Total Equity	\$3,000,000	\$4,000,000	\$2,900,000
Provision for Loan Loss	\$50,000	\$15,000	1,000
EBIT	\$75,000	\$75,000	\$75,000
Tax Rate	21%	21%	21%

Given the above information, which firm has the higher return on equity (ROE)?

公司 A、公司 B 和公司 C 是瓶装水行业的三家。所有公司都有债务和股权的混合资本结构。在 2022 年，三家公司的信用损失都降低了 EBIT。以下是 2022 年他们的财务信息总结。

	Firm A	Firm B	Firm C
Total Assets	\$5,500,000	\$5,000,000	\$3,900,000
Total Debt	\$2,500,000	\$1,000,000	\$1,000,000
Debt Interest Rate	3.40%	3.40%	3.40%
Total Equity	\$3,000,000	\$4,000,000	\$2,900,000
Provision for Loan Loss	\$50,000	\$15,000	1,000
EBIT	\$75,000	\$75,000	\$75,000
Tax Rate	21%	21%	21%

给出以上信息，哪家公司拥有更高的净资产收益率（ROE）？

- A. Firm C has the highest ROE due to its lower equity position.
公司 C 拥有最高的 ROE，因为它有较低的股权地位。
- B. Firm A had the higher ROE because of its large debt position.
公司 A 拥有更高的 ROE，因为它有较大的债务地位。
- C. Without credit losses, firm A would have had the highest ROE.
如果没有信用损失，公司 A 将拥有最高的 ROE。
- D. Without credit losses, firm B would have had the highest ROE.
如果没有信用损失，公司 B 将拥有最高的 ROE。

答案:A

解析:

A 选项正确。 True: Firm C has the highest ROE due to its lower equity position.

- Firm A has a negative ROE (−0.33%) because of its large debt position (as illustrated below). Even after sustaining credit losses the firm still must repay its debt holders. The A sustained heavy credit losses that lowered EBIT. However, even after adjusting for its credit losses Firm A still would have had a smaller ROE than firms B and C.
- Firm B had a positive ROE, but the ROE is very small because of its large equity compared to firms C and A. Firm A sustained a large credit loss that lowered EBIT. However, even after adjusting for its credit losses the firm would have still had a smaller ROE than firm C.
- Firm C had the highest ROE because of its small equity and modest credit loss.

	Firm A	Firm B	Firm C
Total Assets	\$5,500,000	\$5,000,000	\$3,900,000
Total Debt	\$2,500,000	\$1,000,000	\$1,000,000
Debt Interest Rate	3.40%	3.40%	3.40%
Total Equity	\$3,000,000	\$4,000,000	\$2,900,000
Provision for loan loss	\$50,000	\$15,000	1,000
EBIT	\$75,000	\$75,000	\$75,000
Tax Rate	21%	21%	21%
Solution			
EBIT	\$75,000	\$75,000	\$75,000
Interest	\$85,000.00	\$34,000.00	\$34,000.00
Tax	0	\$8,610.00	\$8,610.00
NI	(\$10,000.00)	\$32,390.00	\$32,390.00
ROE	-0.33%	0.81%	1.12%
Without Credit Loss			
EBIT	\$125,000	\$90,000	\$76,000
Interest	\$85,000.00	\$34,000.00	\$34,000.00
Tax	\$8,400.00	\$11,760.00	\$8,820.00
NI	\$31,600.00	\$44,240.00	\$33,180.00
ROE	1.05%	1.11%	1.14%

考察: Finance: Financial Statement Analysis 重要程度: [■ ■ □]

科目: Quantitative Analysis

18. The three lines of defense model (3LoD) has existed in some variation for over 30 years, although it achieved more visibility in 2013 when the Institute of Internal Auditors (IIA) published its initial position paper on the 3LoD, which it updated in 2021 (The IIA’s Three Lines Model, an update of the Three Lines of Defense). GARP recommends firms refer to this update because the IIA “has refined the model to reflect the evolution of risk management within organizations and aim to foster closer collaboration between business functions including internal audit.” In regard to the 3LoD model, each of the following is true EXCEPT which is false?

三道防线模型（3LoD）已经存在了 30 多年，尽管它在 2013 年获得了更多的关注，当时 Institute of Internal Auditors（IIA）发布了其最初的立场文件，该文件在 2021 年进行了更新（The IIA’s Three Lines Model, an update of the Three Lines of Defense）。GARP 建议公司参考这个更新，因为 IIA “已经完善了模型，以反映组织内部风险管理的发展，并旨在促进业务职能之间的更紧密合作，包括内部审计”。关于三道防线（3LoD）模型，下列各项陈述均为真，除了哪一项是假的

- A. The first line defines the organization’s risk appetite but should not attempt to directly influence its risk culture
第一道防线定义了组织的风险偏好，但不应试图直接影响其风险文化
- B. The first line is responsible for implementing and maintaining effective controls to manage material operational risks
第一道防线负责实施和维护有效的控制，以管理重要的运营风险

- C. The second line is part of management (and owns the ORM methodology) but provides support to the first line in order to help the firm create and protect value
第二道防线是管理的一部分（并拥有 ORM 方法论），但为第一道防线提供支持，以帮助公司创造和保护价值
- D. The third line has the clearest boundaries due to its independence
第三道防线由于其独立性，边界最清晰

答案:A

解析:

A 选项正确。 A. False, both clauses. Instead, the following is true: The risk appetite (i.e., risk appetite statement) is developed by senior management and approved by the board. The risk culture is not the sole responsibility but is foundational to the firm’s actual ORMF, such that all employees bear some responsibility for culture.

In regard to (B), (C), and (D), each is TRUE.

考察: Operational Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

19. Liquidity Early Warning Indicators (EWI) can be compared to an automobile’s dashboard signal system where the appearance of a red light points our attention to something that may become a problem if not addressed. Each of the following is a credible red indicator (aka, red flag) in a Liquidity Early Warning Indicator (aka, Liquidity EWI) dashboard EXCEPT which is not a red (red flag) liquidity indicator?

流动性早期预警指标（EWI）可以与汽车仪表盘信号系统进行比较，其中红色灯光的出现提醒我们注意可能成为问题的事情，如果不及早解决。以下各项都是流动性早期预警指标（流动性 EWI）仪表盘中的可信红色指示器（红色标志），除了哪一项不是红色（红色标志）流动性指示器？

- A. Narrowing debt/CDS spreads
债务/CDS 利差缩小
- B. Sudden increase in debt costs
债务成本突然增加
- C. Sudden growth in assets accompanied by volatile liabilities
资产突然增长伴随着波动性负债
- D. Rapid decline in the weighted average maturity of liabilities
负债加权平均期限突然下降

答案:A

解析:

A 选项正确。 A. FALSE. Instead, the liquidity EWI is: a widening debt/CDS spreads (consistent with a sudden increase in debt costs)

In regard to (B), (C), and (D), each is a candidate Liquidity EWI, per the Study Note: “The Basel Committee on Banking Supervision (BCBS) recommends certain measures as EWI. These measures include [collapsed here, new emphasis ours]:

- An unusual growth in assets, particularly when accompanied by volatile liabilities;
- Debt (credit) spreads *widen*, and/or credit default swap (CDS) spreads *widen*.
- Declining diversity in the makeup of assets and liabilities; Growing currency mismatches;
- When the weighted average of liabilities’ maturity declines;
- Positions going beyond or getting close to regulatory limits;
- Certain product line experience negative trends; The financial condition of the bank weakens; Public press that is negative;
- A downgrade in the credit rating; A decline in the stock price;
- Debt costs increase; Retail and/or wholesale funding costs increase;

- Counterparties becoming nervous about the financial condition of the bank;
- Credit lines are lowered;
- Outflows of retail deposits at an increased pace; Certificates of deposit (CDs) are increasingly redeemed;
- Longer-term funding opportunities become more difficult; and Placing short-term liabilities becomes more difficult”

考察: Liquidity Risk Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

20. In factor theory, an asset’s risk premium (aka, expected excess return) is given by $E(r_i) - R_f = \beta(i, m) \times \lambda(m)$ where the asset’s beta is with respect to the stochastic discount factor (SDF; aka, pricing kernel) and (m) in an index of bad time. Let’s make the following assumptions about an asset:

- The riskfree rate is 3.0% such that $E(m) = 1/1.03 = 0.9709$
- The variance (m) equals $30.0\%^2 = 0.090$
- The covariance (r_i, m) equals -0.10680

Which is **nearest** to the asset’s risk premium which refers to the asset’s expected excess return, $E(r_i) - R_f$?

在因子理论中，资产的风险溢价（aka，预期超额回报）由 $E(r_i) - R_f = \beta(i, m) \times \lambda(m)$ 给出，其中资产的 beta 是相对于随机贴现因子（SDF；aka，定价核）和 (m) 中的一个指数的坏时间。让我们对资产做出以下假设：

- 无风险利率为 3.0%，使得 $E(m) = 1/1.03 = 0.9709$
- 方差 (m) 等于 $30.0\%^2 = 0.090$
- 协方差 (r_i, m) 等于 -0.10680

哪一个选项最接近资产的风险溢价，即资产的预期超额回报， $E(r_i) - R_f$ ？

- A. -5.0%
- B. +4.0%
- C. +8.0%
- D. +11.0%

答案:D

解析:

D 选项正确。 The expected excess return, $E(r_i) - R_f = \beta(i, m) \times \lambda(m)$ where $\beta(i, m) = \text{cov}(r_i, m)/\text{var}(m)$ and $\lambda(m) = -\text{var}(m)/E(m)$.

In this case, $\beta(i, m) = \text{cov}(r_i, m)/\text{var}(m) = -0.10680/0.090 = -1.186667$, and $\lambda(m) = -\text{var}(m)/E(m) = -0.090/0.9709 = -0.0926975$.

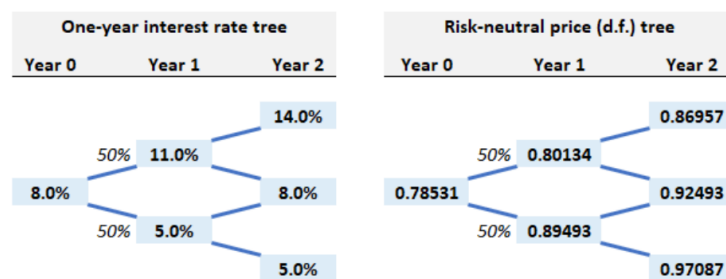
Therefore, $E(r_i) - R_f = \beta(i, m) \times \lambda(m) = -1.186667 \times -0.0926975 = 0.1100011$ or about 11.0%.

考察: Finance: Factor Theory 重要程度: [■ ■ □]

科目: Quantitative Analysis

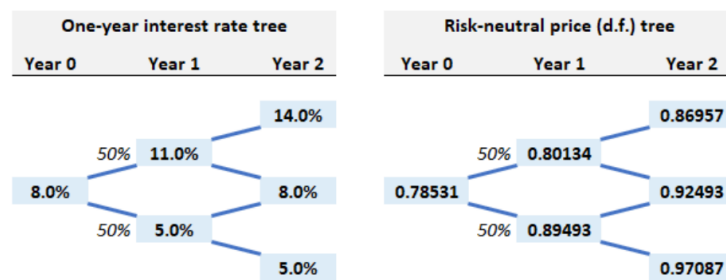
21. Below (in the left-hand panel) is an interest rate tree for the one-year rate where the initial rate of 8.0% jumps 300 basis points to either 11.0% or 5.0% next year. In two years, the one-year rate will be either 2.0%, 8.0%, or 14.0%.

Next, we will assume a three-year zero-coupon bond. Without a risk premium, for example, the date 0, state 2 price would be $0.877193 = 1/(1 + 14\%)$. However, we will instead assume a risk premium. Specifically, investors require compensation of 50 basis points for each year of duration risk. The right-hand panel displays the associated risk-neutral (i.e., , assumes the risk premium) price tree for this three-year zero-coupon bond



In regard to the three-year zero-coupon bond, which of the following is true?

下面（左侧面板）给出了一个一年期利率树，初始利率为 8.0%，在下一年将跳升 300 个基点到 11.0%，或下降至 5.0%。在两年后，一年期利率将变为 2.0%、8.0% 或 14.0%。接下来，假设一只三年期零息债券。在无风险溢价的情况下，例如，时点 0、状态 2 的价格为 $0.877193 = 1/(1 + 14\%)$ 。但是，我们在此假设存在风险溢价：具体来说，投资者要求每一年久期风险获得 50 个基点的补偿。右侧面板展示了这只三年期零息债券的对应风险中性（即考虑风险溢价）的价格树。



关于这只三年期零息债券，下列哪项说法是正确的？

- A. The expected return over the next year is 7.50%
接下来一年的预期回报是 7.50%
- B. The expected return over the next year is 9.00%
接下来一年的预期回报是 9.00%
- C. The implied 3-year spot rate term structure is downward sloping (aka, inverted)
隐含的三年期即期利率期限结构是向下倾斜的（aka，倒置）
- D. Jensen's inequality implies that expected excess return over two years is less than 50 basis points
詹森不等式意味着两年期预期超额回报小于 50 个基点

答案:B

解析:

B 选项正确。 True: The expected return over the next year is 9.00%.
We don't require the formula. The risk premium is 50 basis points for each year of duration risk, and the three-year bond creates two years of duration risk. The expected return, therefore, is 100 basis points above 8.0%.

考察: Fixed Income 重要程度: [■][■][□]

科目: Quantitative Analysis

22. Old Crow Bank had a credit committee meeting yesterday, and the following is a copy of the notes taken during the meeting:

- All members are present Jan (Chief Risk Officer), Eddie (Head of Compliance), Alyssa (Head of Legal), Don (Head of Tax), and Ashton (Chief Financial Officer).
- Jan initiated the meeting by confirming if everyone had received an email containing approval documents from credit officer Doug, who had originated the loan. Everyone confirmed receiving the email with the documents yesterday.

- Jan provided an overview of the loan for approximately five minutes, after which she proceeded to inquire about the opinions of those present. It was the consensus of all individuals, except for Alyssa, that the loan be approved. Alyssa expressed concern regarding the trustworthiness of the counterparty.
- Jan respects Alyssa’s opinion but disagrees and initiates a vote. Two other committee members agree with Jan, resulting in loan approval.

What is the most probable issue with the credit committee’s meeting?

老鹰银行昨天召开了信用委员会会议，以下是会议记录的副本：

- 所有成员都出席了：Jan（首席风险官）、Eddie（合规主管）、Alyssa（法律主管）、Don（税务主管）和 Ashton（首席财务官）。
- Jan 首先确认每个人是否收到了一封包含信用官员 Doug 批准文件的电子邮件。每个人都确认收到了带有这些文件的电子邮件。
- Jan 用大约五分钟的时间介绍了贷款情况，然后询问了与会人员的意见。除了 Alyssa 之外，所有人都同意批准贷款。Alyssa 对交易对手的可靠性表示担忧。
- Jan 尊重 Alyssa 的意见，但不同意，并发起投票。另外两名委员会成员也同意 Jan，导致贷款获得批准。

信用委员会会议最可能存在的问题是什么？

- A. The approval documents should have been distributed earlier.
批准文件应该更早分发。
- B. Jan monopolized the conversation and did not properly facilitate discussion.
Jan 垄断了整个对话，没有起到恰当的引导讨论作用
- C. The meeting was not properly documented.
会议没有恰当的证明文件。
- D. The vote was not unanimous, so the loan should have been rejected.
投票结果并非一致通过，因此这笔贷款本应被否决

答案:C

解析:

Option C is correct. In this scenario, all the approval documents were appropriately distributed to the committee before the meeting. Jan did spend time explaining the investment to the committee, but she also helped facilitate discussion among the committee members and listened to everyone’s opinion. Votes do not have to be unanimous, but they should be held unless the decision is unanimous. However, meetings should be accurately documented in case an investment goes awry. In the case of a bad transaction, auditors would want to know what was discussed in the meeting and if the reasoning is behind the investment decision.

考察: Credit Risk Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

23. A bank is using the VaR and stressed VaR (SVaR) market risk framework according to the Basel 2.5 guidelines. The bank’s internal models for market risk have generated the following risk measures for the current trading book positions (VaR and SVaR are in USD millions):

Multiplicative factors, m and m(S)				3.0
Confidence Level	Previous day		60-day average	
	VaR	SVaR	VaR	SVaR
	(mm)	(mm)	(mm)	(mm)
95.00%	\$110.0	\$190.0	\$127.0	\$238.0
99.00%	185.0	360.0	140.0	270.0
99.90%	283.0	740.0	228.0	490.0

The supervisor has set both multiplication factors, m and m(S), to 3.0. What is the correct capital requirement for general market risk for the bank under Basel 2.5? (note: this question inspired by Question 42 of GARP’s 2020 Part 2 Practice Exam). 一家银行正在按照巴塞尔 2.5 指南使用 VaR 和压力 VaR（SVaR）市场风险框架。该银行的内部市场风险模型为当前交易头寸（VaR 和 SVaR 以百万美元为单位）生成了以下风险衡量：

Multiplicative factors, m and m(S)				3.0
Confidence Level	Previous day		60-day average	
	VaR	SVaR	VaR	SVaR
	(mm)	(mm)	(mm)	(mm)
95.00%	\$110.0	\$190.0	\$127.0	\$238.0
99.00%	185.0	360.0	140.0	270.0
99.90%	283.0	740.0	228.0	490.0

监管机构将乘数 m 和 m(S) 都设置为 3.0。在巴塞尔 2.5 下，该银行的一般市场风险资本要求是多少？（注意：这个问题受到了 GARP 2020 Part 2 模考题第 42 题的启发）

- A. \$630.0 million
- B. \$810.0 million
- C. \$1,230.0 million
- D. \$2,154.0 million

答案:C

解析:

C 选项正确。 True: \$1,230.0 million

The Basel 2.5 market risk capital requirement requires a 99.0% confidence level and is given by:

$$MRC = \max[VaR(t - 1), m \times VaR(60\text{-day avg})] + \max[SVaR(t - 1), m(S) \times SVaR(60\text{-day avg})]$$

In this case,

$$MRC = \max[185.0, 3.0 \times 140.0] + \max[360.0, 3.0 \times 270.0] = \$1,230.0 \text{ million.}$$

考察: Market Risk 重要程度: [■ ■ ■ □]

科目: Quantitative Analysis

24. If a corporation’s marginal tax rate is 30.0% and it purchases an A-rated municipal bond that offers a 4.20% gross yield (aka, yield to maturity), what is the tax equivalent yield (TEY)? 如果一家公司的边际税率是 30.0%，它购买了一只 A 级市政债券，该债券提供 4.20% 的毛收益率（aka，到期收益率），那么税等效收益率（TEY）是多少？

- A. 2.45%
- B. 3.50%
- C. 6.00%
- D. 16.67%

答案:C

解析:

C 选项正确。 TRUE: 6.00%. $TEY = \text{after-tax return on tax-exempt investment} / (1 - \text{marginal tax rate})$; in this case, $TEY = 4.20\% / (1 - 0.30) = 6.00\%$

考察: Finance: Fixed Income 重要程度: [■□□□□]

科目: Foundations of Risk Management

25. Andrew Ang introduces factors and divides them primarily into two types: macro, fundamental-based factors versus investment-style factors. He writes, “Factors drive risk premiums. One set of factors describes fundamental, economywide variables like growth, inflation, volatility, productivity, and demographic risk. Another set consists of tradeable investment styles like the market portfolio, value-growth investing, and momentum investing.” (†)

Ang also claims that the three most important macro factors are growth, inflation, and volatility. His evaluation of these macro factors is based on a long-term historical sample (specifically, 1952:Q1 to 2011:Q4). It is important to qualify the sample window because we cannot be sure that past is prologue; e.g., interest rates experienced two long-term secular trends during this window.

In regard to his historical analysis, each of the following statements is true EXCEPT which is false?

安德鲁·安（Andrew Ang）介绍了因素并将其主要分为两类：宏观、基本面因素 versus 投资风格因素。他写道：“因素驱动风险溢价。一组因素描述了基本面的、经济范围的变量，如增长、通货膨胀、波动性、生产率和人口风险。另一组包括可交易的、投资风格，如市场投资组合、价值增长投资和动量投资。” (†)

安还声称，三个最重要的宏观因素是增长、通货膨胀和波动性。他对这些宏观因素的评估是基于一个长期的历史样本（具体来说，1952 年第一季度至 2011 年第四季度）。qualify the sample window because we cannot be sure that past is prologue; e.g., interest rates experienced two long-term secular trends during this window. 重要的是要限定样本窗口，因为我们不能确定过去是序言；例如，利率在这段时间内经历了两个长期的大趋势。

关于他的历史分析，下列各项陈述均为真，除了哪一项是假的？

- A. Government and investment-grade bonds performed BETTER during economic recessions than during expansions
政府和投资级债券在经济衰退期间表现优于在扩张期间
- B. Both large and small (cap) stocks perform significantly BETTER during economic expansions than during recessions
无论是大型还是小型（cap）股票，在经济扩张期间表现显著优于在经济衰退期间
- C. During periods of high inflation, all five asset classes perform significantly BETTER than during periods of low inflation
在高通货膨胀期间，所有五个资产类别都比在低通货膨胀期间表现显著更好
- D. All five assets classes are much MORE VOLATILE during recessions (or periods of low GDP growth) than during expansions
在衰退期间（或低 GDP 增长期间），所有五个资产类别都比在扩张期间更不稳定

答案:C

解析:

C 选项正确。 False. To be true, this should instead read: During periods of high inflation, all five asset classes perform significantly WORSE than during periods of low inflation.

In regard to (A), (B) and (D), each is TRUE.

考察: Factor Investing 重要程度: [■■□]

科目: Financial Markets and Products

26. Below is the price tree for a stylized constant-maturity Treasury (CMT) swap struck at 2.0% with a notional of \$1.0 million. The risk-neutral probabilities are displayed, specifically: 80.09% (d = 19.91%) from date 0 to date 1; and 65.20% (d = 34.80%)

from date 1 to date 2. Using these risk-neutral probabilities returns an expected discounted value (EDV) of \$3,702.11 for the value of the CMT swap. Please notice this exactly matches Tuckman’s Figure 7.8 in his 4th edition value. This is the so-called model value.

Notional	\$1,000,000	
Strike rate	2.00%	
i-rate evolution	T= 0.5	T= 1.0
		3.00%
	2.50%	2.00%
2.0%	1.50%	1.00%
Risk-neutral	80.09%	65.20%
Probabilities:	19.91%	34.80%
Price tree: 2 dates	T= 1.0	
		\$5,000.00
	\$5,719.52	\$0.00
\$3,702.11	-\$4,227.29	-\$5,000.00
\$3,694.81	← Observed Market Price	
0.25%	← OAS	

Assume the observed market price of this CMT swap is \$3,694.81 (as highlighted in purple), which is \$7.30 less than the market price. This implies an option-adjusted spread (OAS) of 25 basis points. Which of the following is the **best** interpretation of this OAS?

下面给出了一个简化的恒定到期国债（CMT）掉期的价格树，执行利率为 2.0%，名义本金 \$1.0 百万。给出了风险中性概率：从时点 0 到时点 1 为 80.09%（d = 19.91%）；从时点 1 到时点 2 为 65.20%（d = 34.80%）。使用这些风险中性概率，CMT 掉期的预期贴现价值（EDV）为 \$3,702.11。请注意，这与 Tuckman 第四版图 7.8 的数值完全一致。这就是所谓的模型价值。

Notional	\$1,000,000	
Strike rate	2.00%	
i-rate evolution	T= 0.5	T= 1.0
		3.00%
	2.50%	2.00%
2.0%	1.50%	1.00%
Risk-neutral	80.09%	65.20%
Probabilities:	19.91%	34.80%
Price tree: 2 dates	T= 1.0	
		\$5,000.00
	\$5,719.52	\$0.00
\$3,702.11	-\$4,227.29	-\$5,000.00
\$3,694.81	← Observed Market Price	
0.25%	← OAS	

假设这只 CMT 互换的市场价格为 \$3,694.81（紫色突出显示），比市场价格低 \$7.30。这意味着一个选项调整的利差（OAS）为 25 个基点。下列哪一项是对这个 OAS 的最佳解释？

- A. The model returns the market price if we adjust the risk-neutral probabilities to 55.09% and 44.91%
模型返回市场价格，如果我们调整风险中性概率为 55.09% 和 44.91%
- B. The model returns the market price if we assume adjusted cash flows; e.g., \$3,750 instead of \$5,000 on (state 2, date 2)
模型返回市场价格，如果我们假设调整后的现金流；例如，(state 2, date 2) 上的 \$3,750 而不是 \$5,000
- C. The model returns the market price if we use discount rates of 2.25% (date 0, state 0), 1.75% (date 1, state 0), and 2.75% (date 1, state 1)
模型返回市场价格，如果我们使用贴现率为 2.25%（时点 0，状态 0）、1.75%（时点 1，状态 0）和 2.75%（时点 1，状态 1）
- D. We cannot interpret the OAS here because this is a CMT swap, and the OAS requires an embedded option; e.g., a bond with an embedded call option
我们在这里无法解释 OAS，因为这个是 CMT 互换，而 OAS 需要一个嵌入式选项；例如，一个嵌入式看涨期权的债券

答案:C

解析:

C 选项正确。 True: The model returns the market price if we use discount rates of 2.25% (date 0, state 0), 1.75% (date 1, state 0), and 2.75% (date 1, state 1)

考察: Fixed Income Analysis 重要程度: [■ ■ □]

科目: Quantitative Analysis

27. Strongcore Financial Corporation has purchased a long-term at-the-money (ATM) call option from a risky counterparty. The option's stock and strike price are \$25.00 million. The underlying asset's volatility, σ , is estimated to be 20.0% while the

risk-free rate is 3.0% per annum. The European-style option matures in 3.0 years. In regard to the standard normal cumulative distribution function, in this scenario $N(d1)$ equals 0.66750 and $N(d2)$ equals 0.53450.

With respect to its counterparty risk, Strongcore assigns a hazard rate estimate of 6.0% and believes its counterparty could default at any time during the three-year option period; i.e., the cumulative default probability is relevant for this approximation. Further, if its counterparty defaults, Strongcore assigns a conservative recovery rate estimate of 30.0%.

In summary, the assumptions are: $S = K = \$25.0$ million, $\sigma = 20.0\%$, $R_f = 3.0\%$, $T = 3.0$ years; λ (i.e., hazard rate) = 6.0%, and $LGD = 70.0\%$.

In this situation, Strongcore does NOT want to compute the standard credit value adjustment (CVA) formula which is the summation of discounted expected exposures, $EE(t)$, over multiple future periods. Instead, Strongcore wants to rely on the simpler “semi-analytical method” that produces an estimate as a function of the position’s current value, the cumulative default probability, $PD(0,T)$, and the loss given default, LGD . Under this simpler approximation, which of the following is **nearest** to the CVA charge for the long option position?

Strongcore Financial Corporation 从一家存在信用风险的交易对手处购买了一份长期平值（ATM）看涨期权。该期权的标的股票价格和执行价均为 \$25.00 百万。标的资产波动率 σ 估计为 20.0%，无风险利率为年化 3.0%。该欧式期权到期时间为 3.0 年。关于标准正态分布的累积分布函数，在本情境下 $N(d1)$ 等于 0.66750, $N(d2)$ 等于 0.53450。就交易对手风险而言，Strongcore 给出的违约强度（hazard rate）估计为 6.0%，并认为其交易对手可能在三年期权期内的任意时点发生违约；即，对本近似而言需要使用累积违约概率。进一步地，如果交易对手违约，Strongcore 采用 30.0% 的保守回收率估计。总结假设如下： $S = K = \$25.0$ 百万， $\sigma = 20.0\%$ ， $R_f = 3.0\%$ ， $T = 3.0$ 年； λ （即违约强度）= 6.0%，且 $LGD = 70.0\%$ 。在此情况下，Strongcore 不想计算标准的信用估值调整（CVA）公式，即对多个未来时点的贴现预期敞口 $EE(t)$ 求和。相反，Strongcore 希望使用更简化的“半解析方法”，该方法给出一个依赖于头寸当前价值、累积违约概率 $PD(0,T)$ 以及违约损失率 LGD 的估计。在这种更简单的近似下，下列哪一个数值与该多头期权头寸的 CVA 费用 **最接近**？

- A. \$15,600
- B. \$90,000
- C. \$516,000
- D. \$1.1 million

答案:C
解析:
C 选项正确。\$516,000

$CVA = -LGD \times PD(0,T) \times V(t)$. In this case, $V(t) = S \times N(d1) - K \times \exp(-rt) \times N(d2) = 25 \times 0.66750 - 25 \times \exp(-0.030 \times 3) \times 0.53450 = \4.475 million. The cumulative $PD = 1 - \exp(-0.060 \times 3) = 16.473\%$. Therefore the CVA approximation is given by $70\% \times 16.473\% \times 4.475 \text{ million} = \0.5160 million , or \$516,002.

考察: Credit Risk 重要程度:
科目: Quantitative Analysis

28. Andrew is preparing to make a presentation to the board. The presentation’s goal is to introduce a new, proposed operational risk management (ORM) framework and get the board’s feedback on the way to eventual approval. It happens to be the case that his company is publicly traded with a high cost of capital due to its above-average beta. The board members are generally more familiar with market and credit risk but less familiar with operational risk. Therefore, his presentation will review several of the features (and nature) of operational risk, including its dynamic evolution. His presentation will briefly explore the skewed, heavy-tailed feature of operational risk. For candidate frequency distributions, he will illustrate the Poisson and negative binomial. For candidate severity distributions, he will illustrate the log-normal and generalized Pareto Distribution (GPD).

In addition to these features (i.e., dynamic, heavy-tailed), which of the following is TRUE about operational risk?
安德鲁正在准备向董事会做一次汇报。这次汇报的目的是介绍一个新的、拟议的运营风险管理（ORM）框架，并获得董事会的反馈，以便最终获得批准。事实证明，他的公司是一家公开交易的公司，由于其高于平均水平的贝塔系数，其资本成本较高。董事会成员对市场和信用风险比较熟悉，但对运营风险不太熟悉。因此，他的汇报将回顾运营风险的一些特征（及其性质），包括其动态演变。他的汇报将简要探索运营风险的偏斜、重尾特征。对于候选频率分布，他将说明泊松分布和负二项分布。对于候选严重分布，他将说明对数正态分布和广义帕累托分布（GPD）。

除了这些特征（即动态、重尾）外，下列哪一项关于运营风险的说法是正确的？

- A. The easiest way for a high-beta firm to lower its beta is to centralize the operational risk management (ORM) function
高贝塔公司降低其贝塔系数的最简单方法是集中运营风险管理（ORM）功能
- B. Operational risk is highly homogeneous because so-called boundary events define clusters that contain loss events and their shared (aka, common) features
运营风险高度同质化，因为所谓的边界事件定义了包含损失事件和它们的共享（aka，共同）特征的集群
- C. ORM is valuable because it reduces regulatory capital, and regulatory capital is a good indicator of the firm’s quality of risk management
ORM 是有价值的，因为它减少了监管资本，而监管资本是衡量公司风险管理质量的良好指标
- D. Most causes of operational risk trace to control weaknesses; human biases and behavioral failings; and/or operating environment changes
大多数运营风险的原因归因于控制弱点；人类偏见和行为失败；以及/或 operating environment changes

答案:D

解析:

Option D is correct. Most operational risk causes trace to control weaknesses; human biases and behavioral failings; and/or operating environment changes

In regard to (A), (B), and (C), each is FALSE.

- Operational risk is predominantly idiosyncratic and diffuses within the firm. Beta is a systematic risk.
- Operational risk is highly heterogeneous but interconnected. Further, boundary events are “risk events that materialize in a different category than they cause” -Source: GARP Introduction to Operational Risk and Resilience, Section 1.5)
- According to GARP, “regulatory operational risk capital is not sensitive enough to a firm’s risk profile and thus brings no real incentive to optimize the management of operational risks. In fact, regulatory capital is often a poor indicator of risk management quality.” -Source: Chapter (GARP Introduction to Operational Risk and Resilience, Section 1.7)

考察: Operational Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

29. Over the next 24 hours, Greenlux State Bank estimates that the following cash inflows and outflows (all figures in millions) will occur:

Deposit withdrawals	\$70.0
Deposit inflows	\$100.0
Scheduled loan repayments	\$60.0
Acceptable loan requests	\$90.0
Borrowings from the money market	\$80.0
Sales of bank assets	\$30.0
Stockholder dividend payments	\$20.0
Revenues from sale of nondeposit services	\$10.0
Repayment of bank borrowings	\$50.0
Operating expenses	\$40.0

What is the bank’s projected net liquidity position?

通过以下现金流入和流出（所有金额以百万为单位），格林卢克斯州银行预计将在接下来的 24 小时内发生：

Deposit withdrawals	\$70.0
Deposit inflows	\$100.0
Scheduled loan repayments	\$60.0
Acceptable loan requests	\$90.0
Borrowings from the money market	\$80.0
Sales of bank assets	\$30.0
Stockholder dividend payments	\$20.0
Revenues from sale of nondeposit services	\$10.0
Repayment of bank borrowings	\$50.0
Operating expenses	\$40.0

银行预计的净流动性头寸是多少？

- A. -30.0 million
- B. +10.0 million
- C. +40.0 million
- D. +90.0 million

答案:B

解析:

B. TRUE: +10.0 million

Inflows include: Deposit inflows = 100+
Scheduled loan repayments = 60+
Borrowings from the money market = 80+
Sales of bank assets = 30+
Revenues from sale of nondeposit services = 10;
for total inflows of $\$100 + \$60 + \$80 + \$30 + \$10 = \280.0 million.
Outflows include:
Deposit withdrawals = 70+
Acceptable loan requests = 90+
Stockholder dividend payments = 20+
Repayment of bank borrowings = 50+
Operating expenses = 40;
for total outflows of $\$70 + \$90 + \$20 + \$50 + \$40 = \270.0 million
Therefore, the projected net liquidity position = $\$280 - 270 = +10.0$ million.

考察: Finance 重要程度: [■■□]

科目: Quantitative Analysis

30. Emily manages a \$20.0 million portfolio allocated into two positions (aka, two components): a Technology ETF, and a Real Estate Investment Trust (REIT) ETF. She compares four different allocations, where the weight assigned to the Tech ETF component is either 35.0%, 50.0%, 65%, or 80.0%.

	Mix (Allocation) Scenarios			
	#1	#2	#3	#4
Weight in Tech ETF (%)	35.0%	50.0%	65.0%	80.0%
Position (000s)	\$7,000	\$10,000	\$13,000	\$16,000
Individual VaR (000s)	\$2,303	\$3,290	\$4,277	\$5,264
Marginal VaR	0.020	0.118	0.231	0.305
	#1	#2	#3	#4
Weight in REIT ETF (%)	65.0%	50.0%	35.0%	20.0%
Position, USD (000s)	\$13,000	\$10,000	\$7,000	\$4,000
Individual VaR (000s)	\$6,415	\$4,935	\$3,454	\$1,974
Marginal VaR	0.461	0.386	0.232	0.039
Undiversified Portfolio VaR	\$8,718	\$8,224	\$7,731	\$7,237

If her goal is to minimize the portfolio’s diversified value at risk (VaR), which of the above allocations is **best**?
 艾米莉管理着一个 \$20.0 百万的投资组合，分配到两个头寸（aka，两个组件）：一个科技 ETF，和一个房地产投资信托（REIT）ETF。她比较了四种不同的分配，其中分配给科技 ETF 组件的权重分别是 35.0%、50.0%、65.0% 或 80.0%。

	Mix (Allocation) Scenarios			
	#1	#2	#3	#4
Weight in Tech ETF (%)	35.0%	50.0%	65.0%	80.0%
Position (000s)	\$7,000	\$10,000	\$13,000	\$16,000
Individual VaR (000s)	\$2,303	\$3,290	\$4,277	\$5,264
Marginal VaR	0.020	0.118	0.231	0.305
	#1	#2	#3	#4
Weight in REIT ETF (%)	65.0%	50.0%	35.0%	20.0%
Position, USD (000s)	\$13,000	\$10,000	\$7,000	\$4,000
Individual VaR (000s)	\$6,415	\$4,935	\$3,454	\$1,974
Marginal VaR	0.461	0.386	0.232	0.039
Undiversified Portfolio VaR	\$8,718	\$8,224	\$7,731	\$7,237

如果她的目标是最大限度地降低投资组合的分散价值风险（VaR），那么哪一种分配是 **最佳的**？

- A. Mix #1 with 35.0% allocated to the Tech ETF
混合 #1 分配给科技 ETF 35.0%
- B. Mix #2 with 50.0% allocated to the Tech ETF
混合 #2 分配给科技 ETF 50.0%
- C. Mix #3 with 65.0% allocated to the Tech ETF
混合 #3 分配给科技 ETF 65.0%
- D. Mix #4 with 80.0% allocated to the Tech ETF
混合 #4 分配给科技 ETF 80.0%

答案:C

解析:
Option C is correct. Mix #3 with 65.0% allocated to the Tech ETF.

Risk is minimized (i.e., the portfolio’s diversified VaR is minimized) when the marginal VaRs are constant. Among the four mixes, mix #3 is clearly the choice where the marginal VaRs are nearest (i.e., 0.231 versus 0.232 is almost an exact match). Mix #1 has the greatest diversified VaR. Starting from Mix #1, risk is reduced by ADDING to the component with the lower marginal VaR (in this mix, the marginal VaR of the Tech ETF is only 0.020 while the marginal VaR of the REIT ETF is 0.461) and REDUCING the component with the higher marginal VaR. If we thusly start with Mix #1 and add 15.0% to the Tech ETF, then we “arrive” at Mix #2. However, in Mix #2 the Marginal VaR of the REIT ETF is still much higher than the Marginal VaR of the Tech ETF (i.e., 0.386 versus 0.118) so we can continue to shift toward the position with the lower Marginal

VaR (which is still the Tech ETF). If we add another 15% to the Tech ETF, then we arrive at Mix #3: here the Marginal VaRs are almost equal, so we are very close to the allocation that minimizes the portfolio’s diversified VaR. For the record, the diversified portfolio VaR for the four mixes are \$6,131.14 (Mix #1), \$5,043.03 (Mix #2), \$4,621.42 (Mix #3), and \$5,036.58 (Mix #4).

考察: Finance: Risk Management 重要程度: [■■■□]

科目: Quantitative Analysis

31. In comparison to a yield-based DV01 hedge, which of the following is the **best** argument in favor of a regression hedge? 与基于收益率的 DV01 对冲相比，下列哪一项是对回归对冲的最佳论据？

- A. The regression hedge minimizes the profit and loss (P&L) variance of the hedged position
回归对冲最小化了对冲头寸的利润和损失（P&L）方差
- B. The regression hedge is based on economic theory and future developments rather than empirical analysis
回归对冲基于经济理论和未来发展，而不是经验分析
- C. The regression hedge does not require any out-of-sample test(s), which are often unreliable if the sample is small
回归对冲不需要任何样本外测试，如果样本很小，这些测试通常不可靠
- D. The regression hedge solves for a risk-weight which qualifies as a valid risk-weighted asset (RWA) under Basel regulations
回归对冲求解一个风险权重，该权重符合巴塞尔法规下的有效风险加权资产（RWA）

答案:A

解析:

A 选项正确。 True: The regression hedge minimizes the profit and loss (P&L) variance of the hedged position. As Tuckman explains (emphasis ours), “The **best argument for the regression hedge** is actually not the earlier assumption that bond yields change exactly according to the regression model. Write the P&L of the position as ... formula 6.11 ... where the negative signs reflect that a positive face amount with a positive change (i.e., increase) in yield lowers P&L. It can then be shown that the regression hedge in Equation (6.8) minimizes the variance. In other words, **to the extent that P&L variance is an appropriate measure of risk, the regression hedge minimizes the risk of the hedged position.**” -Tuckman, Bruce; Serrat, Angel. Fixed Income Securities (Wiley Finance) (p. 159). Wiley. Kindle Edition.

In regard to (B), (C), and (D), each is FALSE. Instead, the following are true statements:

- The regression hedge is based on empirical analysis but not economic theory and future developments
- The regression hedge typically does require (or at least prefer) out-of-sample test(s)
- The risk weight is the beta coefficient and has nothing to do with risk-weighted asset (RWA) under Basel regulations

考察: Fixed Income: Hedging 重要程度: [■■■□]

科目: Quantitative Analysis

32. Among its financial positions, Zapex Financial Corporation has written a call, entered into an interest rate swap, and written a credit default swap. Among these (the following) four positions, which is the BEST example of wrong-way risk?

在 Zapex Financial Corporation 的金融头寸中，它写了一个看涨期权，进入了一个利率互换，并写了一个信用违约互换。在这四个头寸中（以下），哪一个是最坏的例子？

- A. Zapex has written (aka, sold) a call option to a counterparty in the same industry
Zapex 向同一行业的交易对手出售（aka，卖出）一个看涨期权
- B. Zapex is the fixed-rate payer in an interest rate swap during an economic recovery
Zapex 在经济复苏期间是固定利率支付者
- C. Zapex has written (aka, sold) a credit default swap (CDS) to counterparty whose credit quality is positively correlated with the reference entity
Zapex 向信用质量与参考实体正相关的交易对手出售（aka，卖出）一个信用违约互换（CDS）

D. Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty's default probability
Zapex 有一个远期货币合同头寸，其公允价值收益与交易对手的违约概率正相关

答案:D

解析:

D 选项正确。 Zapex has a position in a forward currency contract where its mark-to-market gain is positively correlated with the counterparty's default probability

Wrong-way (credit) risk is adverse correlation between the exposure and the credit quality, so wrong-way risk is a positive (negative) correlation between exposure and default probability (credit quality). This is wrong-way risk because greater exposure is associated with a lower likelihood of payment. Algebraically, wrong-way (or right-way) risk is some dependence between exposure/EAD and PD. We normally write $EL\% = PD \times EAD$ but that assumes independence between PD and EAD, so a way to think about WWR is simply when $EL > PD \times EAD$, because $EL \neq PD \times EAD$ is also a condition for non-independence. In regard to (A), (B) and (C), each is an example of RIGHT-WAY risk.

考察: Credit Risk: Wrong-Way Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

33. According to the FDIC, model risk management starts with robust model development, implementation, and use (aka, the first element. The second element is a sound validation process. The third element is governance). Which of the following is a TRUE statement about this *first element* (development, implementation and use) of model risk management?
根据 FDIC，模型风险管理始于强大的模型开发、实施和使用（aka，第一个元素。第二个元素是一个健全的验证过程。第三个元素是治理）。下列哪一项是关于模型风险管理 第一个元素（开发、实施和使用）的 TRUE 陈述？

- A. Model development should be a straightforward and routine technical process
模型开发应该是一个简单且常规的技术过程
- B. To exercise conservatism, pick an extreme point on a given model
为了体现审慎性，应在给定的模型分布上选择一个极端点
- C. Banks should never model output with judgmental or qualitative adjustments
行在对模型输出进行调整时绝不应使用主观判断或定性调整
- D. Although the nature of testing varies by the context and type of model, testing is an integral part of model development
尽管测试的性质因上下文和模型类型而异，但测试是模型开发不可或缺的一部分

答案:D

解析:

D. TRUE: Although the nature of testing varies by the context and type of model, testing is an integral part of model development

In regard to (A), (B) and (C), each is FALSE.

- Model development is neither straightforward nor a routine technical process!
- To pick an extreme point on a given modeled distribution is not necessarily conservative if the distribution is mis-specified or mis-estimated in the first place.
- Judgmental or qualitative adjustments are a KEY part of model development/implementation/use: “In addition, a considerable amount of subjective judgment is exercised at various stages of model development, implementation, use, and validation. It is important for decision makers to recognize that this subjectivity elevates the importance of sound and comprehensive model risk management processes.” ... “Banks should ensure that the development of the more judgmental and qualitative aspects of their models is also sound. In some cases, banks may take statistical output from a model and modify it with judgmental or qualitative adjustments as part of model development.”

考察: Operational Risk: Model Risk Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

34. Suppose that Acme Bank reports interest-sensitive assets (ISA) of \$490.0 million and interest-sensitive liabilities (ISL) of \$610.0 million. Recall that the relative interest-sensitive gap (aka, Relative IS GAP) is equal to the Dollar IS GAP divided (aka, scaled) by the size of the bank (where IS assets is a valid measure of size). Respectively, what is the bank’s relative IS GAP; its interest-sensitivity ratio (ISR); and the impact of rising interest rates on its net interest margin (NIM)?
假设 Acme Bank 报告的利息敏感资产（ISA）为 \$490.0 百万，利息敏感负债（ISL）为 \$610.0 百万。请记住，相对利息敏感缺口（aka，相对 IS 缺口）等于利息敏感缺口（Dollar IS GAP）除以银行规模（其中利息敏感资产是一个有效的规模衡量标准）。分别是什么？

- A. The Relative IS GAP is -120.0; the ISR is 1.245; rising interest rates will lower the NIM
相对 IS 缺口为 -120.0；ISR 为 1.245；上升的利率将降低 NIM
- B. The Relative IS GAP is -0.245; the ISR is 0.803; rising interest rates will lower the NIM
相对 IS 缺口为 -0.245；ISR 为 0.803；上升的利率将降低 NIM
- C. The Relative IS GAP is -0.245; the ISR is 1.245; rising interest rates will increase the NIM
相对 IS 缺口为 -0.245；ISR 为 1.245；上升的利率将增加 NIM
- D. The Relative IS GAP is +4.08; the ISR is 0.803; rising interest rates will increase the NIM
相对 IS 缺口为 +4.08；ISR 为 0.803；上升的利率将增加 NIM

答案:B

解析:

B 选项正确。 The Dollar IS GAP = ISA -ISL = \$490.0 – \$610.0 = –120.0. The Relative IS GAP = –120.0/490.0 = –0.245. The ISR = 490.0/610.0 = 0.803. This bank is liability-sensitive such that rising interest rates will lower its net interest margin (NIM).

考察: Market Risk: Interest Rate Risk **重要程度:** [■■■□]

科目: Quantitative Analysis

35. You work for the Private Client division of an international bank. The bank’s Private Client Advisors work with high net worth (NNW) individuals and families. Your boss has tasked you to perform due diligence on a roster of hedge funds; the funds that survive your due diligence will become candidates for presentation (and recommendation) to suitable customers of the bank who will become investors in the fund(s).

Your boss says that your due diligence should include a careful screen for fraud risk on behalf of the investors. In the case of your due diligence, which of the following is the most important ingredient?

你在一个国际银行的私人客户部门工作。该银行的私人客户顾问与高净值（NNW）个人和家庭合作。你的老板让你对一组对冲基金进行尽职调查；通过尽职调查后幸存的基金将成为候选基金，将推荐给合适的银行客户，成为基金的投资者。你的老板说，你的尽职调查应该包括对投资者的欺诈风险进行仔细筛查。在尽职调查中，哪一项是最重要的成分？

- A. Diligent application of a comprehensive checklist including a list of specific questions
仔细应用包括特定问题列表的综合检查清单
- B. Strong pre-existing personal relationships among the fund’s managers and all related parties
基金经理和所有相关方之间强有力的现有个人关系
- C. Guarantees by the fund of exceptional performance (e.g., > 30% per annum) with little or no risk
基金对卓越绩效（例如，年化 > 30%）的保证，风险很小或没有风险
- D. The investors’ membership in an affinity group, preferably a religious group due to high correlation with ethics
投资者在同质群体中的成员资格，最好是宗教群体，因为与伦理的高相关性

答案:A

解析:

A. TRUE: Diligent application of a comprehensive checklist including a list of specific questions

The authors recommend reviewing the SEC website including its list of questions; e.g., Red Flags of Investment Fraud Checklist at <https://www.investor.gov/>.

In regard to (B), (C) and (D), each is false:

- Strong pre-existing personal relationships with related parties is the **opposite of the independence** that investors should require between fund managers and related parties.
- Guarantees of exceptional performance with little or no risk is a classic **red flag**.
- In regard to affinity groups: these are notorious targets of fraud because participants tend to be trusting of their peers.

考察: Finance: Due Diligence 重要程度:

科目: Valuation and Risk Models

36. Which of the following is the BEST summary of the purpose of a copula function?

哪一项是对 copula 函数目的的最佳总结?

- A. To utilize the Gaussian correlation matrix such that tractable pairwise correlation parameters replace intractable dependence values
利用高斯相关矩阵，使得可处理的成对相关参数取代不可处理的依赖值
- B. To specify a single multivariate distribution as a function of various (even unusual) univariate marginal distributions by mapping them to distributions with an accessible or familiar dependence structure
将各种（甚至不寻常）单变量边际分布映射到具有可访问或熟悉依赖结构的分布，指定一个多元分布作为这些单变量边际分布的函数
- C. To translate a vector of outcomes into quantiles and map the quantiles to probabilities such that virtually any combination of multiple outcomes can be expressed as a single vector
将结果向量转换为分位数，并将分位数映射到概率，使得几乎任何多个结果的组合都可以表示为一个单一向量
- D. To decompose a posterior conditional distribution into a combination of the prior probability, the likelihood, and the marginal probability, typically by identifying Cholesky’s decomposition
将后验条件分布分解为先验概率、似然和边际概率的组合，通常通过识别 Cholesky 分解

答案:B

解析:

B 选项正确。To specify a single multivariate distribution as a function of various (even unusual) univariate marginal distributions by mapping them to distributions with an accessible or familiar dependence structure

考察: Quantitative Analysis: Copulas 重要程度: [■□□□□]

科目: Foundations of Risk Management

37. Retail exposures include the following: credit cards, installment loans (e.g., personal finance, educational loans, auto loans, leasing), revolving credits (e.g., overdrafts, home equity lines of credit), and residential mortgages. Each of the following tends to be a stronger feature of retail credit risk rather than corporate credit risk EXCEPT which is more typical of corporate credit risk?

零售风险敞口包括以下内容：信用卡、分期付款（例如，个人财务、教育贷款、汽车贷款、租赁）、循环信贷（例如，透支、房屋净值信贷额度）和住宅抵押贷款。下列哪一项通常是零售信用风险而不是企业信用风险的一个更强特征？

- A. Portfolio tends to behave like a well-diversified portfolio in normal markets
投资组合倾向于在正常市场中表现得像一个分散良好的投资组合
- B. Default by a single customer is never expensive enough to threaten a bank
单个客户违约永远不会昂贵到足以威胁银行
- C. A rise in defaults is often signaled in advance by a change in customer behavior
默认率的上升通常通过客户行为的变化提前发出信号
- D. Portfolio risk is dominated by risk that credit losses will rise to unexpected level due to concentrations of exposures
投资组合风险由由于暴露集中而导致信用损失上升到意外水平的风险主导

答案:D

解析:

D 选项正确。 This tends to be a feature of CORPORATE credit risk: portfolio risk is dominated by risk that credit losses will rise to unexpected level due to concentrations of exposures.

- In regard to (A), (B) and (C), each is a tendency of retail credit risk.

考察: Credit Risk: Retail vs Corporate 重要程度: [■■■]

科目: Quantitative Analysis

38. According to the Basel Committee on Banking Supervision (BCBS, 2018), in regard to cyber-security and cyber-resilience metrics, “Some jurisdictions have methodologies to assess or benchmark regulated institutions’ cyber-security and resilience ... None of these methodologies produce quantitative metrics or risk indicators comparable to those available for financial risks and resilience; e.g., standardized quantitative metrics where established data are available. Instead, indicators provide information on regulated institutions’ approach to building and ensuring cyber-security and resilience more broadly. Supervisory authorities also rely on entities’ own management information, although this differs across entities and is not yet mature.” (†) With respect to current cybersecurity and resilience metrics, each of the following statements is true EXCEPT which is inaccurate?

根据巴塞尔委员会（BCBS, 2018），关于网络安全和弹性指标，“一些司法管辖区有方法来评估或基准化受监管机构的网络安全和弹性... 这些方法没有一个产生可量化的指标或风险指标，与金融风险 and 弹性指标相比，例如，可获得的已建立数据的标准化可量化指标。相反，指标提供有关受监管机构建立和确保网络安全和弹性的一般方法的信息。监管机构还依赖于实体自己的管理信息，尽管这因实体而异，尚未成熟。” (†) 关于当前的网络安全和弹性指标，下列陈述中哪一项是正确的，除了哪一项是不准确的？

- A. No single cybersecurity/resilience metric in isolation is sufficient
没有一个单独的网络安全/弹性指标是足够的
- B. Patch aging (aka, days to patch) is a widespread and comparable metric
补丁老化（aka，补丁到天数）是一个广泛且可比较的指标
- C. Backward-looking metrics are helpful but insufficient; forward-looking metrics are necessary because adversaries dynamically adapt
向后看指标是有帮助的，但不足；向前看指标是必要的，因为对手动态适应
- D. The penetration ratio (i.e., percentage of banks in a jurisdiction that can be penetrated) is a useful metric that compares a supervisor’s effectiveness to other jurisdictions
渗透率（即，一个司法管辖区中可以渗透的银行百分比）是一个有用的指标，用于比较监管机构的有效性与其他司法管辖区

答案:D

解析:

D 选项正确。 False. The displayed penetration ratio is nonsensical (also to compare supervisor effectiveness), but penetration tests (penetration testing) are discussed several times by BCBS as an effective cyber-risk control approach. However, please do note that penetration testing (by type: count and finding rating) is a common and effective cyber-resilience metric. See 4.2.2. Penetration testing.

In regard to (A), (B) and (C), each is TRUE. Specifically:

- No single cybersecurity/resilience metric in isolation is sufficient: “Collectively, these indicators [i.e., cyber-metrics collated by regulated entities] can inform on the broad adequacy of an institution’ s cyber- and operational resilience levels for its business needs and risk appetite. However, no single item taken in isolation is seen as a sufficient metric, and no standard set of indicators has been identified so far to provide a meaningful benchmark.” (†)
- Patch aging (aka, days to patch) is a widespread and comparable metric: “Annex C contains cyber-centric metrics collated by a sample set of regulated institutions for decision-making bodies (boards and board sub-committees). It is notable that the data provided typically allow for trend information so that the reviewer can assess if the situation is getting better or worse. Some metrics track compliance with internal policies while others measure inherent risk. Patch ageing in particular is a widespread and comparable metric.” (†)

- Backward-looking metrics are helpful but insufficient; forward-looking metrics are necessary because adversaries dynamically adapt: “Backward-looking indicators comment on past performance as an indicator of future performance, which is reasonable when institutions’ operations and risk environment are relatively stable over time and more or less independent from outside influences. However, cyber-risk frustrates this because adversaries are dynamic, themselves adapting to institutions’ responses and protective measures, sometimes changing their tactics and strategies even in the space of a single cyber-incident. Distributed denial of service (DDOS) incidents are a good example, where the volume and scale of disrupted internet traffic generated has increased significantly in the last two years and adversaries adapt their techniques in response to an institution’ s defences. While backward-looking metrics continue to be important, jurisdictions are increasingly recognising the need for forward-looking indicators as direct and indirect metrics of resilience, indicating whether a regulated institution is likely to be more or less resilient in the event of a risk crystallising.” (†)

考察: Operational Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

39. Venkat explains that “all risk management frameworks start with a governance structure that defines the roles and responsibilities of various bank employees and committees in overseeing risk-related activities,” and effective governance includes the oversight of intraday liquidity risk. In regard to the governance structure of intraday liquidity risk management which of the following statements is TRUE?

Venkat 解释说, “所有风险管理框架都以一个治理结构开始, 该治理结构定义了各种银行员工和委员会在监督风险相关活动中的角色和责任,” 并且有效的治理包括对日内流动性风险的管理。关于日内流动性风险管理的治理结构, 下列哪一项陈述是正确的?

- A. Intraday liquidity risk should be accepted as a cost of doing business
日内流动性风险应该被接受为经营成本
- B. Roles and responsibilities should be defined within the eight lines (i.e., two times four) of defense model
角色和责任应该在八条防线（即，两条四条）的防御模型中定义
- C. PCS Systems should only be a source of funds; if PCS becomes a use of funds, then a yellow flag should be triggered
PCS 系统应该只是资金的来源; 如果 PCS 成为资金的用途, 那么应该触发一个黄色标志
- D. Intraday liquidity risk should be incorporated in the risk taxonomy and is a component of risk self-assessment including settlement risks.
日内流动性风险应该纳入风险分类, 是风险自评估的一部分, 包括结算风险。

答案:D

解析:

Option D is correct. Intraday liquidity risk should be incorporated in the risk taxonomy and is a component of risk self-assessment including settlement risks.

In regard to (A), (B) and (C), each is FALSE.

- Venkat(†) explains that Active Risk Management is characteristic of the governance of intraday LRM: “Active risk management. In many institutions, intraday liquidity risk is accepted as a cost of doing business and is not as actively managed with the same level of rigor as other types of enterprise risk or even other liquidity risks. The leading banks with large volumes of PCS activities recognize the criticality of understanding and working to reduce their intraday liquidity risks. These institutions classify settlement and systemic risks as components of their risk taxonomy, and critically, incorporate them into the firm’ s risk appetite framework” (†)
- Integration with risk governance includes the THREE lines of defense model: Treasury, Corporate Risk Management, and Internal Audit. Says Venkat(†), “The intraday liquidity risk management framework follows the industry’ s three lines of defense model, with particular emphasis on expertise in the second line of defense to coordinate across the institution ... Corporate Risk Management is the second line of defense, responsible for overseeing funding-related policy and procedures: advising in the development of risk management programs, monitoring the ongoing risk-taking activities across all of a bank’ s funding desks, aggregating reporting across the bank, and providing an independent view of the effectiveness of the bank’ s overall intraday liquidity risk management programs.” (†)

- Payment, clearing, and settlement (PCS) Systems can be either a source or use of funds: “Settlements at PCS systems. Most PCS systems have one settlement per day, with many occurring in the late afternoon timeframe. This may serve as either a source or use of funds depending on the net position of a participant on any given day.” (†)

考察: Liquidity Risk Management 重要程度: [■ ■ □]

科目: Valuation and Risk Models

40. Hedge funds are less regulated than mutual funds. According to the authors (Fung and Hsieh), this lighter regulation of hedge funds offers genuine benefits but also creates unintended consequences. Which of the following is both a negative consequence and a TRUE statement about this relatively light regulation?

对冲基金的监管比共同基金要轻。根据作者 (Fung 和 Hsieh), 这种较轻的对冲基金监管确实带来了真正的利益, 但也产生了意想不到的后果。下列哪一项是相对较轻的监管的负面后果和 TRUE 陈述?

- A. Lack of reliable, standardized, and long-term performance reporting
缺乏可靠、标准化和长期绩效报告
- B. Lack of agreement on the specific definitions of absolute return and alpha
缺乏对绝对回报和 alpha 的具体定义的一致意见
- C. Ability of global macro funds to secretly drift into various strategies and styles with reckless impunity
全球宏观基金能够秘密地进入各种策略和风格, 毫无顾忌
- D. Ability of hedge funds to advertise (and specifically solicit) to individual investors without any restrictions
对冲基金能够秘密地进入各种策略和风格, 毫无顾忌

答案:A

解析:

A 选项正确。 Lack of reliable, long-term performance reporting

According to the authors (emphasis ours): “However, being lightly regulated does have unintended consequences. For one, **performance records of hedge funds are generally not standardized and available reports are prone to measurement errors.** Until the early 1990s, the historical performance of hedge funds was often as private as their investors, and the assessment of hedge fund performance was as much art as science ... the **events of 1994** prompted investors to re-examine the perceived diversification of their hedge fund holdings. In turn, this aided the development of electronically available hedge fund databases with which more formal analysis of hedge fund performance and the attendant risks can be conducted. It was the arrival of electronic databases that made academic research in hedge funds a feasible proposition.” —Source: Chapter 17 of Handbook of the Economics of Finance, Volume 2B, edited by G. Constantinides, M. Harris, and R. Stulz.

- In regard to (B) and (D), these are false statements.
- In regard to (C), it is a feature (aka, advantage) of global macro funds that they can utilize various styles opportunistically.

考察: Finance: Hedge Funds 重要程度: [■ ■ □]

科目: Quantitative Analysis

41. Meissner conducted a detailed study of equity correlations of the 30 stocks in the Dow (DJIA) over 44.5 years (i.e., the 534 months from January 1972 to July 2017) that generated $534 * 30^2 = 480,600$ correlations values. In regard to his empirical findings with respect to equity correlations, each of the following statements is TRUE, except which is false?

Meissner 对道琼斯工业平均指数 (DJIA) 30 只股票的股权相关性进行了详细研究, 持续了 44.5 年 (即, 1972 年 1 月至 2017 年 7 月 534 个月), 产生了 $534 * 30^2 = 480,600$ 个相关值。在关于他对股权相关性的经验发现中, 下列陈述中哪一项是 TRUE, 除了哪一项是 FALSE?

- A. Mean reversion should be included in equity correlation models
均值回归应该包含在股权相关模型中
- B. For equity correlations, the Johnson SB distribution is better than the normal or lognormal
对于股权相关性, Johnson SB 分布比正态或对数正态更好

- C. A stressed scenario of bad economic times should incorporate higher correlation levels and higher correlation volatility
一个压力情景下的糟糕经济时期应该包含更高的相关水平和更高的相关波动
- D. Autocorrelation is a monotonically increasing function of the monthly lag; i.e., higher lags imply higher autocorrelations
自相关是月滞后的单调递增函数；即，更高的滞后意味着更高的自相关

答案:D

解析:

D 选项正确。 D. FALSE. Instead, the autocorrelation tends to decay (i.e., decrease) with greater lags. In regard to (A), (B), and (C), each is TRUE.

考察: Quantitative Analysis: Correlation **重要程度:** [■■■□]

科目: Quantitative Analysis

42. Your colleague has drafted guidance for your firm’s computation of the credit value adjustment (CVA). The guidance begins with four principles: risk-neutrality, discounting, proportionality, and inseparability. The principles are summarized as follows:

- I. Risk-neutrality: CVA can be computed either with risk-neutral (aka, market-implied) or real-world (aka, historical) parameters
- II. Discounting: CVA should be discounted to present value, either via discounted expected exposure (EE) or via explicit discount factors
- III. Proportionality: CVA will be proportional to the likelihood of default (PD), the exposure (EE) and the percentage amount lost in default (LGD).
- IV. Inseparability: It should be impossible to separate the valuation of a transaction from its counterparty risk as measured by CVA

Each of the principles is valid, EXCEPT which of the principles is INCORRECT?

你的同事已经为你公司的信用价值调整（CVA）的计算起草了指导原则。指导原则以四个原则开始：风险中性、折现、比例性和不可分离性。原则总结如下：I. 风险中性：CVA 可以使用风险中性（aka，市场隐含）或实际世界（aka，历史）参数计算 II. 折现：CVA 应该折现到现值，无论是通过折现预期暴露（EE）还是通过显式折现因子 III. 比例性：CVA 将与违约概率（PD）、暴露（EE）和违约损失率（LGD）成比例。IV. 不可分离性：应该不可能将交易的价值与其对手方风险（由 CVA 衡量）分开每个原则都是有效的，除了哪一项原则是不正确的？

- A. Risk-neutrality
风险中性
- B. Discounting
折现
- C. Proportionality
比例性
- D. Inseparability
不可分离性

答案:D

解析:

D 选项正确。 FALSE (Inseparability). Instead, the CVA is separable and this has practical importance. The most fundamental, key relationship is given by: Risk value = Riskfree value – CVA. In regard to (A), (B) and (C) each is TRUE as a principle.

考察: Counterparty Credit Risk **重要程度:** [■■■□]

科目: Quantitative Analysis

43. Acme Financial is a large bank whose Business Indicator (BI), for purposes of operational risk capital under the standardized measurement approach (SMA), is well in excess of EUR 1.0 billion. Therefore, Acme is required to use loss data as a direct input into its operational risk capital calculations. In regard to the general and/or specific criteria recommended by the Basel Committee, which of the following is TRUE about the identification, collection, or treatment of operational loss data?

Acme Financial 是一家大型银行，其业务指标（BI），在标准化测量方法（SMA）下，对于操作风险资本而言，远远超过 EUR 1.0 亿。因此，Acme 被要求将损失数据作为直接输入到其操作风险资本计算中。在巴塞尔委员会推荐的一般和/或特定标准中，哪一项关于识别、收集或处理操作损失数据是 TRUE？

- A. P&L provisions or reserves against potential loss impacts are included in the gross loss computation
损益（P&L）中针对潜在损失影响计提的准备金或储备应计入总损失计算
- B. All internal loss events must be included in the gross loss computation regardless of the size of the loss
所有内部损失事件必须包括在总损失计算中，无论损失的大小如何
- C. Legal expenses are excluded from the gross loss computation in the operational loss data because they represent the realization of legal risk
法律费用被排除在操作损失数据中的总损失计算之外，因为它们代表了法律风险的实现
- D. Receivables (a short-term asset) should be counted as recoveries unless the particular receivable causes the current ratio to exceed an internal limit
应收账款（一个短期资产）应该被计入 recoveries，除非特定的应收账款导致当前比率超过内部限制

答案:A

解析:

A 选项正确。 True: P&L provisions or reserves against potential loss impacts are included in the gross loss computation In regard to (B), (C), and (D), each is FALSE. Rather, the following is true:

- While loss data must capture “all material activities,” losses below the minimum threshold do not need to be included.
- Legal expenses directly related to the event are included (see below). Recall that operational risk INCLUDES legal risk
- Receivables do not count as recoveries (see below).

From Gross loss, net loss, and recovery definitions (under 6. Specific criteria on loss data identification, collection and treatment):
“21. Gross loss is a loss before recoveries of any type. Net loss is defined as the loss after taking into account the impact of recoveries. The recovery is an independent occurrence, related to the original loss event, separate in time, in which funds or inflows of economic benefits are received from a third party.

22. Banks must be able to identify the gross loss amounts, non-insurance recoveries, and insurance recoveries for all operational loss events. Banks should use losses net of recoveries (including insurance recoveries) in the loss dataset. However, recoveries can be used to reduce losses only after the bank receives payment. Receivables do not count as recoveries. Verification of payments received to net losses must be provided to supervisors upon request.

23. The following items must be included in the gross loss computation of the loss data set:

- (a) Direct charges, including impairments and settlements, to the bank’s P&L accounts and write downs due to the operational risk event;
- (b) Costs incurred as a consequence of the event including external expenses with a direct link to the operational risk event (eg legal expenses directly related to the event and fees paid to advisors, attorneys or suppliers) and costs of repair or replacement, incurred to restore the position that was prevailing before the operational risk event;
- (c) Provisions or reserves accounted for in the P&L against the potential operational loss impact;
- (d) Losses stemming from operational risk events with a definitive financial impact, which are temporarily booked in transitory and/or suspense accounts and are not yet reflected in the P&L (“pending losses”). Material pending losses should be included in the loss data set within a time period commensurate with the size and age of the pending item; and
- (e) Negative economic impacts booked in a financial accounting period, due to operational risk events impacting the cash flows or financial statements of previous financial accounting periods (timing losses”). Material “timing losses” should be included in the loss data set when they are due to operational risk events that span more than one financial accounting period and give rise to legal risk.”

(Source: Basel III: Finalizing post-crisis reforms, Basel Committee on Banking Supervision Publication (December 2017), page 132.)

考察: Operational Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

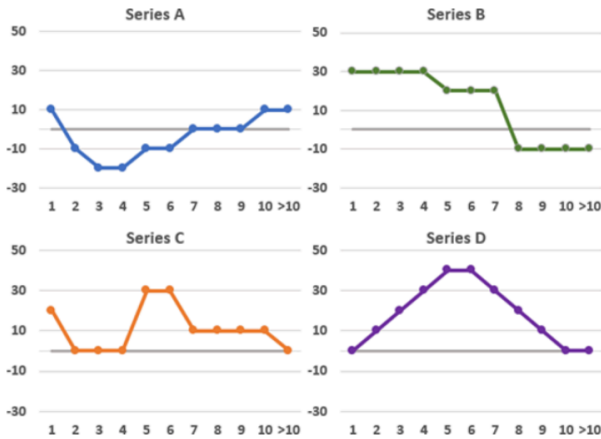
44. Geofinancial Bank currently has the following (very) simplified balance sheet:

Assets		Liabilities	
Bonds	\$30.0	Deposits	\$30.0
Loans	\$70.0	Bonds	\$50.0
		Equity	\$20.0

Further, the maturities of these accounts are as follows:

- **Assets:** The bonds (\$30.0 million) expire in one year. In regard to the loans (\$70.0), \$40.0 million expire in five (5) years, \$10.0 million expire in seven (7) years, and \$20.0 million expire in ten (10) years.
- **Liabilities:** In regard to the deposits (\$30.0 million), \$10.0 million expire in one (1) year, and \$20.0 million expire in two (2) years. In regard to the bonds (\$50.0 million), \$10.0 million expire in five (5) years, \$30.0 million expire in seven (7) years, and \$10.0 million expire in beyond ten (> 10) years.
- **Equity** (\$20.0 million) is presumed to expire in ten (10) years

Consider the following possible term structures of expected cash flows:



Which term structure of expected cash flows is accurate for Geofinancial Bank?

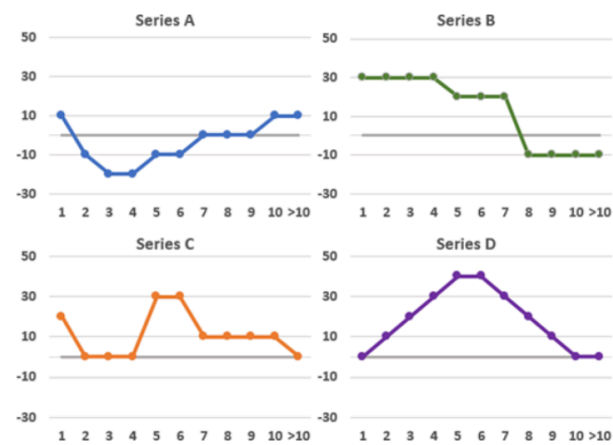
哪一种预期现金流量期限结构适用于 Geofinancial Bank? Geofinancial Bank 目前拥有如下（非常）简化的资产负债表：

Assets		Liabilities	
Bonds	\$30.0	Deposits	\$30.0
Loans	\$70.0	Bonds	\$50.0
		Equity	\$20.0

此外，这些科目的到期情况如下：

- **资产：**债券（\$30.0 百万）在一年后到期。就贷款（\$70.0 百万）而言，\$40.0 百万在五（5）年后到期，\$10.0 百万在七（7）年后到期，\$20.0 百万在十（10）年后到期。
- **负债：**就存款（\$30.0 百万）而言，\$10.0 百万在一（1）年后到期，\$20.0 百万在两（2）年后到期。就债券（\$50.0 百万）而言，\$10.0 百万在五（5）年后到期，\$30.0 百万在七（7）年后到期，\$10.0 百万在十年以上（> 10 年）到期。
- **权益**（\$20.0 百万）假定在十（10）年后到期

考虑以下可能的预期现金流期限结构：



对于 Geofinancial Bank，哪一种预期现金流期限结构是准确的？

- A. Series A
系列 A
- B. Series B
系列 B
- C. Series C
系列 C
- D. Series D
系列 D

答案:A

解析:

A 选项正确。 True: Series A. See below.

Liabil-				
Maturity	Assets	ities	Δ	Net
1	30	10	20	20
2		20	-20	0
5	40	10	30	30
7	10	30	-20	10
10	20	20	0	10
>10		10	-10	0
Sum	100	100		

考察: Finance: Asset Liability Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

45. Acme Industrial Corporation just finished establishing its new financial accounting controls which includes three components: planning, budgeting, and variance monitoring. The company’s next step is to establish a three-legged risk management stool that will analogously include the risk plan, the risk budget, and the risk monitoring process. In regard to the risk plan, each of the following will be an element EXCEPT which is the LEAST likely to be included in Acme’s risk plan?

Acme Industrial Corporation 刚刚完成了其新的财务会计控制体系的建立，其中包括三个组件：计划、预算和差异监控。公司的下一步是建立一个三足风险管理凳子，该凳子将类似地包括风险计划、风险预算和风险监控过程。在关于风险计划时，下列哪一项将是一个元素，除了哪一项是最不可能被包含在 Acme 的风险计划中？

- A. Diversification or risk decomposition policy
多样化或风险分解政策
- B. Strategic responses for events that are low-probability but franchise-threatening
针对低概率但具有威胁性的事件的战略响应
- C. Ample budget for losses that are material, predictable, and avoidable that cascade down
针对可预测且可避免的重大损失的充足预算
- D. Identification of critical dependencies that exist inside and outside the organization
识别存在于组织内部和外部的关键依赖关系

答案:C

解析:

C 选项正确。 FALSE. The risk plan should have zero tolerance for mistakes/losses that are “material, predictable, and avoidable.”

In regard to (A), (B), and (D) each is TRUE. According to the authors, the five guideposts in the risk plan are:

- Expected return and volatility goals; and mile posts to recognize failure.
- Definition of success or failure
- Vision for how risk capital will be deployed; and a diversification or risk decomposition policy
- Bright line definition between “merely disappointing events” and seriously damaging losses. Further, “Strategic responses should exist for any franchise-threatening event—even if such events are low-probability situations. The risk plan should identify those types of losses that are so severe that insurance coverage (e.g., asset class puts) should be sought to cover the downside.” (source: 2020 Financial Risk Management Part II: Risk Management and Investment Management, 10th Edition. Pearson Learning Solutions, 2019. VitalBook file.)
- Identification of critical dependencies inside and external to the organization

考察: Financial Risk Management 重要程度: [■■■]

科目: Quantitative Analysis

46. Bertha has a view about correlation, and she is evaluating several rainbow (aka, multi-asset) options. Her view is that correlation is going to decrease. Among the following trades, which **best** expresses her view?

Bertha 对相关性有一个看法，她正在评估几种彩虹（aka，多资产）选项。她的看法是相关性将会下降。在下列交易中，哪一种最佳表达了她的看法？

- A. Buy (aka, long position in) basket option
购买（aka，多头头寸）篮子选项
- B. Buy (aka, long position in) better-of-two where payoff = $\max(S_1, S_2)$
购买（aka，多头头寸）更好的两个选项，其中 payoff = $\max(S_1, S_2)$
- C. Buy correlation (aka, pay the fixed rate) in a correlation swap
购买相关性（aka，支付固定利率）在相关性交换中
- D. Short position in an exchange option where payoff = $\max(0, S_2 - S_1)$
空头头寸在交换选项中，其中 payoff = $\max(0, S_2 - S_1)$

答案:B

解析:

B 选项正确。 Bertha wants to be “short correlation,” and the better-of-two options become more valuable as the correlation decreases. On the other hand:

- The basket option’ s value becomes less valuable as the correlation decreases, such that buying the basket option is to be long correlation

- Buying correlation (aka, paying the fixed rate) in a correlation swap is long correlation
- An exchange option becomes more valuable as the correlation decreases such that a long position in an exchange option is short correlation, and short position in an exchange option is long correlation

考察: Market Risk: Correlation 重要程度: [■■■□]

科目: Quantitative Analysis

47. Julia the analyst is analyzing the credit risk of a contract to which her firm is considering entering into as a party. If true, which feature of her work (among the choices below) **most likely** signifies or indicates that this is a case of counterparty credit risk rather than (traditional) lending risk or some other type of risk?

Julia 分析师正在分析她的公司考虑作为一方进入的合同中的信用风险。如果为真，她的工作中哪一项特征（在下列选项中）最有可能表明或指示这是一个 counterparty 信用风险而不是（传统）贷款风险或其他类型的风险的情况？

- A. Interest rate (as a market variable) fluctuations have little impact on the amount owed
利率（作为市场变量）波动对欠款的影响很小
- B. The exposure at default (EAD) is 100% of the notional over most of the life of the contract
违约时的暴露（EAD）是合同生命周期的大部分时间内名义金额的 100%
- C. Her firm is the only party assuming credit risk: if her firm defaults, the other party faces no losses
她的公司是唯一承担信用风险的方：如果她的公司违约，其他方不会面临损失
- D. She is employing Monte Carlo simulation because the future value of the contract is highly uncertain
她正在使用蒙特卡洛模拟，因为合同的未来价值高度不确定

答案:D

解析:

D 选项正确。 She is employing Monte Carlo simulation because the future value of the contract is highly uncertain.

According to Jon Gregory, the two key differences between counterparty and lending risk are (1) the lender knows the amount at risk but in the case of counterparty risk the amount is highly uncertain, and (2) lending involves unilateral risk (by the lender) but counterparty risk is bilateral to both counterparties.

In regard to false (a), Herstatt risk is a special case (cross-currency) of settlement risk as opposed to pre-settlement risk.

考察: Counterparty Credit Risk 重要程度: [■■■□]

科目: Quantitative Analysis

48. The Basel regulatory framework has undergone several major renovations but started in the late 1980s when the Basel Committee on Banking Supervision published the first Basel accord, which is now called Basel I; this original Basel I was implemented in 1992. According to Carey (GARP Chapter 19), which BEST summarizes the motivation(s) of the original Basel?

巴塞尔监管框架已经经历了几次重大改革，但始于 1980 年代末，当时巴塞尔银行监管委员会发布了第一份巴塞尔协定，现在称为巴塞尔 I；这个原始的巴塞尔 I 于 1992 年实施。根据 Carey（GARP 第 19 章），哪一项 BEST 总结了原始巴塞尔的动机？

- A. To formalize regulatory capital requirements with a set of regulations (i.e., Basel) that would become legal requirements and therefore carried the force of law; i.e., the motivation was legal
通过一套监管规定（即巴塞尔协议）形式化监管资本要求，使其成为具有法律约束力的法律要求；即其动机是法律层面的考虑
- B. Regulators realized that specific measures of short- and long-term liquidity were necessary given that liquidity crunches had been the essential common denominator of recent financial disasters; i.e., the motivation was liquidity
监管者意识到，有必要对短期和长期流动性进行具体度量，因为流动性紧缩一直是近期金融灾难的共同关键因素；即其动机是流动性管理

- C. The growth of cross-border finance required both the solvency of banks during stressful periods and fair competition (aka, level playing field) between banks competing in each other’s home countries; i.e., the motivations were bank safety and competitive concerns
跨境金融的增长要求银行在压力时期保持偿付能力，并要求在彼此本国市场竞争的银行之间实现公平竞争（即“公平竞争环境”）；因此其动机是银行安全性和竞争方面的考量
- D. In the wake of a global recession, there was a motivation to incentivize banks to increase their lending (which had been historically low) in order to grow the developed and emerging economies; i.e., the motivation was economic recovery especially of emerging economies
在全球经济衰退之后，有动机激励银行增加放贷（此前历史上处于较低水平），以推动发达经济体和新兴经济体的增长；即其动机是促进经济复苏，尤其是新兴经济体

答案:C

解析:

Option C is correct. True: The growth of cross-border finance required both the solvency of banks during stressful periods and fair competition (aka, level playing field) between banks competing in each other’s home countries; i.e., the motivations were bank safety and competitive concerns.

Carey explains, “Two events motivated creation of Basel I.

- First, the growth of cross-border finance continued after Herstatt’s failure and it was evident that the G10 nations had a common interest in ensuring that banks had enough equity to absorb large losses.
- Second, international banks were competing vigorously in each other’s home countries. However, minimum levels of required capital varied significantly across nations, creating a perception that banks headquartered in countries with low minimums had a competitive advantage. In response, members of the BCBS decided to develop a global minimum standard to “level the playing field” and avoid a race to the bottom. That is, while the Basel Accord was partly about ensuring safety and soundness, negotiations also had an element of maneuvering for perceived competitive advantage.”(Source: Chapter 19, 2020 FRM Part II: Operational Risk and Resiliency, 10th Edition. Pearson Learning Solutions.)

考察: Finance: Basel Accords 重要程度: [■■■□]

科目: Quantitative Analysis

49. Which of the following types of liquidity is available in the form of the institution’ s liquidity asset buffer (LAB) and represents liquidity available to meet general financial obligations under a stress scenario?

哪一种流动性形式存在于机构的流动性资产缓冲（LAB）中，并代表在压力情景下可用于满足一般财务义务的流动性？

- A. Operational
运营性
- B. Strategic
战略性
- C. Contingent
偶然性
- D. Restricted
限制性

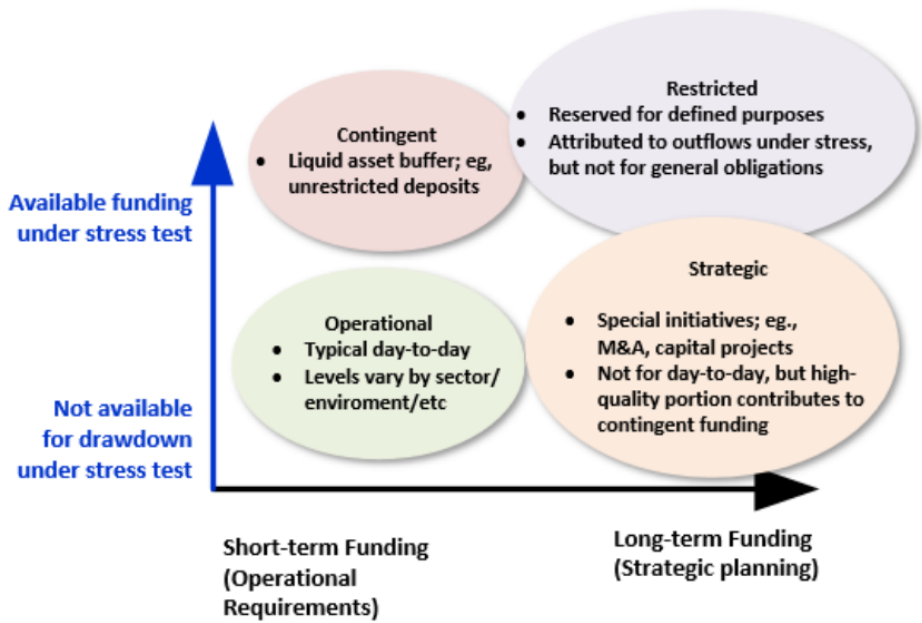
答案:C

解析:

C 选项正确。 Contingent liquidity captures cash or assets that are available when a stressing scenario materializes. The liquidity is contingent on the occurrence of the stress situation. Included within this category are financial assets that can be converted into cash without any material loss in value. Capturing contingent liquidity to cover stressed cash outflows represents the main reason for liquidity stress tests.

See also the diagram below (inspired by Venkat’ s Figure 3.1(†) but rendered by David Harper in MS Visio).

Also, writes Venkat (emphasis ours), “**Liquid asset buffer:** The liquid asset buffer represents the contingent liquidity that is currently in place ... The fundamental characteristics of liquid asset buffer securities should include low credit and market risk, ease and certainty of valuation, trading in an active and sizable market, and low concentration of buyers and sellers. The liquid asset buffer should also meet operational requirements that ensure the liquidity is under the control of the central treasury area of the entity undergoing the stress test.” (†)



考察: Finance: Liquidity Risk 重要程度: [■■□]
科目: Quantitative Analysis

50. Acme Investment Bank recently finished building the third leg of its three-legged risk management stool, after the risk plan and the risk budget: the bank installed its risk monitoring process. The bank shares a belief with the authors (Rosengarten and Zangari) that “risk is the shadow cost behind returns.” Each of the following is probably an element or plank in the bank’s risk monitoring process EXCEPT which is the least likely to be found in the bank’s risk monitoring process?

Acme Investment Bank 最近完成了其三足风险管理凳子的第三条腿的建设，在风险计划和风险预算之后：该银行安装了其风险监控过程。该银行与作者（Rosengarten 和 Zangari）分享了“风险是回报背后的影子成本”的信念。下列哪一项可能是银行风险监控过程中的元素或 plank，除了哪一项最不可能在银行的风险监控过程中找到？

- A. The bank’s monitoring is not dependent on any single tool or metric, but instead relies on an ensemble of various tools and metrics
银行的监控不依赖于任何单一工具或指标，而是依赖于各种工具和指标的集合
- B. The bank measures alpha as a regression intercept where alpha can help evaluate manager skill but unfortunately, its statistical significance often requires more data (i.e., longer history) than is available
银行测量 alpha 作为回归截距，其中 alpha 可以帮助评估经理技能，但不幸的是，其统计显著性通常需要更多数据（即更长的历史），而不是可用的数据
- C. The bank measures return on risk capital (RORC)) where value at risk (VaR) is a measure of risk capital; in addition to the RORC’s level, the bank watches its variability
银行测量风险资本回报（RORC），其中价值 at risk（VaR）是风险资本的度量；除了 RORC 的水平，银行还监控其可变性
- D. After a threshold level of data is collected, subjective articulations of each portfolio manager’s philosophy becomes unnecessary, and should be discarded, in favor of quantitative performance evaluation
在收集到阈值水平的数据后，每个投资组合经理的主观哲学陈述变得不必要，应该被抛弃，取而代之的是定量绩效评估

答案:D
解析:

Option D is correct. D. FALSE. Instead, articulation of each portfolio manager’s philosophy is always necessary as a complement to quantitative tools and to clarify whether style drift occurs.

As GARP explains in the reading, explicit (and subjective) articulation of philosophy is always a necessary complement to quantitative tools if for no other reason than it helps to identify *style drift*. From the reading: “Before discussing these [i.e., commonly used performance tools], however, it is worth noting that performance tools, while necessary, are not a substitute for timely management intervention when there is an indication of abnormal behavior. By the time that abnormal behavior manifests itself in the form of poor performance statistics, the damage might already be irreversible. For this reason, we believe that performance tools must be supplemented with:

- A clear articulation of management philosophy from each portfolio manager. This philosophy statement should identify how the manager expects to extract returns from the market. It should identify ways of knowing when the manager’s process is successful and when it is unsuccessful.
- A routine position and style monitoring process designed to identify deviations from philosophy or process. This is a type of early warning system.” —Source: 2020 Financial Risk Management Part II: Risk Management and Investment Management, 10th Edition. Pearson Learning Solutions, 2019. VitalBook file.

In regard to (A), (B), and (C), each is TRUE. Specifically,

- The bank’s monitoring is not dependent on any single tool or metric, but instead relies on an ensemble of various tools and metrics
- The bank measures alpha as a regression intercept where alpha can help evaluate manager skill but unfortunately, its statistical significance often requires more data (i.e., longer history) than is available
- The bank measures return on risk capital (RORC)) where value at risk (VaR) is a measure of risk capital; in addition to the RORC’s level, the bank watches its variability

考察: Financial Risk Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

51. Acme FinGlobal Bank has engaged consultants to redefine the organization’s risk measures. The project includes value at risk (VaR) mapping across several portfolios (aka, asset classes) at the organization:

- A fund that invests in initial public offerings (IPOs)
- A fund that invests in emergent assets but where with there is little historical data
- Small equity portfolio with less than 30 positions
- Large equity portfolio with 1,000 positions
- A portfolio of corporate bonds that carries six different ratings and has maturities of 3, 5, and 7 years

Each of the following statements is TRUE except which is false?

Acme FinGlobal Bank 聘请了顾问对机构的风险度量进行重新定义。该项目包括在机构内部对若干投资组合（即资产类别）进行价值-at-风险（VaR）映射：

- 一只投资于首次公开募股（IPO）的基金
- 一只投资于新兴资产、但历史数据很少的基金
- 持仓少于 30 个头寸的小型股票投资组合
- 持仓 1,000 个头寸的大型股票投资组合
- 一只公司债券投资组合，包含六种不同评级，期限为 3 年、5 年和 7 年

以下关于这些内容的陈述均为真，除了哪一项是假的？

A. The corporate bond can be mapped to nine general risk factors
公司债券可以映射到九个一般风险因素

- B. The IPO and emergent asset funds cannot be mapped because these instruments lack historical data such that sensitivities cannot be estimated
首次公开募股和新兴资产基金无法映射，因为这些工具缺乏历史数据，因此无法估计敏感性
- C. The large equity portfolio is an ideal candidate for mapping because mapping to general and specific market primitives requires far fewer numerical estimates than specification of a covariance matrix
大型股票投资组合是映射的理想候选，因为映射到一般和特定市场原语需要比指定协方差矩阵更少的数值估计
- D. The equity portfolio VaR mapping process contains sources of model risk, including factor sensitivities (aka, betas) estimation error, subjective risk factor choices, and specific risk omission
股票投资组合价值-at-风险（VaR）映射过程包含模型风险，包括因子敏感性（aka，贝塔）估计误差、主观风险因子选择和特定风险遗漏

答案:B

解析:

Option B is correct. False. Instead, mapping is specifically a solution to the problem of insufficient data. As Jorion explains, “The risk manager would have to replace these positions by exposures on similar risk factors already in the system.” (†)

In regard to (A), (C), and (D), each is TRUE.

(†) Philippe Jorion, Value-at-Risk: The New Benchmark for Managing Financial Risk, 3rd Edition. (New York: McGraw Hill, 2007)

考察: Market Risk Measurement and Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

52. Freecone Financial LLC entered into an interest rate swap some time ago where it agreed to make semi-annual fixed payments at a rate of 3.0% per annum, in exchange for receiving floating payments on a notional amount of \$50.0 million. The floating rate is the six-month secured overnight financing rate (SOFR) and this is also the discount rate used to value the swap. The swap has a remaining life of 1.25 years. The SOFR rate applicable to the exchange in three months was determined three months ago: it was 2.40%. Just as with a LIBOR-based swap, the floating rate is observed at the beginning of each six-month period but paid at the end of the period. Currently, the SOFR term structure is flat at 2.60% per annum with continuous compounding. With respect to its position in this swap, which of the following is nearest to Freecone Financial’s current exposure? Freecone Financial LLC 不久前进入了一个利率互换，其中它同意每半年支付一次固定利率，年利率为 3.0%，以换取收到名义金额为 \$50.0 百万的浮动支付。浮动利率是六个月期有担保隔夜融资利率（SOFR），这也是用于估值互换的贴现率。互换剩余期限为 1.25 年。三个月前确定的适用于交换的 SOFR 利率为 2.40%。与基于 LIBOR 的互换一样，浮动利率在每个六个月期初观察，但在每个六个月期末支付。目前，SOFR 期限结构是平坦的，年利率为 2.60%，连续复利。就其在该互换中的头寸而言，下列哪一项最接近 Freecone Financial 的当前暴露？

- A. Zero
- B. -\$335,900
- C. +\$335,900
- D. +1,335,900

答案:A

解析:

Option A is correct. Current exposure = max(MtM value, 0). In this case, CE = max(−335,900, 0) = 0. Freecone is underwater in the swap; its counterparty has positive current exposure because the counterparty experiences a loss in the event of default. But Freecone does not experience a CURRENT loss in the event of default.

考察: Credit Risk: Current Exposure 重要程度: [■ ■ □]

科目: Quantitative Analysis

53. According to the Guidance on Managing Outsourcing Risk (by the Board of Governors of the Federal Reserve System), with respect to a third-party service provider arrangement, the overall due diligence process by a financial institution should at

least include a review of each of the following EXCEPT which is not essential?
根据联邦储备委员会董事会关于管理外包风险的《指导》，在第三方服务提供商安排方面，金融机构的总体尽职调查过程至少应包括对以下各项的审查，除了哪一项不是必要的？

- A. Ensure service provider has an appropriate background check program for its employees
- B. Financial review of service provider’s most recent annual report and financial statements
对服务提供商最近年度报告和财务报表的财务审查
- C. Evaluate the service provider’s performance along environmental, social, and governance (ESG) factors
评估服务提供商在环境、社会和治理（ESG）因素方面的表现
- D. Evaluate the adequacy of standards, policies and procedures including adherence to applicable laws, regulations and supervisory guidance
评估标准、政策和程序的充分性，包括遵守适用法律、法规和监管指导

答案:C

解析:

- C 选项正确。 A review of ESG factors is not mentioned. According to the Board, the three broad review areas are:
- 1. *Business background, reputation, and strategy*: “Financial institutions should review a prospective service provider’s status in the industry and corporate history and qualifications; review the background and reputation of the service provider and its principals; and ensure that the service provider has an appropriate background check program for its employees.”
 - 2. *Financial performance and condition*: “Financial institutions should review the financial condition of the service provider and its closely-related affiliates. The financial review may include: The service provider’s most recent financial statements and annual report with regard to outstanding commitments, capital strength, liquidity and operating results; The service provider’s sustainability, including factors such as the length of time that the service provider has been in business and the service provider’s growth of market share for a given service ...”
 - 3. *Operations and internal controls*.

考察: Operational Risk: Outsourcing Risk 重要程度: [■■■□]

科目: Quantitative Analysis

54. The staff at Umbrella Street Bank, which is a commercial bank (it could also be a savings institution or credit union), produces several liquidity risk reports on a daily, weekly, monthly, and quarterly basis. Among these liquidity risk reports is a deposit tracker report. Among the following metrics, which is MOST LIKELY to appear in their deposit tracker report?
Umbrella Street Bank 的员工（这是一家商业银行，也可以是一家储蓄机构或信用合作社）会按日、周、月和季度频率编制多种流动性风险报告。在这些流动性风险报告中包含一份存款追踪报告。在下列指标中，哪一项最有可能出现在他们的存款追踪报告中？

- A. Loan-to-deposit (LTD) ratio (current and forecast) versus board-approved lower floor (limit) of 120.0%
贷款-存款（LTD）比率（当前和预测）与董事会批准的最低地板（限制）120.0%
- B. Loan-to-deposit (LTD) ratio (current and forecast) versus board-approved upper ceiling (limit) of 80.0%
贷款-存款（LTD）比率（当前和预测）与董事会批准的最高天花板（限制）80.0%
- C. Market-to-book ratio of common equity (current and forecast) versus investor-communicated target of 1.30
普通股的市场-账面比率（当前和预测）与投资者沟通的目标 1.30
- D. The leverage-adjusted duration gap (current and forecast) versus board-approved upper ceiling of 3.5 years
杠杆调整后的期限缺口（当前和预测）与董事会批准的最高天花板 3.5 年

答案:B

解析:

B 选项正确。 True: Loan-to-deposit (LTD) ratio (current and forecast) versus board-approved upper ceiling (limit) of 80.0%

In regard to (A), (C) and (D), each is FALSE.

考察: Liquidity Risk Management 重要程度: [■■■]

科目: Quantitative Analysis

55. You work as a pension fund administrator and the Investment Committee asks for your input. The pension fund seeks to increase its allocation to alternative investments and, in particular, will add a new hedge fund strategy; aka, style. You are limited to four strategies. You are perusing the brochures of four hedge funds (each named after Greek gods per a popular marketing convention!). The four strategies are global macro, merger arb, long/short equity, and equity market neutral. Below are the four funds. Each fund has a name, a strategy (aka, style), a tagline, and a very brief description.

- *Acheilus* is a Global Macro strategy. We are your effective asset (factor) allocator! We opportunistically make bets in different markets utilizing a range of strategies; you can think of us as betting on a range of factors such as commodities, interest rates, exchange rates.
- *Boreas* is a Merger Arb (aka, Risk Arb) strategy. We collect nickels (and dimes) in front of the proverbial steamroller! Like our sibling event-driven strategy (i.e., Distressed), we can offer you nonlinear returns but with tail exposure to extreme market conditions.
- *Chimera* is a Long/Short Equity strategy. We are friendly but will never hug a benchmark! Although we have some directional market exposure, we can be classified as relative value because we are stock pickers who seek alpha by way of a diversity of opinions and ability.
- *Daphne* is an Equity Market Neutral strategy. We bring you a basket of betas! As an event-driven strategy, our returns are highly linked to several common market risk factors including equity indices, bond spreads, and volatility factors.

Each of these descriptions is consistent with its strategy (aka, style) category EXCEPT for one. Select the fund that is NOT consistent with its description.

你在一家养老金基金担任管理人，投资委员会向你征求意见。该养老金基金计划提高对另类投资的配置，特别是将新增一种对冲基金策略（即风格）。你最多只能选择四种策略。你正在翻阅四只对冲基金的宣传册（每只基金都以希腊神命名，这是常见的营销惯例！）。这四种策略分别是全球宏观、并购套利、多空股票以及股票市场中性。下面是这四只基金。每只基金都有名称、策略（即风格）、一句宣传语以及非常简短的说明。

- *Acheilus* 是一只全球宏观（Global Macro）策略基金。我们是你高效的资产（因子）配置者！我们会在不同市场中以机会主义方式进行押注，并运用多种策略；你可以将我们视为在大宗商品、利率、汇率等一系列因子上进行押注。
- *Boreas* 是一只并购套利（Merger Arb，也称风险套利 Risk Arb）策略基金。我们是在俗称的“蒸汽压路机”前面捡镍币（和一角钱）的人！与我们的兄弟事件驱动策略（即困境投资 Distressed）类似，我们可以为你提供非线性的收益，但同时在极端市场环境下暴露于尾部风险。
- *Chimera* 是一只多空股票（Long/Short Equity）策略基金。我们很友好，但绝不会去“拥抱”基准！尽管我们有一定方向性的市场敞口，但可以被归类为相对价值策略，因为我们是通过择股来获取阿尔法，依靠的是多样化的观点与能力。
- *Daphne* 是一只股票市场中性（Equity Market Neutral）策略基金。我们给你带来一篮子贝塔！作为一只事件驱动策略基金，我们的收益与若干共同的市场风险因子高度相关，包括股票指数、债券利差以及波动率因子。

在上述描述中，除了其中一只基金以外，其余每只基金的描述都与其策略（即风格）类别相一致。请选择那只与其描述不一致的基金。

A. Acheilus
阿基里斯

B. Boreas
波瑞阿斯

C. Chimera
奇美拉

D. Daphne
达芙妮

答案:D

解析:

D 选项正确。 The tag/description is thrice inconsistent: Equity market neutral is unlinked to observable market risk factors (in the cited research, it is the ONLY strategy found without links to market risk factors). Consequently, an equity market-neutral fund would not “bring a basket of betas:” its intention is to neutralize betas; specifically, zero beta with respect to the common market factor. Further, it is not event-driven; equity market-neutral is non-directional and classified as a “niche strategy” by the authors, but can also be classified as “equity hedge.”

In regard to (A), (B), and (C), each of the other funds (Acheilus, Boreas, and Chimera) have consistent strategies, taglines, and descriptions; i.e., they are conceivably true.

考察: Finance: Hedge Fund Strategies 重要程度: ■■■

科目: Quantitative Analysis

56. Betty’s firm wants to allocate an economic capital cushion based on a one-day 99.5% value at risk (VaR) model. Further, she will conduct a basic backtest (aka, exception or frequency test) where the frequency of tail losses fit a binomial distribution. Her backtest window is limited to two years, $T = 500$ days. In regard to this backtest, which of the following statements is TRUE?

Betty 的公司想要基于一天的 99.5% 价值 at risk (VaR) 模型分配经济资本缓冲。此外, 她将进行一个基本回测 (aka, 异常或频率测试), 其中尾部损失的频率拟合二项分布。她的回测窗口限制为两年, $T = 500$ 天。就这个回测而言, 下列哪一项陈述是正确的?

- A. To increase the power of her backtest, she can decrease the VaR confidence level and apply a higher (k) capital multiplier
为了增加回测的威力, 她可以降低 VaR 置信水平并应用更高的 (k) 资本乘数
- B. At this high VaR confidence level and relatively short backtest window, the normal approximation is a better fit to test the model’s bias than the binomial distribution
在这个高 VaR 置信水平和相对较短的回测窗口下, 正态近似比二项分布更适合测试模型的偏差
- C. She can achieve a more powerful tests of the same hypothesis, and without the need for any additional information, by utilizing any of several extensions approved by regulators
她可以通过利用任何由监管机构批准的扩展, 在没有额外信息的情况下, 实现更强大的同一假设测试
- D. Although VaR is always a one-tailed quantile (e.g., the 95.0% standard normal VaR is always 1.645 and never 1.960), the backtest is always a two-tailed (2T) backtest such that Betty must have a lower and an upper cutoff
尽管 VaR 总是单尾分位数 (例如, 95.0% 标准正态 VaR 总是 1.645 且从不 1.960), 但回测总是双尾 (2T) 回测, 因此 Betty 必须有一个下限和一个上限

答案:A

解析:

A 选项正确。 To increase the power of her backtest, she can decrease the VaR confidence level and apply a higher (k) capital multiplier.

A key statistical challenge for the VaR backtest is that a high confidence level implies a small sample of exceptions. A remedy is to lower the confidence level, but apply a multiplier (to the lower-confidence VaR) in order to approximate a higher-confidence VaR.

As Jorion explains, “The lack of power of this framework is due to the choice of the high VAR confidence level (99 percent) that generates too few exceptions for a reliable test. Consider instead the effect of a 95 percent VAR confidence level. (To ensure that the amount of capital is not affected, we could use a larger multiplier k.)” —Philippe Jorion. Value at Risk, 3rd Ed.: The New Benchmark for Managing Financial Risk (p. 150). Kindle Edition.

In regard to (B), (C) and (D), each is FALSE. Specifically,

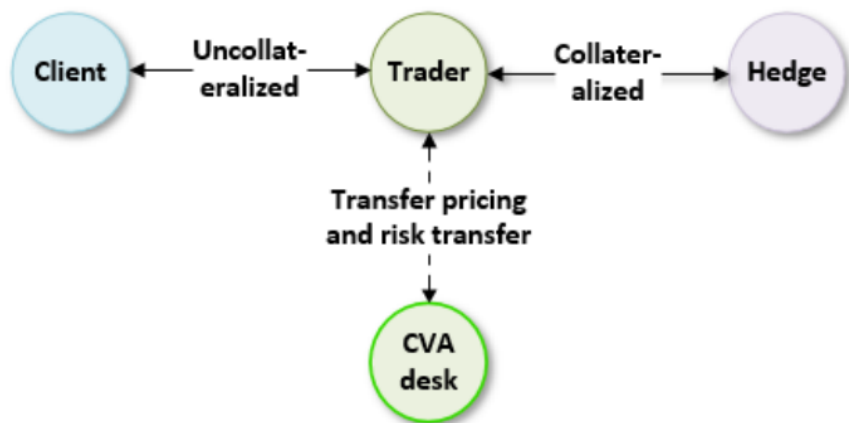
- A general guideline for the normal approximation of the binomial is that it can be used when $n \times p$ and $n \times (1 - p)$ are greater than 10; some would say greater than 5. In this case, $0.5\% \times 500 = 2.5$, such that the binomial probably has too much skew.

- The readings (Jorion, Hull or even Dowd) do not reference power gains via regulator-approved extensions
- VaR is always a one-tailed quantile (e.g., the 95.0% standard normal VaR is always 1.645 and never 1.960), but the backtest is a statistical test and can be either one- or two-tailed (2T)

考察: Value at Risk (VaR) 重要程度: [■■■□]

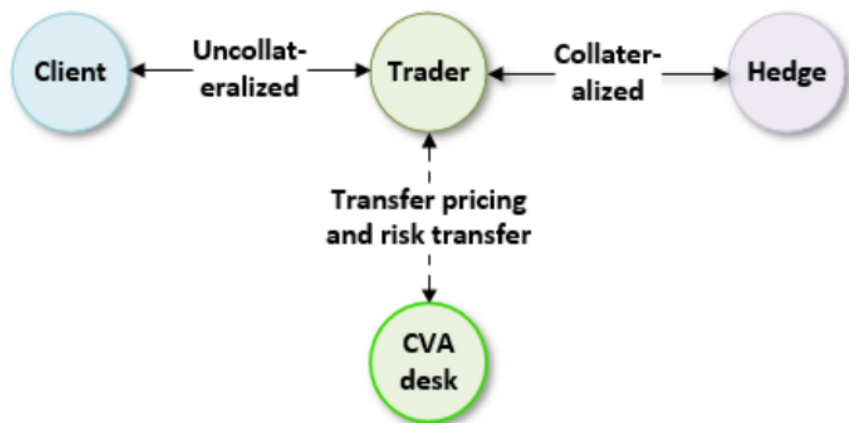
科目: Quantitative Analysis

57. The figure below illustrates the general role of the credit value adjustment (CVA) desk at Heroic Multifunction Bank (aka, the bank). Assume the trader is a market maker. The uncollateralized trade with the customer (on the left) is hedged with a collateralized trade (on the right):



Assume the uncollateralized transaction (with the client) has a positive mark-to-market (MTM) value such that, by definition, the collateralized hedge position, therefore, has a negative MTM value. If the market moves such that, with respect to the position with the client, the position's positive MTM value INCREASES and becomes even more positive, then which of the following implications is most likely to be TRUE?

下图展示了 Heroic Multifunction Bank（即该银行）中信用估值调整（CVA）交易台的一般作用。假设交易员是做市商。与客户之间的无抵押交易（左侧）通过一笔有抵押交易（右侧）进行对冲：



假设与客户之间的无抵押交易具有正的市值（MTM），则按定义，相应的有抵押对冲头寸必然具有负的市值。如果市场发生变动，使得客户头寸的正向市值进一步上升、变得更加为正，则下列哪一项推论最有可能为真？

- A. Funding cost (FVA) is positive
融资成本（FVA）为正
- B. From the trader's (market maker's) perspective, CVA decreases
从交易员（做市商）的角度来看，CVA 减少
- C. Because the position is hedged, there is no associated regulatory cost of capital to the bank
由于头寸得到对冲，银行没有与监管资本相关的成本

D. The trader (market maker) writes a contingent credit default swap (CCDS) to transfer its counterparty risk
交易员（做市商）写一份或有信用违约互换（CCDS）来转移其对手方风险

答案:A

解析:

A 选项正确。 A. TRUE: Funding cost (FVA) is positive. It will be necessary to post collateral on the hedge and this amount will need to be funded. In regard to (B), (C) and (D), each is FALSE.

- In regard to false (B), CVA is likely to increase as exposure to the further positive position increases
- In regard to false (C), their hedge will not eliminate the need for regulatory capital! Says Gregory, “Capital: There will be capital requirements and these will change – which is important because capital is a cost. The capital for the uncollateralised transaction will increase as it becomes in-the-money. The capital for the hedge will not offset this due to being collateralised –and even if it was uncollateralised, there would likely be a convexity impact (the capital for one transaction would increase by more than the capital reduction on the other).”
- In regard to false (D), the problem here is that the trader needs to buy (purchase) not write (sell) a CCDS. Says Gregory: “More tailored credit derivative products such as contingent CDS (CCDS) have been designed to hedge counterparty risk even more directly. CCDSs are essentially CDSs but with the notional of protection indexed to the exposure on a contractually specified derivative (or even portfolio of derivatives). They allow the synthetic transfer of counterparty risk linked to a specific trade and counterparty to a third party. However, the CCDS market has never developed any significant liquidity.”

(Source: Jon Gregory, The xVA Challenge: Counterparty Credit Risk, Funding, Collateral, and Capital, 3rd edition (West Sussex, UK: John Wiley & Sons, 2015))

考察: Finance: Counterparty Credit Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

58. Andrew Ang makes an important, provocative statement when he writes “Reported illiquid asset returns are not returns.” He claims that people overstate the expected returns and understate the risk of illiquid assets, and he attributes this to three key biases. According to Ang, each of the following is a bias that overstates the expected returns (and/or understates the risk) of illiquid assets EXCEPT which is not accurate?

Andrew Ang 写了一句重要、有争议的话：“报告的流动性较差资产的回报不是回报。”他声称人们高估了流动性较差资产的预期回报，低估了其风险，并将此归因于三个关键偏差。根据 Ang，下列哪一项是高估了流动性较差资产的预期回报（和/或低估了风险）的偏差，除了哪一项不准确？

- A. Survivorship bias can inflate returns by 4.0% or more
生存偏差可以将回报提高 4.0% 或更多
- B. Infrequent sampling (aka, infrequent trading) artificially reduces risk and risk-related metrics such as volatility, correlation and beta
不频繁采样（aka，不频繁交易）人为地降低了风险和与风险相关的指标，如波动性、相关性和 beta
- C. Turnover bias decreases the typical time between transactions and tends to artificially increase the expected return by 5.0% or more
turnover bias 降低了交易之间的典型时间，并倾向于人为地增加预期回报 5.0% 或更多
- D. Selection bias (aka, reporting bias) is a distortion of the sample that artificially increases (ie, overestimates) alpha and artificially decreases (ie, underestimates) beta
选择偏差（aka，报告偏差）是样本的扭曲，人为地增加（即高估）alpha 和人为地减少（即低估）beta

答案:C

解析:

C 选项正确。False. Turnover bias is made-up. In regard to (A), (B) and (D), each is TRUE. According to Ang , with respect to illiquid assets, the three biases that prompt investor to overstate expect returns while underestimating risk are: survivorship bias; infrequent sampling; and selection bias.

考察: Alternative Investments: Illiquid Assets 重要程度: [■ ■ □]

科目: Quantitative Analysis

59. Which of the following provides the BEST explicit link(s) between a bank’s contingency funding plan (CFP) and its liquidity stress testing framework?

哪一项提供了银行 contingency funding plan（CFP）与其流动性压力测试框架之间最佳的显式链接？

- A. Communication plan
沟通计划
- B. Limit structure and escalation levels
限制结构和升级级别
- C. Funding sources concentration and non-core funding
资金来源集中度和非核心资金
- D. Industry factors such as profitability trends in the financial sector
行业因素，如金融行业的盈利趋势

答案:B

解析:

B 选项正确。 True: limit structure and escalation levels

Explain Lai and Tuosto (emphasis ours), “II. Integrate with Broader Risk Management Frameworks: The CFP is not a stand-alone tool, but rather, an integrated part of the institution’s liquidity risk management and firm-wide risk management frameworks ... The CFP should be explicitly linked to the liquidity risk measurement framework and the liquidity stress test, in particular through its limit structure and escalation levels. For example, the liquidity risk measures used in the institution’s BAU risk management activities serve as a foundation from which the CFP defines its early warning indicators (EWIs). Additionally, linkages to the business continuity and crisis management frameworks will reinforce key operational and communication protocols during times of crisis.” —Chi Lai and Richard Tuosto, Contingency Funding Planning in Venkat’s Liquidity Risk Management

考察: Liquidity Risk Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

60. Bodie, Kane and Marcus explain, based on an analogy originally developed by Robert C. Merton, that we can attempt to quantify the value of market timing by recognizing that foresight is similar to holding a call option on the owned equity portfolio (without paying the premium): the perfect market timer exercises her option, so to speak, by switching from the riskfree asset to the equity portfolio (or vice-versa) depending on which offers the better return. Consequently, the Black-Scholes (BSM) formula can be used, but when the strike price is given by $\$1.00 \cdot \exp(rT)$, the BSM call option formula conveniently reduces to $c = 2N[0.5\sigma(e)\sqrt{T}] - 1$.

If we assume annual forecasts ($T = 1.0$ year), and the volatility of the equity portfolio, $\sigma(e) = 36.0\%$ per annum, which of the following is **nearest** to the value of perfect market timing, as a percentage of the equity portfolio’s value?

Bodie, Kane 和 Marcus 基于 Robert C. Merton 最初发展的类比，解释了我们可以尝试量化市场时机价值，通过认识到远见卓识类似于持有 owned equity portfolio（即股票投资组合）的看涨期权（无需支付溢价）：完美的市场时机者行使她的期权，换句话说，根据哪一方提供更好的回报，从无风险资产切换到股票投资组合（或反之亦然）。因此，Black-Scholes（BSM）公式可以被使用，但当敲定价格为 $\$1.00 \cdot \exp(rT)$ 时，BSM 看涨期权公式方便地简化为 $c = 2N[0.5\sigma(e)\sqrt{T}] - 1$ 。

如果我们假设年度预测（ $T = 1.0$ 年），以及股票投资组合的波动率， $\sigma(e) = 36.0\%$ 每年，哪一项最接近完美市场时机的价值，作为股票投资组合价值的百分比？

- A. 7.39%
- B. 14.28%
- C. 46.50%

D. 83.77%

答案:B

解析:

B 选项正确。 In this case, $0.5\sigma(e)\sqrt{T} = 0.5 \times 36.0\% \times \sqrt{1} = 18.0\% = 0.180$ and $N(0.18) = 0.57140$ if rounded per the lookup table. Then $c = 2 \times 0.57140 - 1 = 14.28\%$.

考察: Portfolio Management 重要程度: [■ ■ □]

科目: Quantitative Analysis

61. Patricia has been asked by the Chief Risk Officer (CRO) to estimate value at risk (VaR) at high confidence levels. Initially, she fits a generalized extreme value (GEV) distribution to the loss data. However, some members of the ultimate audience (the board) do not understand the math. She observes that Dowd(†) explains, "[t]here are also short-cut ways to estimate VaR (or ES) using EV theory. These are based on the idea that if $\xi > 0$, the tail of an extreme loss distribution follows a power-law times a slowly varying function: $F(x) = k(x)x^{-1/\xi}$ where $k(x)$ varies slowly with x . For example, if we assume for convenience that $k(x)$ is approximately constant, then [this equation] becomes: $F(x) \approx kx^{-1/\xi}$ ". Notice the assumption that $k(x)$ is approximately constant.

The exponent in Dowd's power law maintains consistency with the generalized extreme value (GEV) distribution by referring to the exponent as a negative reciprocal, $-1/\xi$, but given this exponent is also a constant along with $k(x)$, we can further simplify the approximation: instead of $F(x) \approx kx^{-1/\xi}$, we can say that $F(x) \approx kx^{-\xi}$ which is sometimes represented as $F(x) \approx kx^{-\alpha}$ where $\alpha = 1/\xi$. For example, Patricia assumes the exponent's value is -4.0. In the former syntax, $\xi = 0.25$ such that $(-1/\xi) = -4.0$; in the latter version, $\alpha = 4.0$ such that $(-\alpha) = -4.0$, to the same effect. For our purpose here, we will simply refer to $F(x) \approx kx^{-\xi}$.

Consequently, she decides to estimate extremely confident VaRs with a power-law shortcut. Specifically, when Patricia sets xi, $\xi = 4.0$, she observes that the 99.50% VaR is \$6.0 million. With respect to the exponent, she assumes $F(x) \approx kx^{-\xi} = F(x) \approx kx^{-4.0}$. If the loss tail follows the power law, which is **nearest** to the implied 99.90% VaR?

Patricia 被首席风险官 (CRO) 要求估计高置信水平的价值 at risk (VaR)。最初, 她将广义极值 (GEV) 分布拟合到损失数据。然而, 最终受众 (董事会) 中的一些成员不理解数学。她观察到 Dowd(†) 解释说: “还有其他基于 EV 理论的简便方法来估计 VaR (或 ES)。这些是基于这样的想法: 如果 $\xi > 0$, 极端损失分布的尾部遵循幂律乘以缓慢变化函数: $F(x) = k(x)x^{-1/\xi}$ 其中 $k(x)$ 随 x 缓慢变化。例如, 如果我们假设为了方便起见 $k(x)$ 是大约恒定的, 那么 [这个方程] 变为: $F(x) \approx kx^{-1/\xi}$ ”。注意假设 $k(x)$ 是大约恒定的。

幂律的指数与广义极值 (GEV) 分布的指数一致, 通过将指数称为负倒数, $-1/\xi$, 但鉴于这个指数也与 $k(x)$ 一起是常数, 我们可以进一步简化近似: 代替 $F(x) \approx kx^{-\xi}$, 我们可以说 $F(x) \approx kx^{-\xi}$ 有时表示为 $F(x) \approx kx^{-\alpha}$ 其中 $\alpha = 1/\xi$ 。例如, Patricia 假设指数的值是 -4.0。在前一种语法中, $\xi = 0.25$ 使得 $(-1/\xi) = -4.0$; 在后一种版本中, $\alpha = 4.0$ 使得 $(-\alpha) = -4.0$, 效果相同。为了我们的目的, 我们将简单地参考 $F(x) \approx kx^{-\xi}$ 。

因此, 她决定使用幂律捷径估计极其自信的 VaR。具体来说, 当 Patricia 设置 xi, $\xi = 4.0$ 时, 她观察到 99.50% VaR 为 \$6.0 百万。关于指数, 她假设 $F(x) \approx kx^{-\xi} = F(x) \approx kx^{-4.0}$ 。如果损失尾部遵循幂律, 哪一项最接近 implied 99.90% VaR?

- A. \$5.0 million
- B. \$9.0 million
- C. \$13.0 million
- D. \$21.0 million

答案:B

解析:

B 选项正确。 We first need to solve for k . Dowd(†) uses an exponent of $(-1/\xi)$ but we shall refer to its simpler reciprocal $(-\xi)$ in the exponent. In this way, the power law says the following: $\text{Prob}(v > x) = F(x) \approx kx^{-\xi}$. Solving for the constant k , it must be that: $k = F(x)x^{\xi}$. In this case where we are given that $x = 6.0$ when $F(x) = 0.50\%$ and $\xi = 4.0$, we can find $k = 0.50\% \cdot 6.0^4 = 6.480$. Now we know, because we are assuming constant (k) per the approximation, that our distribution is $F(x) \approx kx^{-\xi} = 6.480x^{-4.0}$. We want the VaR implied by $F(x) = 0.10\%$ such that $0.10\% = 6.480x^{-4.0}$ where $x = (6.480/0.10\%)^{1/4.0} = 8.97$ which rounds to \$9.0 million.

考察: Quantitative Analysis: Extreme Value Theory 重要程度: [■■■]

科目: Quantitative Analysis

62. Acme Financial has a position in five derivative contracts with its counterparty. The current mark-to-market values of the positions are the following: +12, -9, +20, -6, and +8. The five positions are all included in a single netting set. Let’ s define the “netting benefit” as the reduction in current exposure due to netting; that is, the difference between the total current exposure with netting and the current exposure “as if” netting did not apply. What is the netting benefit in this set?

Acme Financial 在其对手方有五个衍生合约头寸。这些头寸的当前市值如下: +12, -9, +20, -6, 和 +8。这五个头寸都包含在一个单一的净额结算集中。让我们定义“净额收益”为由于净额结算而减少的当前敞口; 即, 净额结算后的总当前敞口与假设没有净额结算时的当前敞口之间的差异。这个集合中的净额收益是多少?

- A. Zero
- B. 3
- C. 15
- D. 25

答案:C

解析:

C 选项正确。 True: 15. Without netting, total exposure = $\max(0, +12) + \max(0, -9) + \max(0, 20) + \max(0, -6) + \max(0, +8) = 40$. With netting, total exposure = $+12 - 9 + 20 - 6 + 8 = 25$. Notice this is the same as $-(-9 - 6)$.

考察: Counterparty Credit Risk 重要程度: [■■■]

科目: Quantitative Analysis

63. Patricia is a risk analyst who works for a large international bank. Her team is developing a new risk management framework for the bank and is currently working on risk identification. As part of extreme risk identification, Patricia has been assigned the important task of designing and performing the firm’s scenario analysis. Soon she will facilitate a brainstorming session for which she has prepared the following documents: external loss data, internal loss data (including near misses), risk and control self-assessment (RCSA) results, key risk indicator scores, audit issues, and concentrated exposures. Assuming she continues to follow best (or good) practice with respect to scenario analysis, which of the following is a TRUE statement?

Patricia 是一名风险分析师, 在大银行工作。她的团队正在为银行开发新的风险管理框架, 目前正在进行风险识别。作为极端风险识别的一部分, Patricia 被分配了重要任务, 即设计和执行公司的场景分析。不久她将主持一个头脑风暴会议, 她准备了以下文件: 外部损失数据、内部损失数据 (包括近似事件)、风险和控制自我评估 (RCSA) 结果、关键风险指标得分、审计问题以及集中敞口。

假设她继续遵循场景分析的最佳 (或良好) 实践, 下列哪一项陈述是正确的?

- A. Because most large operational losses result from external causes, her primary focus should be scenarios driven by external causes
因为大多数大型运营损失来自外部原因, 她的主要焦点应该是由外部原因驱动的场景
- B. She should compare her firm’s generated scenarios to industry-specific list(s) published by external sources, but such a comparison should be made after her firm’s own internal generation exercise
她应该比较她公司生成的场景与外部来源发布的行业特定列表, 但这样的比较应该在她公司自己的内部生成练习之后进行
- C. The initial output of scenario identification should be limited to approximately 10 or 12 scenarios in order to avoid the fallacy of missing the forest for the trees
场景识别的初始输出应限制在约 10 或 12 个场景, 以避免忽视森林的谬误
- D. Because the ultimate goal of scenario analysis is to predict the future with a sufficient degree of certainty, her process should acknowledge and elevate the role of human subjectivity
因为场景分析的最终目标是预测未来, 具有足够的确定性, 她的过程应该承认并提升人类主观性的作用

答案:B

解析:

B 选项正确。 B. True: She should compare her firm’s generated scenarios to industry-specific list(s) published by external sources, but such a comparison should be made after her firm’s own internal generation exercise. According to GARP (section 3.2), “Finally, it can be useful for a firm to compare the scenarios it has generated with an industry list of scenarios, to determine whether it has omitted any relevant scenarios or risk drivers. The Operational Risk Consortium for Insurance Companies (ORIC) and ORX are examples of industry-specific organizations that provide ready-made scenario lists to their members. It is recommended, however, for an organization to compare its scenario list to a list of scenarios from an external industry source only after the scenario generation exercise and not before, so that the externally procured scenarios do not bias the generation process.” (†)

In regard to (A), (C), and (D), each is FALSE. Instead:

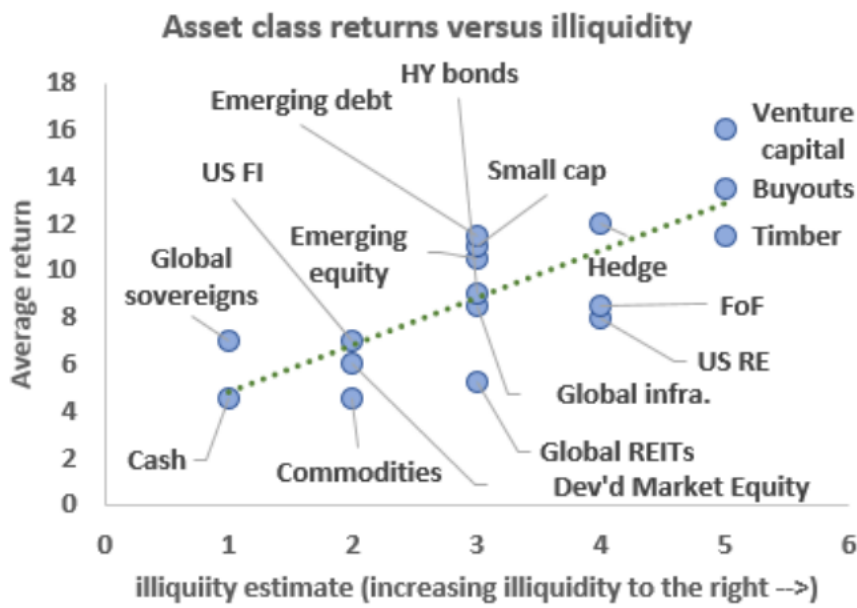
- Most large operational losses result from internal causes
- GARP recommends more scenarios, e.g., at least 20 to 30 scenarios for mid-sized firms. For large international banks, ~50 scenarios are not uncommon
- Scenario analysis is not about predicting the future but identifying a range of scenarios. Further, says GARP, “regulators demand that firms minimize subjectivity and biases as much as possible when conducting scenario identification and assessment.” (†)

(†) 2023 GARP FRM Part II: Operational Risk and Resilience, Pearson Learning Solutions

考察: Operational Risk 重要程度: [■ ■ □]

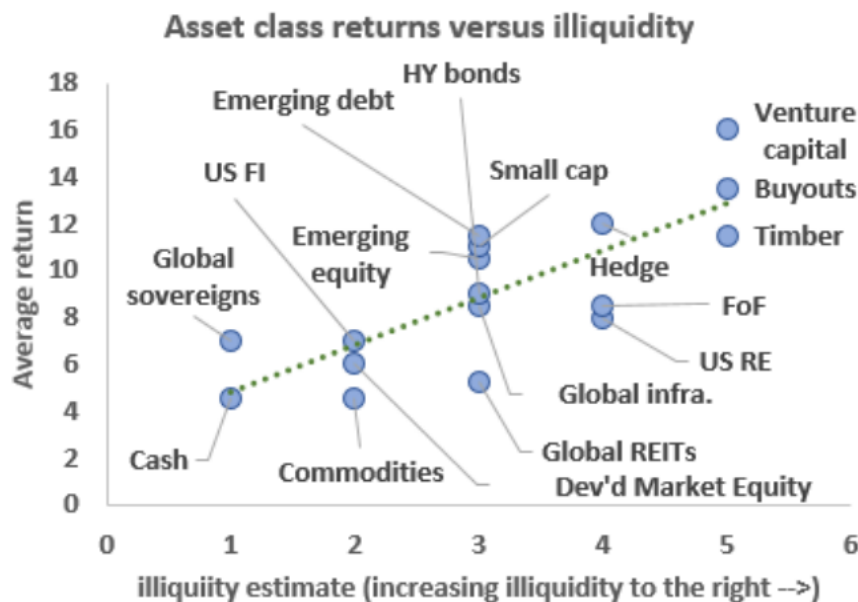
科目: Quantitative Analysis

64. In order to identify the presence of illiquidity risk premium(s), Andrew Ang(†) references data presented by Antti Ilmanen in his well-regarded book Expected Returns (An Investor’s Guide to Harvesting Market Rewards). This data is displayed below as a scatterplot where the y-axis is the long-run average return of the asset class and the x-axis is an index of illiquidity. A higher index (ie, to the right) implies less liquidity. For example, the venture capital as an asset class is assigned to the least liquid (most illiquid) asset class but it also plots the highest long-run average return.



In regard to the illiquidity risk premium, which of the following statements is TRUE?

为了识别流动性风险溢价, Andrew Ang(†) 引用了 Antti Ilmanen 在其广受好评的书籍 Expected Returns (An Investor’s Guide to Harvesting Market Rewards) 中呈现的数据。这些数据以散点图形式显示, 其中 y 轴为资产类别的长期平均回报, x 轴为流动性指数。一个更高的指数(即向右)意味着更少的流动性。例如, 风险投资作为一种资产类别被归类为流动性最差(最不流动性)的资产类别, 但它也绘制了最高的长期平均回报。



就流动性风险溢价而言，下列哪一项陈述是正确的？

- A. In general illiquid asset classes offer high risk-adjusted returns
一般来说，流动性较差资产类别的风险调整回报率较高
- B. These charts demonstrate that there do exists large illiquidity risk premiums ACROSS asset classes
这些图表表明，确实存在跨资产类别的较大流动性风险溢价
- C. There do exist large illiquidity risk premiums WITHIN many asset classes, but not ACROSS asset classes
许多资产类别内部确实存在较大的流动性风险溢价，但不是跨资产类别
- D. Illiquid equities earn the same returns as liquid equities; and illiquid bonds earn the same returns as liquid bonds
流动性较差股票与流动性较好股票获得相同的回报；流动性较差债券与流动性较好债券获得相同的回报

答案:C

解析:

C 选项正确。 True: There do exist large illiquidity risk premiums WITHIN many asset classes, but not ACROSS asset classes. In regard to (A), (B) and (D), each is FALSE.

考察: Finance: Portfolio Management 重要程度: ■■□

科目: Quantitative Analysis

65. Sally represents an investor who likes the strategy and style of a new, entrepreneurial start-up hedge fund. At her client's request, she is going to conduct due diligence. According to Kevin Mirabile (author of the hedge fund due diligence reading), as Sally conducts her diligence of the hedge fund on behalf of the investor, which of the following is the most important principle or ingredient?

Sally 代表一位喜欢新创业对冲基金策略和风格的投资者。在客户的要求下，她将进行尽职调查。根据 Kevin Mirabile（对冲基金尽职调查阅读的作者），作为 Sally 代表投资者进行对冲基金尽职调查，哪一项是最重要的原则或成分？

- A. She should observe strong evidence that the fund has eliminated business model risk via the SEC's Zeta Sigma AIML filter
她应该观察到 strong evidence 表明该基金通过 SEC 的 Zeta Sigma AIML 过滤器消除了商业模型风险
- B. She should confirm that the CEO is exceptionally good at all three top-level skills sets: investment, operational, and business
她应该确认 CEO 在所有三个顶级技能集：投资、运营和业务方面都非常出色
- C. She should follow all the steps in the process but with flexibility as due diligence is also an art and each fund is unique with its own strategy
她应该遵循所有步骤，但要有灵活性，因为尽职调查也是一种艺术，每个基金都有自己独特的策略

D. She should clarify that the fund’s incentives are actual equity ownership grants rather than phantom equity and that they are granted throughout the organization to all full-time employees.
她应该澄清该基金的激励措施是实际的股票所有权授予，而不是幻影股票，并且它们在整个组织中授予所有全职员工。

答案:C

解析:

C 选项正确。 She should follow all the steps in the process but with flexibility as due diligence is also an art and each fund is unique with its own strategy.

GARP explains that “performing due diligence on a hedge fund is as much art as it is science. An investor must use a comprehensive checklist to ensure that nothing is left out or omitted yet remain free to ask open-ended questions that provide insights into a firm’ s philosophy or culture. The guidelines discussed here for investment, operational, and business model risk assessment were for the most part generic. When performing due diligence, investors must modify their approach to be strategy specific.” (Summary, 2020 Financial Risk Management Part II: Risk Management and Investment Management, 10th Edition. Pearson Learning Solutions, 2019. VitalBook file.)

In regard to false (A), (B) and (D):

- The Zeta Sigma AIML filter is fictional and business model risk cannot be eliminated
- The authors point out that only “dinosaurs” (as founders) attempt to manage all three because managers are rarely good at all three (nevermind exceptionally good)! Managers should not choose to be both CEO and CIO.
- Incentive structures vary by organization.

考察: Hedge Funds: Due Diligence 重要程度: [■■■□]

科目: Quantitative Analysis

66. Bertha is a risk manager who estimates the market risk of one of her firm’ s portfolios. Because geometric returns are time-additive and arithmetic returns are cross-additive, she measures performance with both arithmetic and geometric returns. Her assumptions are gathered below:

- The portfolio value is \$100.0 million and she assumes there are 250 trading days per year.
- Using geometric returns, the average return +12.0% per annum with a standard deviation of 28.0%
- Using arithmetic returns, the average return +14.0% per annum with a standard deviation of 32.0%

She wants to measure the one-day 95.0% value at risk (VaR) under both approaches: assuming arithmetic returns are normal (aka, normal VaR) and assuming geometric returns are normal; aka, lognormal VaR. She assumes the returns are serially independent. Which of the following is **nearest** to the difference between the one-day 95.0% normal VaR and the one-day 95.0% lognormal VaR?

Bertha 是一位风险经理，估计她公司一个投资组合的市场风险。由于几何回报是时间可加的，而算术回报是交叉可加的，她使用算术和几何回报来衡量绩效。她的假设如下：

- 投资组合价值为 \$100.0 百万，她假设每年有 250 个交易日。
- 使用几何回报，平均回报为 +12.0% 每年，标准差为 28.0%
- 使用算术回报，平均回报为 +14.0% 每年，标准差为 32.0%

她想测量两种方法下的一日 95.0% 价值 at risk (VaR)：假设算术回报是正态的（aka，正态 VaR）和假设几何回报是正态的； aka，对数正态 VaR。她假设回报是序列独立的。哪一项最接近一日 95.0% 正态 VaR 与一日 95.0% 对数正态 VaR 之间的差异？

- A. The one-day 95% normal VaR is approximately \$37,700 less than the one-day 95% lognormal VaR
一日 95.0% 正态 VaR 大约比一日 95.0% 对数正态 VaR 少 \$37,700
- B. The one-day 95% normal VaR is approximately \$380,100 less than the one-day 95% lognormal VaR
一日 95.0% 正态 VaR 大约比一日 95.0% 对数正态 VaR 少 \$380,100

- C. The one-day 95% normal VaR is approximately \$29,400 greater than the one-day 95% lognormal VaR
一日 95.0% 正态 VaR 大约比一日 95.0% 对数正态 VaR 多 \$29,400
- D. The one-day 95% normal VaR is approximately \$448,800 greater than the one-day 95% lognormal VaR
一日 95.0% 正态 VaR 大约比一日 95.0% 对数正态 VaR 多 \$448,800

答案:D
解析:

Option D is correct. The one-day 95% normal VaR is approximately \$448,800 greater than the one-day 95% lognormal VaR. The one-day 95% normal VaR(%) is given by:

$$-14\%/250 + 32\% \times 1/\sqrt{250} \times 1.645 = 3.273246\%$$

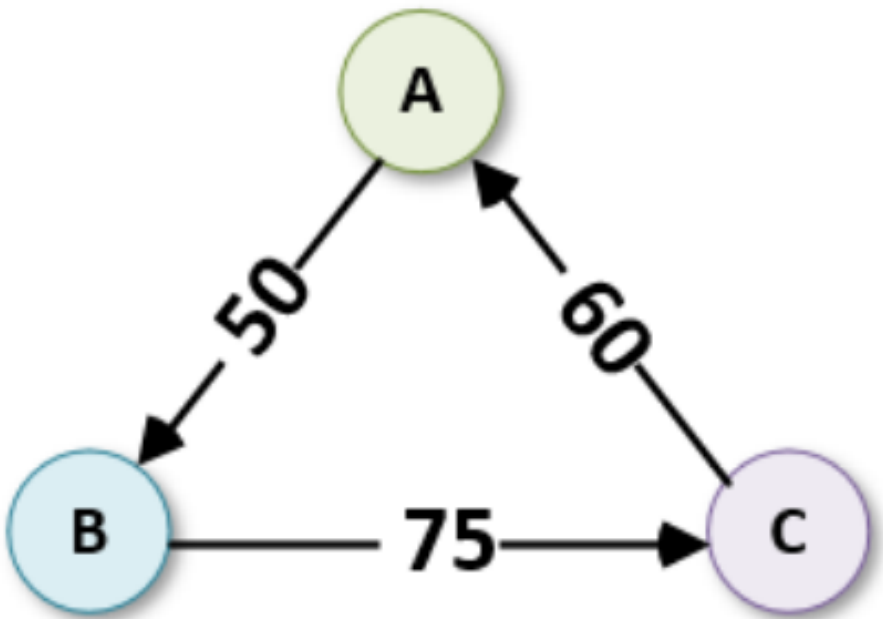
The one-day 95% lognormal VaR(%) is given by:

$$1 - \exp(12\%/250 - 28\% \times 1/\sqrt{250} \times 1.645) = 2.824436\%$$

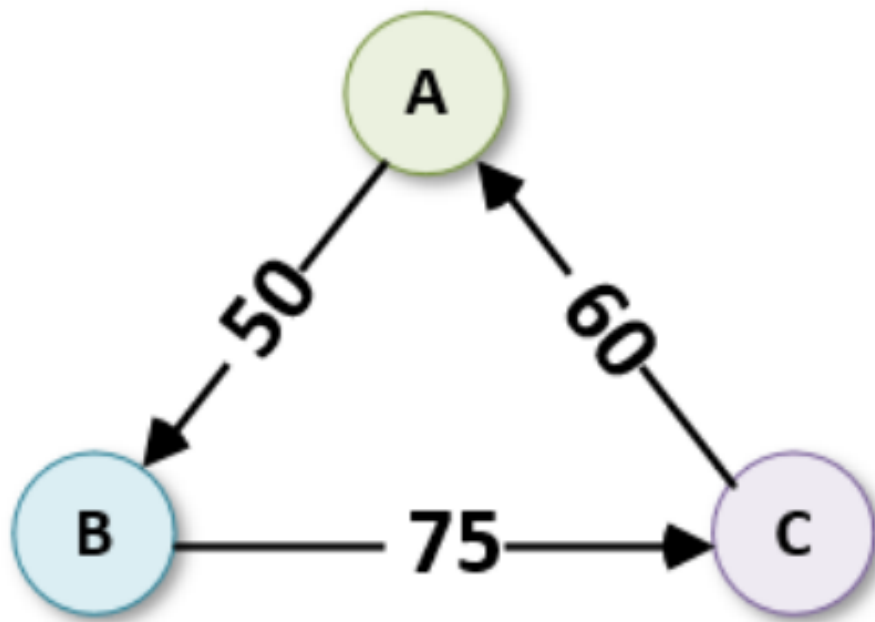
The difference is $3.273246\% - 2.824436\% = 0.448810\%$, or about $0.448810\% \times \$100.0 \text{ million} = \$448,810$; with an exact quantile, the difference is \$448,765.91.

考察: Quantitative Analysis: Value at Risk 重要程度: [■■■]
科目: Quantitative Analysis

67. The graph below represents a trilateral scenario between three different counterparties (called A, B, and C) with position sizes in a certain fungible (aka, interchangeable) product. The values represent notional sizes and the direction represents exposures:



In this ring, the current total notional is 370. If trade compression can be achieved, which of the following is an efficient solution?
这个图展示了三个不同对手方（称为 A、B 和 C）在某种可互换产品中的头寸大小。值表示名义金额，方向表示敞口：



在这个环中，当前总名义金额为 370。如果可以实现交易压缩，哪一项是最有效的解决方案？

- A. A -> 25 -> B -> 60 -> C (new total notional = 110)
选项 A: A -> 25 -> B -> 60 -> C (新总名义金额 = 110)
- B. A -> 30 -> B -> 45 -> C (new total notional = 150)
选项 B: A -> 30 -> B -> 45 -> C (新总名义金额 = 150)
- C. B -> 35 -> C -> 30 -> A (new total notional = 65)
选项 C: B -> 35 -> C -> 30 -> A (新总名义金额 = 65)
- D. B -> 25 -> C -> 10 -> A (new total notional = 70)
选项 D: B -> 25 -> C -> 10 -> A (新总名义金额 = 70)

答案:D

解析:

Option D is correct. D. True: B -> 25 -> C -> 10 -> A (new total notional = 70)

The easiest approach is to identify the lowest exposure, in this case 50, and subtract it from all three. In this way,

- current A -> 50 -> B is eliminated;
- current B -> 75 -> C is reduced to B -> (75 - 50 = 25) -> C; and
- current C -> 60 -> A is reduced to C -> (60 - 50 = 10) -> A

考察: Finance 重要程度: [■ ■ □]

科目: Quantitative Analysis

68. Sally is a risk analyst at an international bank who is leading a project to conduct a causal analysis of the bank’s operational risk. The project will coordinate with, and inform modifications to, the bank’s loss database. One of the project’s goals is to analyze the organization’s controls, control layers, and key risk indicators (KRIs), and key control indicators (KCIs).

She prefers to use a factor model, and her candidates include the following approaches:

- *Fault tree analysis (FTA)*: this deductive method will build a series of and/or conditions that can happen simultaneously or alternatively
- *Root-cause analysis with bowtie tool*: the visual tool will employ the 5-why analysis to probe deeply into the root causes of events

- *FAIR methodology*: this structured framework will generate scenarios, where each scenario will assume an asset at risk, a threat community, a threat type, and an effect
- *Swiss cheese model*; aka cumulative act effects: by anticipating active failures and identifying latent conditions, this model will promote effective control layering and identifying active failures

Each of these four approaches has advantages and disadvantages. For ANY of these approaches, which of the following paired statements (i.e., a single advantage paired with a single disadvantage) is a TRUE statement?

Sally 是一名国际银行的风险分析师，领导一个项目，进行银行运营风险的有因分析。该项目将与银行损失数据库的修改协调，并告知修改。该项目的目标是分析组织的控制、控制层和关键风险指标（KRIs）以及关键控制指标（KCIs）。她偏好使用因子模型，她的候选方法包括以下几种：

- *Fault tree analysis (FTA)*: 这个演绎方法将构建一系列同时或替代发生的事件或条件
- *Root-cause analysis with bowtie tool*: 这个可视化工具将使用 5-why 分析深入探究事件的根源原因
- *FAIR methodology*: 这个结构化框架将生成场景，每个场景假设一个风险资产、威胁社区、威胁类型和影响
- *Swiss cheese model*; aka cumulative act effects: 通过预测主动失败和识别潜在条件，这个模型将促进有效的控制层级和识别主动失败

每种方法都有优点和缺点。对于任何一种方法，下列哪一对陈述（即单个优点与单个缺点）是正确的陈述？

- A. Fault tree analysis (FTA) is simple and non-reliant on domain expertise; however, it is unrealistic because it can only assume independent events
故障树分析（FTA）简单且不依赖于领域专业知识；然而，它是不现实的，因为它只能假设独立事件
- B. The Bowtie tool is proactive because it assesses the likelihood and/or impact of risks before they materialize by examining controls and indirect causes; however, it natively lacks a quantitative framework such that prioritization is harder
Bowtie 工具弓形图工具是前瞻性的，因为它通过检视控制和间接原因，在风险尚未发生前评估风险的可能性和/或影响；但它本身缺少定量框架，因此优先级排序更困难
- C. The FAIR model is inclusive because it emphasizes subjective identification (aka, findability) of factors; however, it is not a probabilistic approach and may be insufficient when distributions are required
FAIR 模型是包容的，因为它强调对因素的主观识别（即可发现性）；然而，它不是一种概率方法，在需要分布时可能不足
- D. The Swiss cheese model is elegant because clusters and their centroids (aka, holes) are quantitatively retrieved via k-means clustering; however, it struggles when incorporating human factors into a holistic risk perspective
he Swiss cheese 模型很优雅，因为聚类及其中心（即“洞”）是通过 k-means 聚类定量获得的；但是，当需要把人因纳入整体风险视角时，它会遇到困难

答案:B

解析:

B 选项正确。 True: The Bowtie tool is proactive because it assesses the likelihood and/or impact of risks before they materialize by examining controls and indirect causes; however, it natively lacks a quantitative framework such that prioritization is harder
In regard to (A), (C), and (D), each is false.

考察: Operational Risk 重要程度: [■ ■ □]

科目: Financial Markets and Products

69. According to Rose and Hudgins, the factors affecting the choice of non-deposit funding sources include the following: the relative costs of each funding source; the risk of each funding source, including the dependability and volatility; the length of time for which funds will be needed; the size of the financial institution needing the funds; and regulations on the use of other sources of funds. In regard to the various sources of non-deposit funding, which of the following statements is TRUE?

根据 Rose 和 Hudgins，影响非存款资金来源选择的因素包括以下几个：每个资金来源的相对成本；每个资金来源的风险，包括可靠性和波动性；资金需求的时间长度；需要资金的金融机构的规模；以及对使用其他资金来源的监管。在各种非存款资金来源方面，下列哪一项陈述是正确的？

- A. The Fed Funds market is the least popular because its effective interest rate is higher than the alternatives
联邦基金市场最不受欢迎，因为它比其他选择的有效利率更高
- B. Repurchase agreements are more popular than Fed Funds because they are simpler, however, they incur greater credit risk
回购协议比联邦基金更受欢迎，因为它们更简单，但它们承担更大的信用风险
- C. The advantages of the commercial paper (CP) market are the lack of credit risk and consistently high (i.e., low volatility)
credit availability, however, volume tends to be low and relative cost is high
商业票据（CP）市场的优势是缺乏信用风险和持续高（即低波动性）的信用可用性，但交易量 tends to be low and relative cost is high
- D. Advances from Federal Home Loan Banks (aka, FHLBanks) are highly popular among thrifts and small banks as a stable
source of flexible (with respect to maturity) funding at below-market interest rates
联邦住房贷款银行（aka, FHLBanks）的贷款在储蓄机构和中小银行中非常受欢迎，作为稳定来源的灵活（相对于到期日）
资金，以低于市场的利率

答案:D

解析:

D 选项正确。 Advances from Federal Home Loan Banks (aka, FHLBanks) are highly popular among thrifts and small banks as a stable source of flexible (with respect to maturity) funding at below-market interest rates.

From the Study Notes: “Advanced from Federal Home Loan Banks: In recent years the Federal Home Loan Bank (FHLB) System has moved into lending money to home mortgage lenders. Comprising 12 regional banks, the FHLB was created in 1932 as a source of stable funding for depository institutions facing potential runs by depositors. According to many scholars, the involvement of FHLB in the “sources of funds” arena has not only increased the amount of available funds for financial institutions that use FHLB as a source (with mortgages as collateral), but FHLB has also increased the liquidity of home loans and provided an incentive for more financial institutions to provide credit to the housing market.

The presence of FHLB has quickly expanded in recent years, with over 6,000 commercial banks, over 1,300 thrift institutions, over 700 credit unions, and over 80 insurance companies now eligible to borrow from the FHLB. The draw to FHLB funds stems from it being a stable source of finance and offering below-market interest rates. Financial institutions can borrow from the FHLB for a flexible time period, ranging from overnight to over 20 years, with fixed or variable interest rates. Part of the reason the FHLB can offer below-market interest rates is because its federal charter allows it to borrow money on the cheap and pass those savings on to member banks, who are stockholders and receive dividends. The Federal Home Loan Banks System also has a favorable position in the event of failure. Legally, the FHLB is first in line, even in front of the FDIC, to recover funds if a failure occurs.”

In regard to (A), (B) and (C), each is FALSE. Instead:

- The Fed Funds market is the most popular and its effective interest rate tends to be LOWER than the alternatives, although it is VOLATILE (is its downside)
- Repurchase agreements are less popular and more complex than Fed Funds; also, they incur less credit risk because they are secured by collateral (they are often viewed as “collateralized Fed funds transactions”).
- Commercial paper (CP) is high in volume, moderate in cost (advantages) but volatility in available capacity and subject to credit risk (disadvantages)

考察: Finance: Non-deposit Funding Sources 重要程度: [■■■□]

科目: Quantitative Analysis

70. We can add a momentum factor to the Fama-French so that it becomes a four-factor model. This momentum factor is denoted by WML (i.e., past winners minus past losers) or UMD (i.e., stocks that have gone up minus stocks that have gone down). At least with respect to the historical window analyzed, which is the long period from January 1965 to December 2011, which of the following statements is TRUE about the momentum factor?

我们可以将动量因子添加到 Fama-French 模型中，使其成为四因子模型。这个动量因子由 WML（即过去赢家减去过去输家）或 UMD（即上涨股票减去下跌股票）表示。至少在分析的历史窗口中，从 1965 年 1 月到 2011 年 12 月，哪一项关于动量因子的陈述是正确的？

- A. Momentum is a negative feedback strategy which is inherently stabilizing
动量是一种负反馈策略，本质上具有稳定性
- B. The momentum factor is observed in equities but is NOT observed in bonds, commodities, and real estate
动量因子在股票中被观察到，但在债券、商品和房地产中没有被观察到
- C. Momentum investing by definition is an anti-value strategy; correlations between HML and WML are strongly negative
动量投资定义上是一种反价值策略；HML 和 WML 之间的相关性 strongly negative
- D. The cumulated profits on momentum strategies have been an order of magnitude larger than cumulated profits on either size or value
动量策略的累积利润比规模或价值策略的累积利润大一个数量级

答案:D

解析:

Option D is correct. The cumulated profits on momentum strategies have been an order of magnitude larger than cumulated profits on either size or value.

In regard to (A), (B) and (C), each is false.

考察: Factor Investing 重要程度: ■■□

科目: Quantitative Analysis

71. Peter has collected the historical sample of daily profit and loss (P/L) observations for his firm’s new equity portfolio but his dataset only contains 250 days. Ideally, he would prefer 1,000 observations because the board asked for a presentation that displays value at risk (VaR) at confidence levels of (99.0%, 99.1%, ..., 99.8%, and 99.9%). The board also expressed a preference for a non-parametric approach; apparently, some directors are concerned that parametric approaches might be too difficult to communicate and understand. In addition to granular VaR confidence levels (e.g., 99.1%), if possible, the board would like to see confidence INTERVALS around the VaR estimates in order to gauge their precision; e.g., a 9X% confidence interval around the 99.1% VaR.

Therefore, Peter considers the following four approaches: basic historical simulation (BHS), bootstrapped historical simulation (BOOT), historical simulation using non-parametric density estimation (NPDE), and age-weighted historical simulation (AWHS). Given the realities of his relatively small, empirical dataset, which of the following statements is TRUE?

Peter 收集了他公司新股票组合的每日损益（P/L）历史样本，但他的数据集只有 250 天。理想情况下，他希望能有 1,000 个观察值，因为董事会要求一个显示价值 at risk（VaR）的图表，置信水平为 (99.0%, 99.1%, ..., 99.8%, 和 99.9%)。董事会还表达了对非参数方法的偏好；显然，一些董事担心参数方法可能太难沟通和理解。除了细粒度 VaR 置信水平（例如 99.1%），如果可能的话，董事会希望看到 VaR 估计周围的置信区间，以衡量其精度；例如，一个 9X% 置信区间围绕 99.1% VaR。因此，Peter 考虑了以下四种方法：基本历史模拟（BHS），引导历史模拟（BOOT），使用非参数密度估计的历史模拟（NPDE），和年龄加权历史模拟（AWHS）。鉴于他相对较小的经验数据集的现实情况，哪一项陈述是正确的？

- A. Peter can easily eliminate the BHS approach because basic historical simulation cannot even retrieve a 99.10% (or 99.20% etc.) VaR quantile due to the fact that 1/250 is greater than 0.10%
Peter 可以很容易地消除基本历史模拟（BHS）方法，因为基本历史模拟甚至无法检索到一个 99.10%（或 99.20% 等）VaR 分位数，因为 1/250 大于 0.10%
- B. Peter has a dilemma (i.e., none of the four approaches is feasible) as the construction of confidence levels around VaR estimates will require him to employ a parametric (or semi-parametric at least) approach
Peter 有一个困境（即，没有任何一种方法可行），因为构建 VaR 估计周围的置信水平将需要他采用参数（或至少半参数）方法
- C. Peter has a dilemma (i.e., no approach is good here) simply because his sample size is too small to meet the board’s preferences under any of the four approaches
Peter 有一个困境（即，没有任何一种方法可行），因为他的样本量太小，无法满足董事会在这四种方法下的偏好
- D. Non-parametric density estimation (NPDE) can retrieve VaRs at granular confidence levels (e.g., 99.10%) as if the distribution is a continuous probability density function (pdf)
Non-参数密度估计（NPDE）可以在细粒度置信水平（例如 99.10%）下检索 VaR，就像分布是一个连续概率密度函数（pdf）

答案:D

解析:

D 选项正确。Non-parametric density estimation (NPDE) can retrieve VaRs at granular confidence levels (e.g., 99.10%) as if the distribution is a continuous probability density function (pdf)

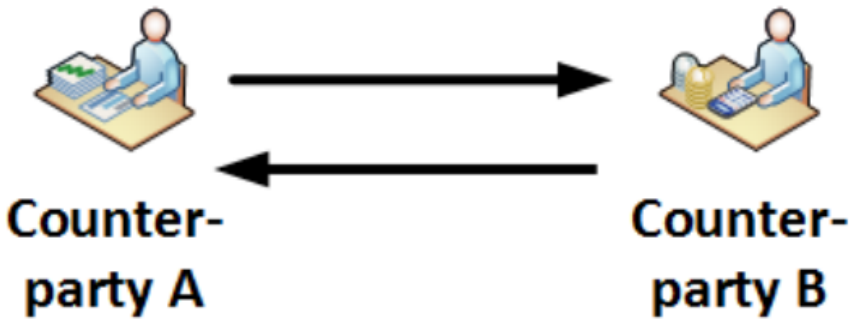
In regard to (A), (B), and (C), each is FALSE. Instead, the following are TRUE statements:

- Basic historical simulation definitely does return a 99.1% VaR here it will (probably) be the same as the 99.0% VaR
- Due to the convenient power of order statistics, confidence intervals can be generated under all four approaches; or really, confidence intervals can be built for any parametric or non-parametric VaR or ES estimate!
- The small sample size might favor the bootstrap or the non-parametric density estimation

考察: Market Risk Measurement and Management 重要程度: [■■■]

科目: Quantitative Analysis

72. Two counterparties (A and B, illustrated below) have between them a set of OTC derivative transactions.



Associated with the positions, these counterparties also entered into a bilateral collateral agreement prior to initiation of the underlying transactions. Each of the following is true EXCEPT which is false?

双方交易对手（A 和 B，见下图）之间存在一组场外（OTC）衍生品交易。针对这些头寸，双方在发起基础交易之前还签订了双边抵押协议。以下陈述均为真，除了哪一项为假？

- A. If they followed best practice, collateral posting should be continuous for both of them
如果他们遵循最佳实践，抵押品应连续存入双方
- B. As a counterparty risk mitigant, collateral posting is generally superior to a reset feature
作为对手方风险缓解措施，抵押品存入通常优于重置功能
- C. Variation margin is a type of collateral that aims to track the change in valuation (aka, change in MTM) of the position over time
变动保证金是一种抵押品，旨在跟踪时间内的头寸估值（aka，MTM 变化）变化
- D. As a bilateral agreement, in the case of a positive MTM, the party will call for collateral; and in the case of negative MTM, the party will be required to themselves post collateral
作为双方协议，在正 MTM 情况下，一方将要求抵押品；在负 MTM 情况下，一方将被迫自己存入抵押品

答案:A

解析:

A 选项正确。False. Continuous posting of collateral is operationally too expensive In regard to (B), (C) and (D), each is TRUE.

考察: Counterparty Credit Risk 重要程度: [■■■]

科目: Quantitative Analysis

73. MetroScape Bank prides itself on its modern banking solutions and customer-centric approach. During the last quarter, however, a series of events occurred that suggest the bank’s controls should (or at least can) be improved. Specifically, each of the four major types of human error was observed:

- I. Jane, a seasoned cashier at MetroScape, had been handling cash transactions for years. One busy Friday afternoon, while multitasking between customer queries and cash handling, she inadvertently gave out \$500 more to a customer than he had withdrawn. Her muscle memory, developed from years of handling similar transactions, failed her in that moment of distraction.
- II. MetroScape recently redesigned its compensation and incentive programs. The incentive program rewards cross-selling, especially when a valuable customer adds an additional account and/or utilizes a new program. Due to this incentive plan, some branches aggressively cross-sold products to existing customers without sufficient regard for product suitability. Several customers called headquarters to complain about the sales tactics.
- III. A new investment product was introduced at MetroScape, and the bank’s financial advisors were tasked with guiding interested clients. Tom, one of the advisors, faced a unique situation where a client wanted to merge this new product with another existing one. Unsure and without prior experience or training in this scenario, Tom made an educated guess. Unfortunately, his decision led to significant tax implications for the client.
- IV. Lucy, a senior executive at MetroScape, was aware of the bank’s policy against insider trading. However, tempted by the prospect of significant personal gain, she used confidential information about an upcoming merger to buy shares in a company that MetroScape was about to acquire. Her actions, driven by personal greed, were a clear violation of both the bank’s policy and federal regulations.

In regard to these events and their associated control implications, each of the following statements is true EXCEPT which is false?

MetroScape 银行自豪于其现代银行业解决方案和客户中心方法。然而，在最后一个季度，一系列事件表明银行的控制应该（或至少可以）得到改进。具体来说，观察到了四种主要类型的人为错误：

- I. Jane，一位经验丰富的出纳员，多年来一直在处理现金交易。一个繁忙的周五下午，她在处理客户查询和现金处理时，无意中多给了客户 \$500。她的肌肉记忆，来自于多年处理类似交易的经验，在那一刻的分心中失败了。
- II. MetroScape 最近重新设计了其薪酬和激励计划。激励计划奖励交叉销售，特别是在一个重要客户添加另一个账户和/或使用新程序时。由于这个激励计划，一些分支机构对现有客户进行了激进的交叉销售，而没有充分考虑产品的适用性。许多客户打电话到总部抱怨销售策略。
- III. MetroScape 引入了一种新的投资产品，银行的财务顾问被要求引导感兴趣的客户。Tom，一位顾问，面临一个独特的情况，一个客户想要将这个新产品与其他现有产品合并。不确定且没有之前在这个场景中的经验或培训，Tom 做出了一个有根据的猜测。不幸的是，他的决定导致了客户的重要税务影响。
- IV. Lucy，MetroScape 的一位高级执行官，知道银行的政策禁止内部交易。然而，被可能的重大个人收益所诱惑，她使用了关于即将进行的合并的机密信息，购买了 MetroScape 即将收购的公司的股票。她的行为，由个人贪婪驱动，清楚地违反了银行的政策和联邦法规。

针对这些事件及其相关的控制影响，下列陈述均为真，除了哪一项为假？

- A. The fourth (IV.) is a violation which is best mitigated by supervisory controls
第四（IV.）是一种违反，最佳缓解由监管控制
- B. The third (III.) is a type of knowledge-based (KB) mistake which can be mitigated by directive controls
第三（III.）是一种知识基础（KB）错误，可以通过指令控制缓解
- C. The second (II.) is a quality-based (QB) mistake where the best mitigation is typically lean Six Sigma response
第二（II.）是一种质量基础（QB）错误，最佳缓解通常是精益六西格玛响应
- D. The first (I.) is a skill-based (SB) slip, and root-cause analysis may imply process re-engineering, environmental factors, and/or workflow changes
第一（I.）是一种技能基础（SB）滑倒，根因分析可能暗示流程重新工程、环境因素和/或工作流变化

答案:C

解析:

C 选项正确。 False. This is a knowledge-based (KB) mistake. According to the reading (GARP), the relevant response(s) to knowledge-based mistakes are directive controls.

In regard to (A), (B), and (D), each is TRUE.

考察: Operational Risk: Human Error 重要程度: [■■■□]

科目: Quantitative Analysis

74. What is the primary (valid) reason that banks need proper liquidity transfer pricing (LTP)?

银行需要适当流动性转移定价（LTP）的主要（有效）理由是什么？

- A. Because banks engage in maturity transformation
因为银行参与期限转换
- B. Because there is a lack of liquidity transformation products in the marketplace
因为市场上缺乏流动性转换产品
- C. Because the true economic cost of liquidity risk is approximately zero
因为流动性风险的真正经济成本大约为零
- D. Because markets are volatile; e.g., volatility 1-year LIBOR/Swap spread
因为市场是波动的；例如，波动 1 年 LIBOR/Swap 利差

答案:A

解析:

A 选项正确。 True: Because banks engage in maturity transformation (maturity transformation is the recurring causal theme in the reading).

As Grant explains (in 3.1), Why banks need LTP: In their daily operations, banks make money by funding long-term loans (assets) with short-term deposits (liabilities), a process that is commonly referred to as maturity transformation. As pointed out by the BCBS (September 2008), this “makes banks inherently vulnerable to liquidity risk, both of an institution-specific nature and that which affects markets as a whole” (p 3). But provided banks use LTP to account for the costs, benefits and risks of liquidity in product pricing, new product approval processes and profit and performance assessments, they should not be discouraged from engaging in maturity transformation. Banks with poor LTP practices are more likely to accrue larger amounts of long-term illiquid assets, contingent commitments and shorter-dated volatile liabilities, substantially increasing their vulnerability to funding shortfalls.” (Joel Grant, 2011. “Liquidity Transfer Pricing: A Guide to Better Practice,” Occasional Paper, Financial Stability Board, Bank for International Settlements.)

In regard to (B), (C) and (D) each is FALSE.

考察: Liquidity Risk Management 重要程度: [■■■□]

科目: Quantitative Analysis

75. In the paper “Finding Bernie Madoff” (by Dimmock and Gerken), the authors produce several findings as they attempt to test the predictability of investment fraud. In regard to their findings specifically, each of the following is TRUE except which is not?

在论文“寻找 Bernie Madoff”（由 Dimmock 和 Gerken 撰写）中，作者们试图测试投资欺诈的可预测性，并产生了几个发现。具体来说，下列陈述均为真，除了哪一项为假？

- A. Disclosed soft dollars do not significantly predict fraud
披露的软美元不能显著预测欺诈
- B. Direct Theft and Self Dealing are the most common types of fraud
直接盗窃和自我交易是最常见的欺诈类型
- C. Current disclosures required by the SEC have significant power for predicting future fraud
当前 SEC 要求的披露对预测未来欺诈具有显著作用

D. Investors (on average) are compensated for fraud risk through superior performance and/or higher fees 投资者（平均）通过卓越表现和/或更高的费用来补偿欺诈风险

答案:D

解析:

D 选项正确。 FALSE. Instead, the authors find that investors are NOT compensated for fraud risk, either through superior performance and/or higher fees

In regard to (A), (B), and (C), each is TRUE.

考察: Operational Due Diligence 重要程度: [■■■]

科目: Quantitative Analysis

76. Ralph employed the peaks-over-threshold (POT) approach under extreme value theory (EVT) to the estimation of value at risk (VaR) for his firm’ s portfolio. He has four years of data, $n = 1,000$ days. He would prefer to apply a maximum likelihood estimation method (aka, MLE) to parameter estimation, but he thinks he needs to choose the threshold first. When he selects a threshold of $u = 4.0\%$, he observes $N(u) = 28$ exceedances such that $N(u)/n = 2.80\%$. Fitting the data generates these parameters: scale, $\beta = 0.80\%$; and tail index, $\xi = 0.310$. Based on these assumptions, the 99.90% confident VaR is 8.670%.

He subsequently decides to increase the threshold from 4.0% to 5.0% while retaining the tail index at 0.310 and the scale parameter at 0.80%. The number of exceptions declines to $N(u) = 17$ such that $N(u)/n = 17/1,000 = 1.70\%$. His revised 99.90% VaR is similar at 8.630%. In regard to choosing a threshold, which of the following statements is TRUE?

Ralph 在其公司投资组合的价值-at-风险（VaR）估计中，采用了极值理论（EVT）下的阈值超越（POT）方法。他拥有四年的数据，即 $n = 1,000$ 个交易日。他更倾向于使用极大似然估计（MLE）进行参数估计，但他认为首先需要选择阈值。当他选择阈值 $u = 4.0\%$ 时，观察到 $N(u) = 28$ 次超越，因此 $N(u)/n = 2.80\%$ 。对数据进行拟合得到如下参数：尺度参数 $\beta = 0.80\%$ ，尾指数 $\xi = 0.310$ 。在这些假设下，99.90% 置信 VaR 为 8.670%。随后，他决定将阈值从 4.0% 提高到 5.0%，同时保持尾指数为 0.310、尺度参数为 0.80% 不变。此时超越次数下降到 $N(u) = 17$ ，因此 $N(u)/n = 17/1,000 = 1.70\%$ 。调整后得到的 99.90% VaR 与之前相近，为 8.630%。在阈值选择方面，下列哪一项陈述是正确的？

- A. The higher threshold offers the benefit of lower bias but the drawback of higher variance
更高的阈值提供了降低偏差但提高方差的优点
- B. The higher threshold offers the benefit of lower variance but the drawback of higher bias
更高的阈值提供了降低方差但提高偏差的优点
- C. Because the desired confidence is 99.90% and $0.10\% = 1/1,000$, the (optimal) threshold should be set equal to the dataset’ s second-worst return
因为所需的置信水平为 99.90% 和 $0.10\% = 1/1,000$ ，（最优）阈值应设置为等于数据集的第二差回报
- D. He can avoid a subjective choice of the threshold by applying an analytical maximum log-likelihood estimation (MLE) function to all three parameters simultaneously
他可以通过应用分析极大似然估计（MLE）函数同时对所有三个参数进行计算，避免主观选择阈值

答案:A

解析:

A 选项正确。 We confront a bias-variance tradeoff in either EVT approach. In the blocks approach, the tradeoff is a reflection of the size (or number) of blocks. In the POT approach, it is a reflection of the arbitrary threshold choice, which is analogous to the block size choice. If the threshold is too low, then our model will not comport with EVT’ s asymptotic properties and the model is likely to be inaccurate (aka, biased). A higher threshold implies lower bias (aka, greater accuracy) but this necessarily fits to fewer data points such that we can expect higher variance (aka, less precision).

Dowd_(†) references the bias-variance tradeoff in the context of semi-parametric estimation methods (e.g., Hill estimator): “... procedure based on the fact that the choice of k implies a trade-off between bias and variance. If we increase k , we get more data and so move to the centre of the distribution. This increases the precision of our estimator (and therefore reduces its variance), but also increases the bias of the tail estimator by placing relatively more weight on observations closer to the centre of our distribution. Alternatively, if we decrease k and move further out along the tail, we decrease bias but have fewer data to work with and get a higher variance.”

考察: Quantitative Analysis: Extreme Value Theory 重要程度: [■ ■ □]

科目: Quantitative Analysis

77. Consider two counterparties who are about to enter into bilateral framework under an ISDA Master Agreement. They are currently negotiating the credit support annex (CSA) which will govern the terms of collateral posting. Each of the following is true EXCEPT which is inaccurate?

考虑两个交易对手，他们即将根据 ISDA 主协议进入双边框架。他们目前正在谈判信用支持附件（CSA），该附件将规定抵押品存入的条款。下列陈述均为真，除了哪一项为假？

- A. If the goal is to minimize funding costs, both parties prefer to avoid rehypothecation rights
如果目标是降低融资成本，双方都倾向于避免再抵押权
- B. With respect to VARIATION margin, both parties probably prefer rehypothecation rights; and probably will NOT require segregation
在变动保证金方面，双方可能更喜欢再抵押权；可能不会要求分离
- C. With respect to INITIAL margin, both parties probably are likely to insist on non-rehypothecation; and probably will require segregation
在初始保证金方面，双方可能倾向于坚持非再抵押；可能需要分离
- D. Whoever is the collateral poster (aka, collateral giver) prefers the method of Security Interest rather than the method of Title Transfer (which is preferred by the collateral receiver)
谁是抵押品存入者（aka，抵押品提供者）更喜欢安全利益方法而不是标题转移方法（由抵押品接收者更喜欢）

答案:A

解析:

A 选项正确。 A. FALSE. Rehypothecation minimizes funding cost. The trade-off is that rehypothecation minimizes funding cost but increases counterparty risk.

In regard to (B), (C) and (D), each is TRUE.

考察: Counterparty Risk 重要程度: [■ ■ □]

科目: Quantitative Analysis

78. Reporting on operational risks presents a unique set of challenges. As GARP explains, “Unlike other risk factors, assessing the volatility of operational risk is especially challenging given this is a relatively new discipline. Credit and market risk metrics often have over a hundred years of historical trend information. Operational risk measurement is relatively new, requiring more attention to the qualitative and quantitative tracking measures used to evaluate changes in risk profiles.”(†)

Let’ s assume the organization’ s central operational risk function (ORF) has primary responsibility for operational risk reporting. In regard to the challenges to reporting operational risks, each of the following statements is true EXCEPT which is false?

报告操作风险面临一系列独特挑战。正如 GARP 所解释的那样：“与其他风险因素不同，评估操作风险的波动性尤为困难，因为这是一个相对较新的领域。信用风险和市场风险度量往往拥有一百多年历史趋势信息。操作风险计量则相对较新，因此在用以评估风险状况变化的定性和定量跟踪指标方面需要投入更多关注。” (†) 假设机构的中央操作风险职能部门（ORF）对操作风险报告负有首要责任。关于操作风险报告所面临的挑战，下列陈述均为真，除了哪一项为假？

- A. The ORF should focus scarce resources on preventing and addressing major incidents rather than getting distracted by daily volatility
中央操作风险职能部门应将稀缺资源集中在预防 and 解决重大事件，而不是被日常波动分散注意力
- B. When the purpose is benchmarking across business units, losses should be reported in dollar terms (i.e., rather than percentage or basis points) because low percentages could mask high dollar losses
当目的是跨业务单位进行基准测试时，损失应以美元（即，而不是百分比或基点）报告，因为低百分比可能掩盖高美元损失
- C. The ORF should regularly review small, frequent losses to spot potential control breaches or process flaws; if these losses are consistent and random, consider including their average cost in the service price

中央操作风险职能部门应定期审查小、频繁的损失，以发现潜在的控制违规或流程缺陷；如果这些损失是连续且随机的，应考虑将它们的平均成本包含在服务价格中

D. The ORF can address the challenge of compiling qualitative risk data through either (i) conversion and summation, (ii) categorization, and (iii) worst-case reporting.

中央操作风险职能部门可以通过（i）转换和汇总，（ii）分类，和（iii）最坏情况报告来应对编译定性风险数据的挑战。

答案:B

解析:

B 选项正确。 False. Instead, percentages are preferred to dollars.

Says GARP, “Reporting operational losses as a percentage of gross income, of total cost (for a cost center), total budget (for a project), or in basis points of capital facilitates the comparability of similar entities across business units. Reporting in percentages instead of absolute values also makes the reporting scalable and applicable to entities of all sizes. Ideally, operational risk managers will also better understand the cost-benefit assessment and trade-offs related to implementing the appropriate controls and operating restrictions.”(†)

In regard to (A), (C) and (D), each is TRUE.

- The ORF should focus scarce resources on preventing and addressing major incidents rather than getting distracted by daily volatility. Says GARP, “A proper allocation of risk management resources is to focus on the prevention and remediation of large incidents and not to be dispersed and distracted by the containment of daily volatility”(†)
- The ORF should regularly review small, frequent losses to spot potential control breaches or process flaws; if these losses are consistent and random, consider including their average cost in the service price. Says GARP, “Small and frequent losses should be identified and analyzed regularly to detect any pattern that could signal a breach in control or a structural flaw in a process that would warrant an action plan. If these losses occur randomly, without any specific structure, and are structurally limited, stable, and repetitive, their average cost can be passed through to the customers as part of the cost of services.”(†)
- The ORF can address the challenge of compiling qualitative risk data through either (i) conversion and summation, (ii) categorization, and (iii) worst-case reporting.

(†) GARP Chapter 6, Risk Reporting, 2023 FRM Part 2, Topic 7.

考察: Operational Risk Reporting 重要程度: ☒☒☐

科目: Quantitative Analysis

79. According to Patrick McGuire, which of the following best summarizes the proximate cause of the US dollar shortage during the global financial crisis (GFC)?

根据 Patrick McGuire，哪一项最准确地总结了全球金融危机（GFC）期间美元短缺的直接原因？

- A. Depositors initiated a classic run on the banks and converted their domestic deposits to foreign currency
存款人发起了一场经典的银行挤兑，并将其国内存款转换为外国货币
- B. The effective maturity of banks’ dollar funding lengthened while their dollar assets shortened, and this decreased their maturity mismatch.
银行美元资金的有效期限延长，而美元资产期限缩短，这减少了它们的期限错配。
- C. The effective maturity of banks’ dollar funding shortened while their dollar assets lengthened, and this increased their maturity mismatch.
银行美元资金的有效期限缩短，而美元资产期限延长，这增加了它们的期限错配。
- D. Banks shifted their portfolios to non-bank assets that included banks’ retail and corporate lending, lending to hedge funds, and holdings of securities ranging from US Treasury and agency securities to structured finance products
银行将其投资组合转移到非银行资产，包括银行零售和公司贷款、对冲基金贷款以及从美国国债和机构证券到结构化金融产品的证券持有。

答案:C

解析:

Option C is correct. The effective maturity of banks’ dollar funding shortened while their dollar assets lengthened, and this increased their maturity mismatch.

McGuire (emphasis ours): “Events during the crisis led to severe disruptions in banks’ sources of short-term funding. Interbank markets seized up, and dislocations in FX swap markets made it even more expensive to obtain US dollars via swaps. Banks’ funding pressures were compounded by instability in non-bank sources of funds as well, notably dollar money market funds and dollar-holding central banks. The market stress meant that the effective maturity of banks’ US dollar funding shortened just as that of their US dollar assets lengthened, since many assets became difficult to sell in illiquid markets. This endogenous rise in maturity mismatch, difficult to hedge ex ante, generated the global US dollar shortage. Our estimate of the size of banks’ US dollar funding gaps at the onset of the crisis shed light on why the international policy response was necessary, and why it took the form of a global network of central bank swap lines.” (Patrick McGuire, Gotz von Peter, 2009. “The US Dollar Shortage in Global Banking and the International Policy Response,” BIS Working Papers, Bank for International Settlements.)

考察: Finance: Global Banking 重要程度: [■ ■ □]

科目: Quantitative Analysis

80. Janice is preparing to conduct factor regressions in order to discover the factor loadings in her firm’ s equity portfolio. She will begin by regressing three explanatory (aka, independent) variables: the market factor (MKT) plus two additional factors. Each of the following pairs of factors are good candidates for these additional factors EXCEPT which of the following pair is NOT a good candidate?

Janice 准备进行因子回归，以发现其公司股票投资组合中的因子载荷。她将首先回归三个解释变量：市场因子（MKT）加上两个额外因子。下列每一对因子中，哪一对不是好的候选因子？

- A. Quality and momentum (style factors)
质量与动量（风格因子）
- B. Strategy and reputation (style factors)
- C. Economic growth and inflation (macro factors)
经济增长与通胀（宏观因子）
- D. Real interest rates and liquidity (macro factors)
实际利率与流动性（宏观因子）

答案:B

解析:

B 选项正确。 False. In reality, it is difficult to invest directly in reputation, and it seems almost impossible to imagine how to operationalize strategy as an investable factor.

According to Andrew Ang (see <https://www.blackrock.com/us/individual/investment-ideas/what-is-factor-investing>) there are two primary types of factors: macroeconomic factors and style factors. Macroeconomic factors include economic growth, credit (default risk), real interest rates, emerging markets (political and sovereign risk), inflation, and liquidity. Style factors include value, quality, minimum volatility, size, momentum, and carry.

In this regard, each of (A), (C) and (D) is TRUE:

- Quality and momentum are genuine style factors
- Economic growth and inflation are genuine macro factors

考察: Factor Investing 重要程度: [■ ■ □]

科目: Quantitative Analysis